THEMATIC REPRESENTATION OF SOIL LOSS POTENTIAL ZONES USING MCDM TECHNIQUE: A CASE STUDY ON DARAKESWAR RIVER BASIN IN BANKURA, WEST BENGAL

1. INTRODUCTION:

Soil erosion poses a significant environmental challenge, particularly in regions where human activities and natural processes converge to destabilize the landscape. Understanding the spatial distribution of soil loss potential is crucial for sustainable land management and conservation planning. This task involves evaluating various physical, climatic, and anthropogenic factors that contribute to soil erosion, integrating them into a comprehensive framework for identifying vulnerable zones. The interplay of these factors is complex, necessitating a multi-criteria approach to accurately delineate areas at high risk of soil degradation.

Soil texture plays a pivotal role in determining a region's susceptibility to erosion, as finer particles are more prone to detachment and transport by water or wind. Coupled with this, the geomorphology of an area—characterized by landforms and surface features—provides insights into the natural predisposition of the terrain to erosion. Hydrological factors such as the Stream Power Index and Sediment Transport Index further enhance our understanding by quantifying the potential energy of water to erode and carry sediment downstream. These indices, derived from topographic and hydrological data, are critical for evaluating areas where erosion processes are most intense.

The influence of slope cannot be overstated, as both steepness and length directly impact the velocity of surface runoff and its capacity to erode soil. Rainfall intensity acts as a catalyst, with higher precipitation levels exacerbating the detachment of soil particles. Land use and land cover (LULC) changes, driven by agricultural expansion, deforestation, or urbanization, modify the protective vegetation cover, often accelerating erosion. Human interventions, such as the construction of roads, settlements, and drainage networks, alter natural water flow and contribute to localized erosion hotspots.

Integrating these diverse factors using geospatial tools allows for a detailed and systematic assessment of soil loss potential. High-resolution data and spatial analysis techniques provide the capability to map vulnerable zones with precision. The inclusion of proximity to roads, rivers, and settlements ensures a holistic perspective, accounting for both natural and human-induced drivers of soil erosion. This approach not only identifies areas at risk but also aids in prioritizing regions for intervention and resource allocation.

The outcome of such an analysis is instrumental for policymakers, land managers, and environmental planners. By pinpointing critical zones, it becomes possible to design targeted erosion control measures, optimize land use practices, and mitigate the adverse impacts of soil erosion on agriculture, water quality, and ecosystem health. This endeavour is a step towards sustainable development, ensuring that the balance between environmental conservation and human activities is maintained.

2. STUDY AREA:

The study area is confined to the portion of the Darakeswar River Basin that lies within the Bankura district of West Bengal, India. This region is part of the sub-basin of the larger Damodar River system, characterized by its undulating terrain, diverse geomorphological features, and a semi-arid climatic condition. Geographically, the area falls approximately between 22°52'N to 23°38'N latitude and 86°58'E to 87°34'E longitude, encompassing a diverse landscape shaped by natural processes and anthropogenic activities.

The Darakeswar River, a perennial stream, is one of the significant tributaries contributing to the hydrology of the region. The basin within Bankura exhibits a mix of rural settlements, agricultural land, and forested zones, which collectively influence its hydrological and erosion patterns. The terrain is marked by lateritic uplands, low-lying floodplains, and gently sloping areas, which together dictate surface runoff, sediment transport, and soil erosion dynamics. The region is also characterized by a network of seasonal streams that contribute to the main river system during the monsoon season.

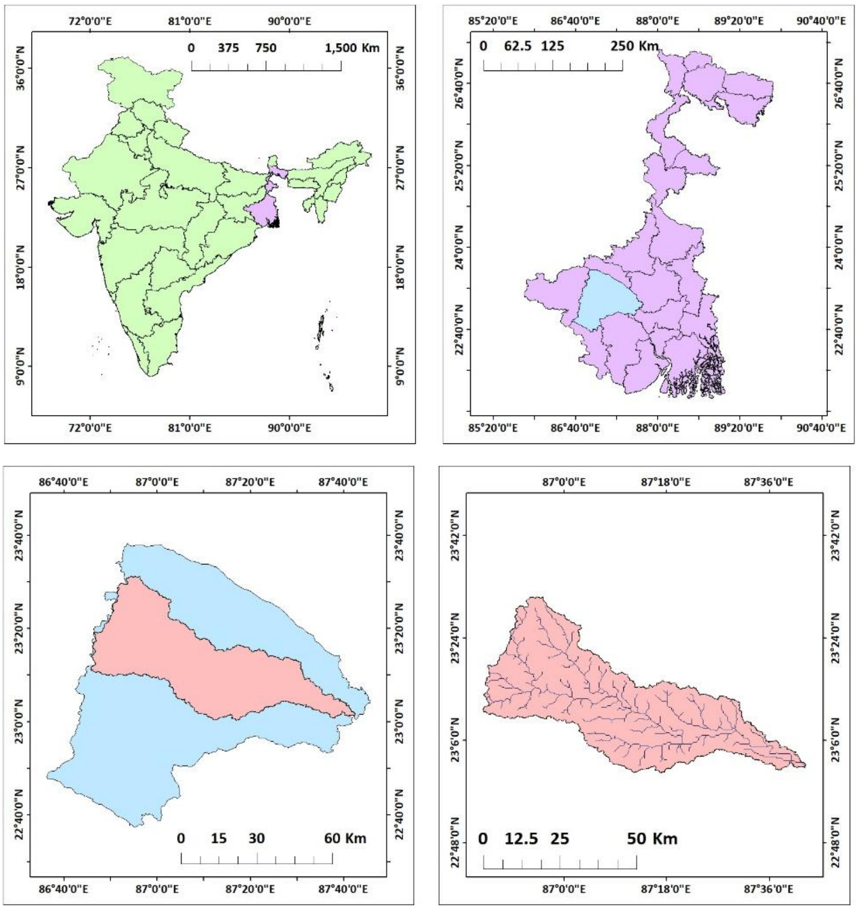


Figure 1: Location map of Darakeswar River Basin in Bankura district

The climate is predominantly tropical, with distinct summer, monsoon, and winter seasons. The monsoon season contributes to the majority of the annual rainfall, often leading to high-intensity rainfall events that exacerbate soil erosion. The land use and land cover (LULC) in the study area primarily include agricultural fields, scrubland, dense forest patches, and settlements. These diverse land types, coupled with the area's geomorphology and drainage density, make it a critical zone for assessing soil loss potential.

This section of the Darakeswar River Basin is an ideal site for studying soil erosion, given its varied topography, human interventions, and climatic influences. The findings from this region can inform effective land and water management strategies, ensuring environmental sustainability and resource optimization for local communities.

3. DATA SOURCES AND METHODOLOGY:

DATA SOURCES:

- **Soil Texture**:
  - Source: National Bureau of Soil Survey and Land Use Planning (NBSSLUP).
  - Details: Provides soil texture classification, essential for assessing erodibility.
- **Stream Power Index (SPI)**:
  - Derived from: DEM (USGS).
  - Calculated using the formula:

$$SPI = A \cdot \tan(\beta)$$

  - where:
    - A: Upslope contributing area (m^2).
    - β: Slope gradient in radians.
- **Sediment Transport Index (STI)**:
  - Derived from: DEM (USGS).
  - Formula:

$$STI = \left(\frac{A}{22.13}\right)^m \cdot \left(\frac{\sin\theta}{0.0896}\right)^n$$

- **Geomorphology**:
  - Source: Bhukosh portal, Geological Survey of India (GSI).
  - Details: Provides geomorphological maps for classifying terrain features.
- **Slope Length and Steepness Factor (LS)**:
  - Derived from: Digital Elevation Model (DEM) obtained from the USGS Earth Explorer.
  - Calculated using ArcGIS with the formula:

$$LS = \left(\frac{\lambda}{22.13}\right)^m \cdot (\sin\theta \cdot 0.0896)^n$$

  - where:
    - λ: Slope length (m).
    - θ: Slope angle in radians.
    - m and n: Empirical constants (typically m=0.5, n=1.3m)

- **Rainfall Intensity**:
    - Source: Climate Hazards Group InfraRed Precipitation with Station Data (CHRS data portal).
    - Details: Provides NETCDF data.
- **Land Use and Land Cover (LULC)**:
    - Source: ESRI Sentinel-2 Land Cover Explorer.
    - Details: Classified LULC data for assessing human and natural influences on soil erosion.
- **Drainage Density**:
    - Source: Downloaded spatial datasets from BB Bike.
    - Derived from: Processed in ArcGIS using line density feature.
- **Distances from Road, River, and Settlement**:
    - Source: Downloaded spatial datasets from BB Bike.
    - Derives from: Processed in ArcGIS to calculate Euclidean distances for each feature.

These diverse data sources and derived parameters provide a comprehensive framework for analysing soil loss potential using geospatial tools.

METHODOLOGY:

The methodology for assessing soil loss potential integrates advanced geospatial techniques and multi-criteria decision-making approaches to achieve precise results. Initially, all the spatial data layers representing various factors influencing soil erosion were collected and processed using ArcGIS. Each criterion, including soil texture, geomorphology, slope steepness and length, Stream Power Index (SPI), Sediment Transport Index (STI), rainfall intensity, land use and land cover (LULC), drainage density, and proximity to roads, rivers, and settlements, was standardized into a uniform spatial resolution to ensure compatibility for analysis.

The Frequency Ratio (FR) method was applied to quantify the relationship between each factor and soil loss susceptibility. For each criterion, the spatial data were classified into discrete categories, and the frequency ratio was calculated by comparing the distribution of observed erosion-prone areas to the total distribution within each class. These weights were used to assess the relative contribution of each factor to soil erosion potential.

Simultaneously, the Fuzzy Analytical Hierarchy Process (Fuzzy AHP) was employed to derive weights for the criteria based on expert judgment and pairwise comparisons. This method incorporated fuzziness to handle uncertainties in decision-making, producing a consistent and reliable hierarchy of factor importance. The weights from the Fuzzy AHP method were combined with the FR outputs to develop a composite soil loss potential map.

Table for Frequency ratio for Soil loss potentiality

Rk	NAME	Rj	n-Rj+1	n-Rk+1	Wij	PERCENTAGE
1	Geomorphology	11	2	12	0.025641	2.564102564
2	Slope steepness &Slope length	3	10	11	0.128205	12.82051282
3	Slope	4	9	10	0.115385	11.53846154
4	Stream power index	2	11	9	0.141026	14.1025641
5	Sediment transfer index	1	12	8	0.153846	15.38461538
6	Dist.frm.road	12	1	7	0.012821	1.282051282
7	Dist.frm.settlement	9	4	6	0.051282	5.128205128
8	LULC	10	3	5	0.038462	3.846153846
9	Soil texture	5	8	4	0.102564	10.25641026
10	Drainage density	7	6	3	0.076923	7.692307692
11	Rainfall	6	7	2	0.089744	8.974358974
12	Lineament	8	5	1	0.064103	6.41025641
N=12				78	1	100

TABLE FOR FUZZY AHP

Sl. No.	Parameters	Geomorphology	LS	Slope length	SPI	STI	Dist.frm.road	Dist.frm.settlement	Dist.frm.river	LULC	Soil texture	Drainage density	Rainfall	Lineament
1	Geomorphology	1,1,1	7,8,9	6,7,8	4,5,6	2,3,4	2,3,4	1,1,1	1,1,1	3,4,5	5,6,7	2,3,4	6,7,8	5,6,7
2	Slope steepness	1/9,1/8,1/7	1,1,1	9,9,9	1,2,3	4,5,6	1,1,1	1,2,3	1,1,1	2,3,4	6,7,8	1,1,1	4,5,6	5,6,7
3	Slope length	1/8,1/7,1/6	1/9,1/9,1/9	1,1,1	1,1,1	2,3,4	1,1,1	1,1,1	2,3,4	1,1,1	1,1,1	2,3,4	4,5,6	6,7,8
4	Stream power index	1/6,1/5,1/4	1/3,1/2,1/1	1,1,1	1,1,1	2,3,4	1,1,1	1,1,1	2,3,4	2,3,4	1,1,1	3,4,5	3,4,5	3,4,5
5	Stream transfer index	1/4,1/3,1/2	1/6,1/5,1/4	1/4,1/3,1/2	1/2,1/3,1/4	1,1,1	1,1,1	1,1,1	3,4,5	2,3,4	1,1,1	2,3,4	4,5,6	4,5,6
6	Dist.frm.road	1/4,1/3,1/2	1,1,1	1,1,1	1,1,1	1,1,1	1,1,1	6,7,8	4,5,6	1,1,1	1,1,1	1,1,1	2,3,4	2,3,4
7	Dist.frm.settlement	1,1,1	1/3,1/2,1/1	1,1,1	1,1,1	1,1,1	1/8,1/7,1/6	1,1,1	2,3,4	2,3,4	2,3,4	1,1,1	1,1,1	2,3,4
8	Dist.frm.river	1,1,1	1,1,1	1/4,1/3,1/2	1/4,1/3,1/2	1/5,1/4,1/3	1/6,1/5,1/4	1/4,1/3,1/2	1,1,1	4,5,6	1,1,1	4,5,6	1,1,1	4,5,6
9	LULC	1/5,1/4,1/3	1/4,1/3,1/2	1,1,1	1/4,1/3,1/2	1/4,1/3,1/2	1,1,1	1/4,1/3,1/2	1/6,1/5,1/4	1,1,1	5,6,7	1,1,1	1,1,1	1,2,3
10	Soil texture	1/7,1/6,1/5	1/8,1/7,1/6	1,1,1	1,1,1	1,1,1	1,1,1	1/4,1/3,1/2	1,1,1	1/7,1/6,1//5	1,1,1	1,1,1	2,3,4	4,5,6
11	Drainage density	1/4,1/3,1/2	1,1,1	1/4,1/3,1/2	1/5,1/4,1/3	1/4,1/3,1/2	1,1,1	1,1,1	1/6,1/5,1/4	1,1,1	1,1,1	1,1,1	4,5,6	5,6,7
12	Rainfall	1/8,1/7,1/6	1/6,1/5,1/4	1/6,1/5,1/4	1/5,1/4,1/3	1/6,1/5,1/4	1/4,1/3,1/2	1,1,1	1,1,1	1,1,1	1/4,1/3,1/2	1/6,1/5,1/4	1,1,1	2,3,4
13	Lineament	1/7,1/6,1/5	1/7,1/6,1/5	1/8,1/7,1/6	1/5,1/4,1/3	1/6,1/5,1/4	1/4,1/3,1/2	1/4,1/3,1/2	1/6,1/5,1/4	1/3,1/2,1/1	1/6,1/5,1/4	1/7,1/6,1/5	1/4,1/3,1/2	1,1,1

TABLE FOR GEOMETRIC MEAN

	A	B	C
A,B,C	45	55	66
A1,B1,C1	37.11	43.125	50.142
A2,B2,C2	22.236	27.253	32.277
A3,B3,C3	20.5	26.7	33.25
A4,B4,C4	20.166	25.2	30.5
A5,B5,C5	22.25	26.333	30.5
A6,B6,C6	14.708	18.976	22.666
A7,B7,C7	17.889	21.45	25.083
A8,B8,C8	12.366	14.783	17.583
A9,B9,C9	13.66	15.809	18.066
A10,B10,C	13.116	15.222	20.583
A11,B11,C	6.491	6.729	7.5
A12,B12,C	3.109	3.992	5.55
TOTAL	248.601	300.572	359.7

TABLE FOR MEAN WEIGHT

Parameters	A	B	C	Wi	PERCENTAGE
Geomorphology	0.125104	0.182984	0.265486	0.396584	18.24688753
Slope steepness	0.103169	0.143476	0.201697	0.313878	14.44157447
Slope length	0.061818	0.09067	0.129835	0.195767	9.007261848
Stream power index	0.056992	0.088831	0.133748	0.190405	8.760581372
Stream transfer index	0.056063	0.08384	0.122687	0.180799	8.318592307
Dist.frm.road	0.061857	0.08761	0.122687	0.190362	8.758596463
Dist.frm.settlement	0.04089	0.063133	0.091174	0.134414	6.184408716
Dist.frm.river	0.049733	0.071364	0.100897	0.154729	7.119116789
LULC	0.034379	0.049183	0.070728	0.107137	4.929411675
Soil texture	0.037976	0.052596	0.072671	0.114796	5.281783119
Drainage density	0.036464	0.050643	0.082795	0.114706	5.277622484
Rainfall	0.018046	0.022387	0.030169	0.050489	2.323015162
Lineament	0.008643	0.013281	0.022325	0.029366	1.351148062
TOTAL				2.173433	100

To finalize the analysis, a fishnet grid was created across the study area, dividing it into uniformly spaced points. These fishnet points were clipped to the study area boundary, ensuring that they represented the entire spatial extent of the region accurately. Inverse

Distance Weighting (IDW) interpolation was then applied to the clipped fishnet points to produce a continuous surface representing soil loss potential. This final output provides an intuitive and spatially explicit representation of zones vulnerable to soil erosion, facilitating effective planning and mitigation strategies.

4. RESULTS AND DISCUSSION:

Soil texture map:

The map depicts the spatial distribution of soil texture within the Darakeswar River Basin. Soil texture, characterized by the relative proportions of sand, silt, and clay, significantly influences soil erodibility. Sandy clay loam soils, with a balanced mixture of sand, silt, and clay, exhibit moderate erodibility. Sandy loam soils, with a higher proportion of sand, are more susceptible to erosion due to their lower cohesion. Loam soils, with a balanced mixture of all three particle sizes, generally have good structural stability and moderate erodibility. Understanding the spatial distribution of soil texture is crucial for assessing soil erosion risk and implementing appropriate conservation measures.

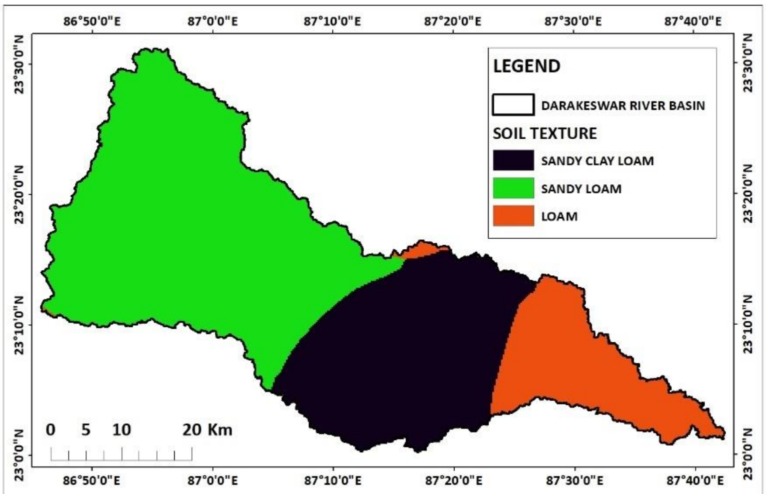


Figure 2: Soil texture map

Stream Power Index (SPI) map:

The map depicts the spatial distribution of Stream Power Index (SPI) values within the Darakeswar River Basin. SPI is a geomorphic index that quantifies the erosive power of flowing water. Higher SPI values indicate areas with greater erosive potential, while lower values indicate areas with lower erosive potential. The map reveals a significant spatial variation in SPI values across the basin. Areas with higher SPI values, often associated with steeper slopes and higher flow velocities, are more prone to erosion. Conversely, areas with lower SPI values, typically characterized by gentle slopes and lower flow velocities, are less susceptible to erosion. Understanding the spatial distribution of SPI

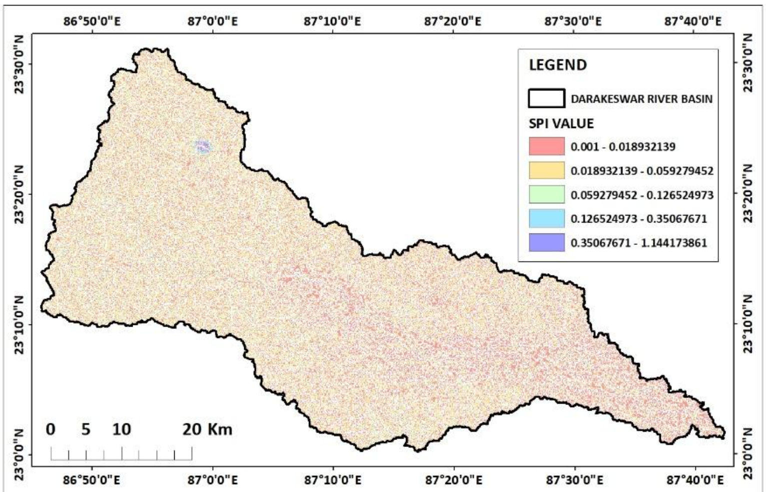


Figure 3: SPI map

is crucial for assessing soil erosion risk and implementing appropriate conservation measures.

Sediment Transport Index (STI) map:

The map depicts the spatial distribution of Sediment Transport Index (STI) values within the Darakeswar River Basin. STI is a geomorphic index that quantifies the potential for sediment transport by flowing water. Higher STI values indicate areas with greater sediment transport potential, while lower values indicate areas with lower sediment transport potential. The map reveals a significant spatial variation in STI values across the basin. Areas with higher STI values, often associated with steeper slopes and higher flow velocities, are more prone to soil erosion and sediment transport. Conversely, areas with lower STI values, typically characterized by gentle slopes and lower flow velocities, are less susceptible to erosion and sediment transport. Understanding the spatial distribution of STI is crucial for assessing soil erosion risk and implementing appropriate conservation measures.

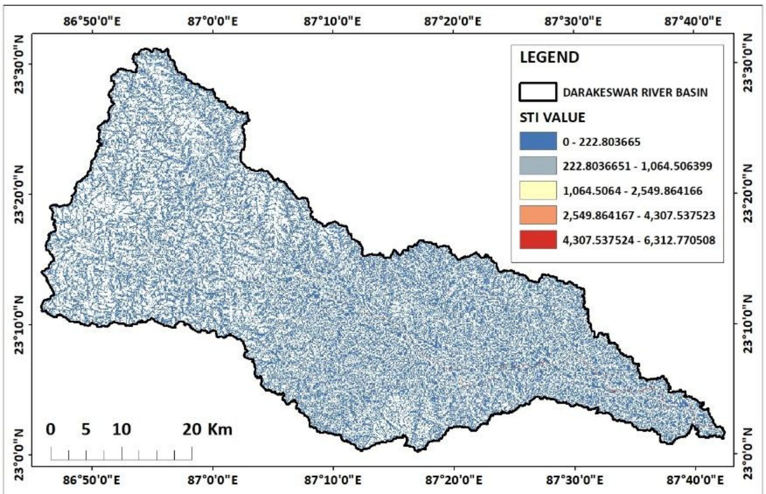


Figure 4: STI map

Geomorphology:

The map depicts the geomorphological characteristics of the Darakeswar River Basin. Geomorphology plays a crucial role in soil erosion processes as it influences factors like slope steepness, drainage density, and sediment transport. The basin exhibits a diverse range of geomorphological features, including alluvial plains, floodplains, moderately dissected hills and valleys, and pediment-pediplain complexes. Alluvial plains and floodplains, characterized by gentle slopes and low erodibility, are less prone to soil erosion. In contrast, moderately dissected hills and valleys, with steeper slopes and higher drainage density, are more susceptible to erosion. Pediment-pediplain complexes, with their low relief

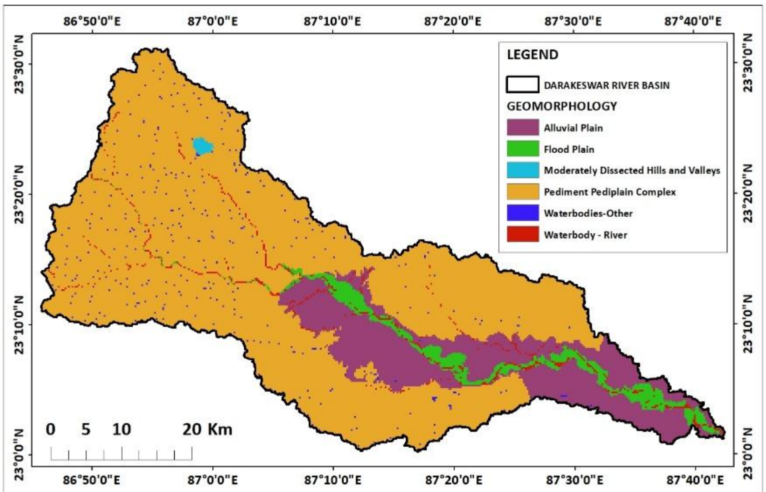


Figure 5: Geomorphology map

and gentle slopes, generally exhibit moderate erodibility. Understanding the spatial distribution of these geomorphological features is essential for assessing soil erosion risk and implementing appropriate conservation measures.

Slope steepness and length map:

The map depicts the spatial distribution of slope steepness and length within the Darakeswar River Basin. Slope steepness and length are crucial factors influencing soil erosion as they determine the potential for surface runoff and soil detachment. Steeper slopes and longer slopes are more prone to soil erosion, especially during intense rainfall events.

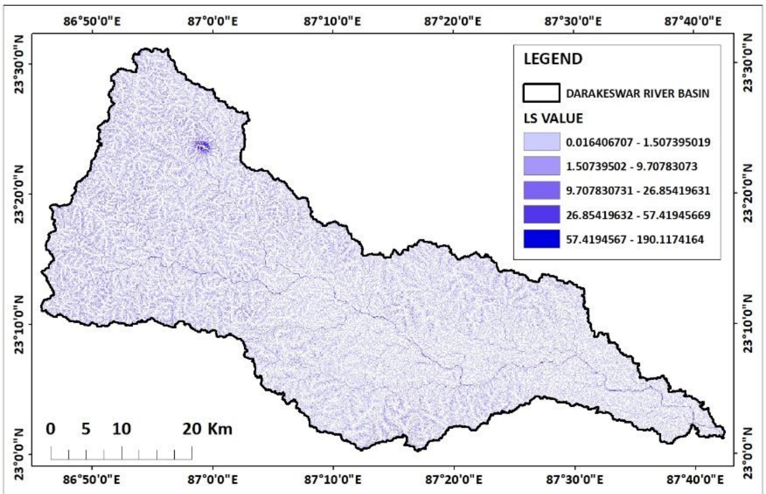


Figure 6: LS map

The map reveals a significant variation in slope steepness and length across the basin. Areas with steeper slopes and longer slopes are more susceptible to soil erosion. Conversely, areas with gentle slopes and shorter slopes are less prone to erosion. Understanding the spatial distribution of slope steepness and length is crucial for assessing soil erosion risk and implementing appropriate conservation measures.

Rainfall intensity map:

The map depicts the spatial distribution of rainfall intensity within the Darakeswar River Basin. Rainfall intensity is a crucial factor influencing soil erosion as it determines the erosive power of raindrops. Higher rainfall intensity can lead to increased surface runoff and soil detachment. The map reveals a significant variation in rainfall intensity across the basin. Areas with higher rainfall intensity are more prone to soil erosion, especially in regions with steep slopes and low vegetation cover. Conversely, areas with lower rainfall intensity are less susceptible to erosion. Understanding the spatial distribution of rainfall intensity is crucial for assessing soil erosion risk and implementing appropriate conservation measures. By identifying these areas, policymakers and land managers can prioritize

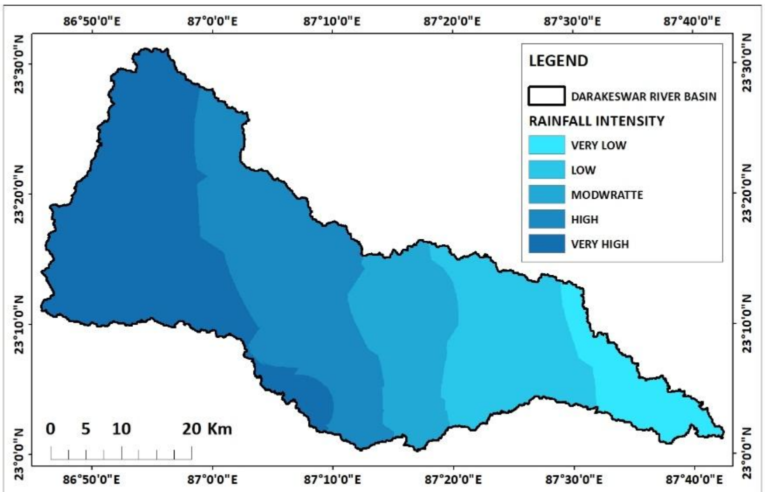


Figure 7: Rainfall intensity map

conservation efforts and implement targeted interventions to mitigate soil erosion and protect valuable soil resources.

LULC map:

The map depicts the spatial distribution of land use and land cover (LULC) classes within the Darakeswar River Basin. LULC plays a crucial role in soil erosion as different land cover types have varying degrees of soil protection. The basin exhibits a diverse range of LULC classes, including dense forest, flooded vegetation, agricultural land, built-up areas, bare land, and scrub field. Dense forests and flooded vegetation provide excellent soil cover and reduce erosion. Agricultural land, depending on the type of crop and management practices, can have varying degrees of erosion potential. Built-up areas and bare land, with their low vegetation cover, are highly susceptible to erosion.

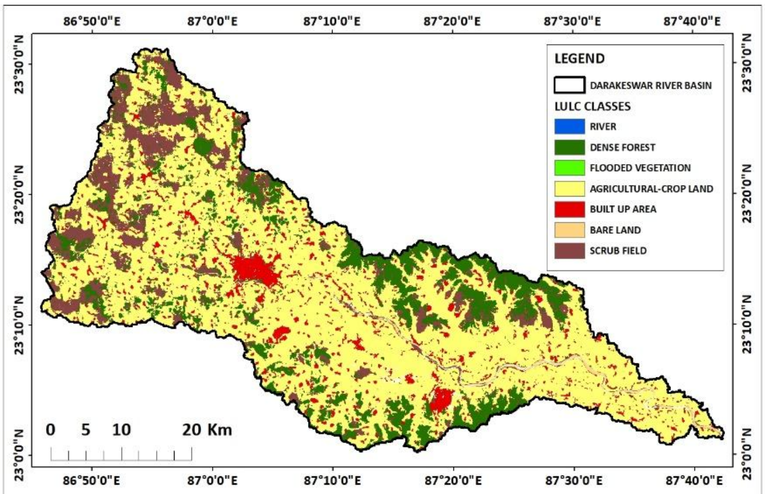


Figure 8: LULC map

Drainage density map:

The map depicts the spatial distribution of drainage density within the Darakeswar River Basin. Drainage density is a measure of the total length of streams in a given area, and it significantly influences soil erosion processes. Higher drainage density can lead to increased surface runoff and soil erosion, especially in areas with steep slopes and erodible soils. The map reveals a significant variation in drainage density across the basin. Areas with higher drainage density are more prone to soil erosion, while areas with lower drainage density are less susceptible. Understanding the spatial distribution of drainage density is crucial for assessing soil erosion risk and implementing appropriate conservation measures.

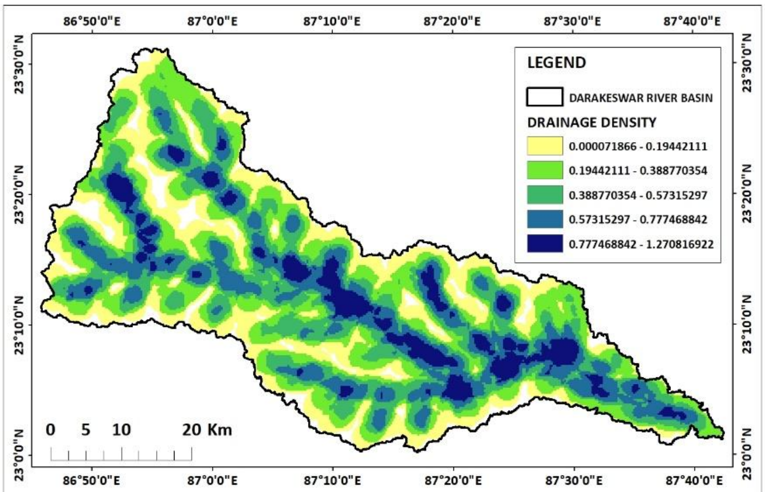


Figure 9: Drainage density map

Distance from road map:

The map depicts the spatial distribution of distances from roads within the Darakeswar River Basin. Distance from roads is a crucial factor influencing soil erosion as it affects accessibility for agricultural practices and conservation measures. Areas closer to roads may have better access to agricultural inputs and services, potentially leading to improved land management practices and reduced erosion. However, proximity to roads can also increase the risk of soil erosion due to factors like road construction and traffic-related activities. The map reveals a significant variation in distances from roads across the basin. Areas closer to roads are generally more accessible and may have better infrastructure for soil conservation. However, it is important to consider other factors like slope, soil type, and land use to assess the overall soil erosion risk. Understanding the spatial distribution of distances from roads is crucial for implementing targeted conservation measures and promoting sustainable land management practices.

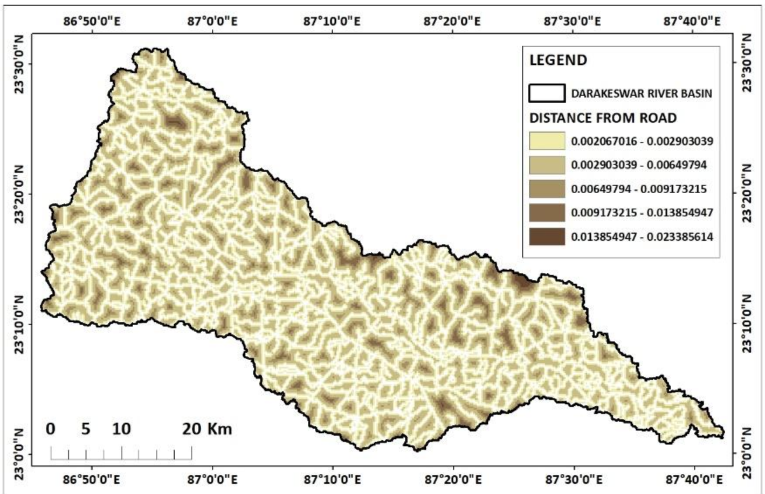


Figure 10: Distance from road map

Distance from river map:

The map depicts the spatial distribution of distances from rivers within the Darakeswar River Basin. Areas closer to rivers may have higher soil moisture content, reduced surface runoff, and increased vegetation cover, which can help mitigate soil erosion. However, proximity to rivers can also increase the risk of riverbank erosion and flooding, which can contribute to soil loss. The map reveals a significant variation in distances from rivers across the basin. Areas closer to rivers are generally more influenced by riverine processes and may have lower soil erosion rates. Conversely, areas farther from rivers may be more susceptible to erosion due to factors like lower soil moisture content and increased surface runoff. Understanding the spatial distribution of distances from rivers is crucial for assessing soil erosion risk and implementing appropriate conservation measures.

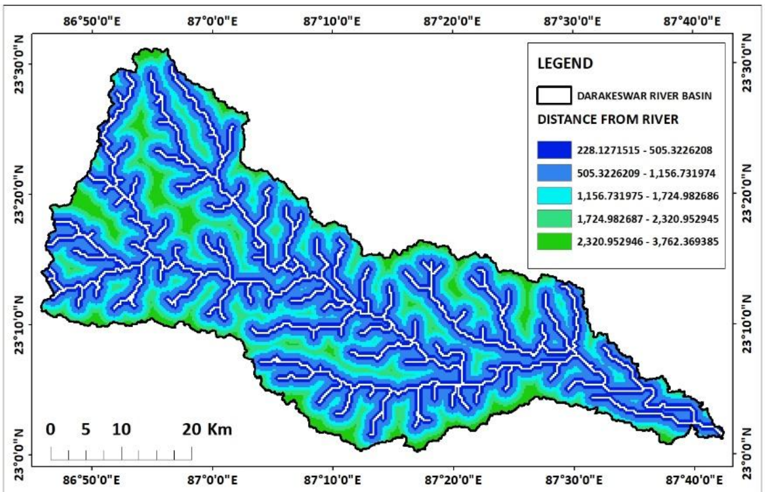


Figure 11: Distance from river map

Distance from settlement map:

The map depicts the spatial distribution of distances from settlements within the Darakeswar River Basin. Distance from settlements is a crucial factor influencing soil erosion as it affects the intensity of human activities and land use practices. Areas closer to settlements may experience higher levels of disturbance, such as deforestation, agriculture, and construction, which can increase soil erosion. Conversely, areas farther from settlements may be less impacted by human activities and retain more natural vegetation cover, which can help protect soils from erosion. The map reveals a significant variation in distances from settlements across the basin. Areas closer to settlements are generally more prone to soil erosion due to increased human activities. Conversely, areas farther from settlements may be less susceptible to erosion due to lower levels of disturbance. Understanding the spatial distribution of distances from settlements is crucial for assessing soil erosion risk and implementing appropriate conservation measures.

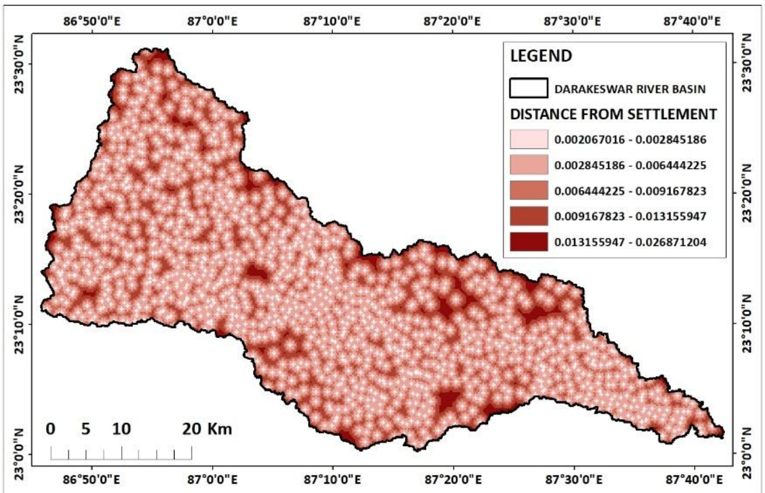


Figure 12: Distance from settlement map

Weighted Sum map:

The map depicts the spatial distribution of soil loss potential within the Darakeswar River Basin. The soil loss potential is calculated using a weighted sum index, which combines multiple factors such as soil texture, slope, rainfall intensity, land use/land cover, and drainage density. The map reveals a significant variation in soil loss potential across the basin. Areas with higher values indicate higher soil loss potential, while areas with lower values indicate lower soil loss potential. Understanding the spatial distribution of soil loss potential is crucial for prioritizing conservation efforts and implementing appropriate soil conservation measures.

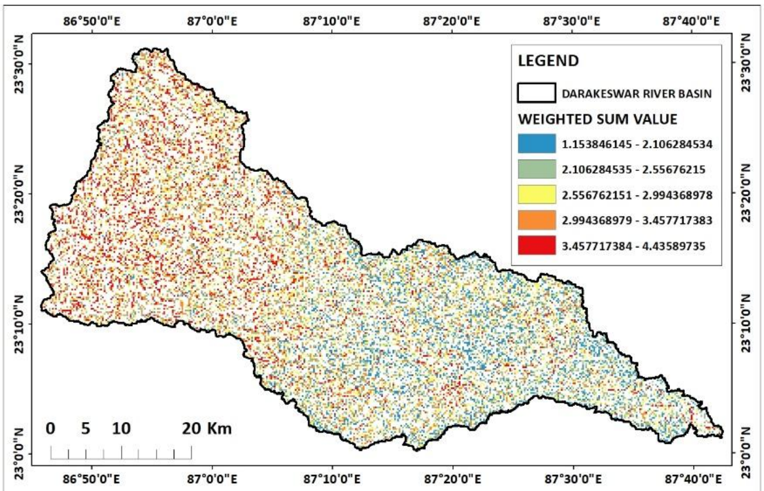


Figure 13: Weighted Sum map

IDW map based on soil loss potential zones:

The areas with high soil loss potential, represented by the red zones on the map, are predominantly concentrated along the steep slopes and near the river channels. These regions are characterized by higher Stream Power Index (SPI) and Sediment Transport Index (STI) values, indicating intense water flow and sediment mobility. The combination of steep gradients and high runoff energy accelerates soil detachment and transport in these areas. Additionally, these zones are often associated with exposed soil surfaces due to sparse vegetation cover, as observed in the LULC classification, further contributing to higher erosion risks.

Moderate soil loss zones, marked in yellow, cover substantial portions of the basin. These areas typically exhibit a mix of agricultural land and scrubland, with moderate slope gradients and less intense hydrological activity. While the rainfall intensity in these zones may still contribute to soil erosion, the presence of some vegetation or crop cover provides partial protection against surface runoff. However, unsustainable agricultural practices in these areas could potentially escalate the erosion risk if not managed appropriately.

The low soil loss zones, represented in blue, are generally found in relatively flat terrain with dense vegetation cover, such as forests or stable agricultural land. These areas exhibit low drainage density and minimal surface runoff energy, which significantly reduce the potential for soil erosion. The protective vegetation cover in these zones plays a crucial role in stabilizing the soil and minimizing erosion rates. Such areas are critical for maintaining the ecological balance of the region and should be prioritized for conservation efforts.

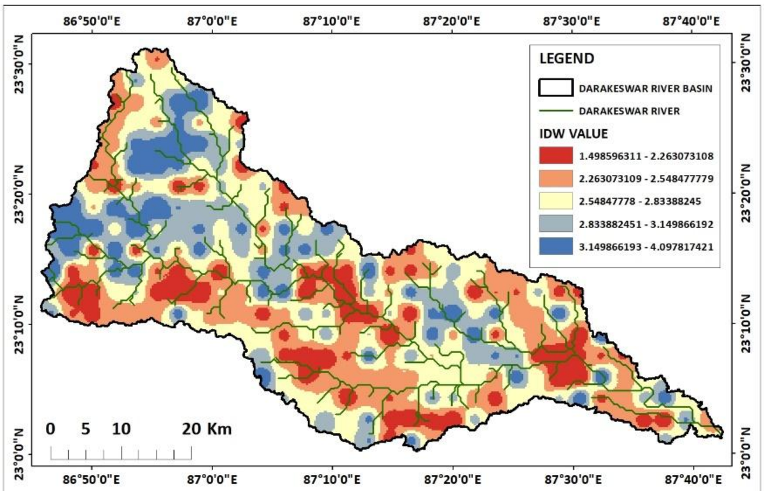


Figure 14: IDW map based on soil loss potential zones

The proximity analysis further highlights the influence of anthropogenic factors on soil erosion. Areas closer to roads, settlements, and rivers demonstrate higher soil loss potential due to increased human interventions, such as deforestation, construction, and agricultural expansion. These activities disrupt the natural landscape and intensify runoff, leading to localized erosion hotspots. Proper land use planning and management are essential to mitigate these impacts in vulnerable zones.

The integration of Frequency Ratio (FR) and Fuzzy AHP methods with geospatial tools has proven effective in identifying and prioritizing soil loss potential zones. While the FR method provided quantitative insights into the relationship between individual factors and erosion susceptibility, the Fuzzy AHP method added a layer of decision-making robustness by incorporating expert judgment. The IDW interpolation applied to fishnet points ensured a continuous and spatially detailed representation of the erosion-prone areas, making the results highly actionable.

Despite the robustness of the methodology, certain limitations must be acknowledged. The accuracy of the results depends heavily on the quality and resolution of the input datasets, such as DEM, rainfall intensity, and LULC. Additionally, the use of IDW interpolation, while effective, assumes a linear spatial relationship, which may not capture complex erosion processes in some areas. Future studies could explore more advanced machine learning approaches or dynamic models to enhance prediction accuracy.

The results of this study hold significant implications for land and water resource management in the Darakeswar River Basin. High-priority zones identified on the map can serve as focal areas for implementing soil conservation measures, such as afforestation, contour farming, or check dams. Moreover, the insights gained from this analysis can inform local policymakers and stakeholders about the critical need to balance development activities with environmental sustainability. By addressing the key drivers of soil erosion, this research contributes to safeguarding the ecological health and agricultural productivity of the region.

5. CONSLUSION:

The analysis of soil loss potential in the Darakeswar River Basin within Bankura district has provided a comprehensive understanding of erosion susceptibility across the region. By integrating multiple criteria, the study effectively delineates areas with varying erosion risks. The application of Frequency Ratio (FR) and Fuzzy AHP methods, combined with geospatial tools like ArcGIS and IDW interpolation, ensured a robust and precise spatial representation of soil erosion potential.

The results revealed that high soil loss zones are primarily located near river channels, steep slopes, and areas with sparse vegetation, often exacerbated by human activities such as agriculture and settlement expansion. Moderate erosion zones are associated with mixed land use, while low-risk zones are characterized by flat terrain and dense vegetation, highlighting the protective role of forested areas.

This study underscores the critical need for targeted soil conservation measures in high-risk areas, such as afforestation, contour farming, and sustainable land management practices. The findings provide valuable insights for policymakers and stakeholders, facilitating informed decisions to mitigate soil erosion and promote long-term environmental sustainability in the region.

THEMATIC REPRESENTATION OF GROUND WATER POTENTIAL ZONES USING MCDM TECHNIQUE: A CASE STUDY ON PHANSIDEWA BLOCK, DARJEELING, WEST BENGAL

1. INTRODUCTION:

Groundwater, a vital natural resource, is essential for various human activities, including agriculture, industry, and domestic use. However, its availability and quality vary significantly across different regions. To effectively manage and utilize groundwater resources, it is crucial to identify and delineate groundwater potential zones. These zones represent areas with high potential for groundwater occurrence and extraction. Several factors influence groundwater potential, including geological, hydrological, and topographical characteristics. Geological formations, such as porous and permeable sedimentary rocks like sandstone and limestone, are ideal for groundwater storage and transmission. These formations act as aquifers, which are underground layers of water-bearing rock or sediment. Hydrological factors, such as rainfall, infiltration rates, and evapotranspiration, play a significant role in groundwater recharge and discharge. Topographical features, including slope, elevation, and drainage patterns, influence groundwater flow and accumulation.

To delineate groundwater potential zones, a multi-criteria approach is commonly employed. This involves integrating various parameters, such as geology, hydrogeology, geomorphology, soil type, land use/land cover, rainfall, and drainage density. These parameters are often analysed using Geographic Information Systems (GIS) to create thematic maps. By overlaying these maps, it is possible to identify areas with favourable conditions for groundwater occurrence. One of the key factors in groundwater potential assessment is the identification of recharge areas. These are areas where rainwater infiltrates the ground and replenishes groundwater aquifers. Factors such as soil permeability, slope, and vegetation cover influence the rate of infiltration.

Another important aspect is the delineation of discharge areas. These are areas where groundwater discharges to surface water bodies, such as rivers and lakes. Understanding discharge areas is crucial for assessing the impact of groundwater extraction on surface water flow and water quality.

In conclusion, the identification and delineation of groundwater potential zones is essential for sustainable groundwater management. By considering various factors and employing a multi-criteria approach, it is possible to assess groundwater availability, quality, and vulnerability. This information can be used to develop appropriate groundwater management strategies, including groundwater extraction limits, recharge enhancement measures, and pollution prevention measures. By adopting sound groundwater management practices, we can ensure the long-term sustainability of this precious resource.

2. STUDY AREA:

The study area, Phansidewa block, is located in the Darjeeling district of West Bengal, India. It lies within the geographical coordinates of 26° 53' N to 27° 10' N latitude and 88° 15' E to 88° 35' E longitude. This region is characterized by its hilly terrain, dense forest cover, and abundant water bodies.

The study area, Phansidewa block of Darjeeling district in West Bengal, is located between 26° 53' 30" N to 27° 00' 00" N latitude and 88° 15' 00" E to 88° 22' 30" E longitude. It encompasses a diverse range of physiographic features, including steep hills, valleys, and river basins. The region is characterized by a humid subtropical climate with abundant rainfall, particularly during the monsoon season.

Geologically, the Phansidewa block is predominantly composed of metamorphic rocks, such as schist, gneiss, and quartzite. These rocks, formed under high pressure and temperature, exhibit varying degrees of permeability and porosity, influencing groundwater occurrence and flow. The complex geological structure, with numerous faults and fractures, further complicates the groundwater hydrology of the region.

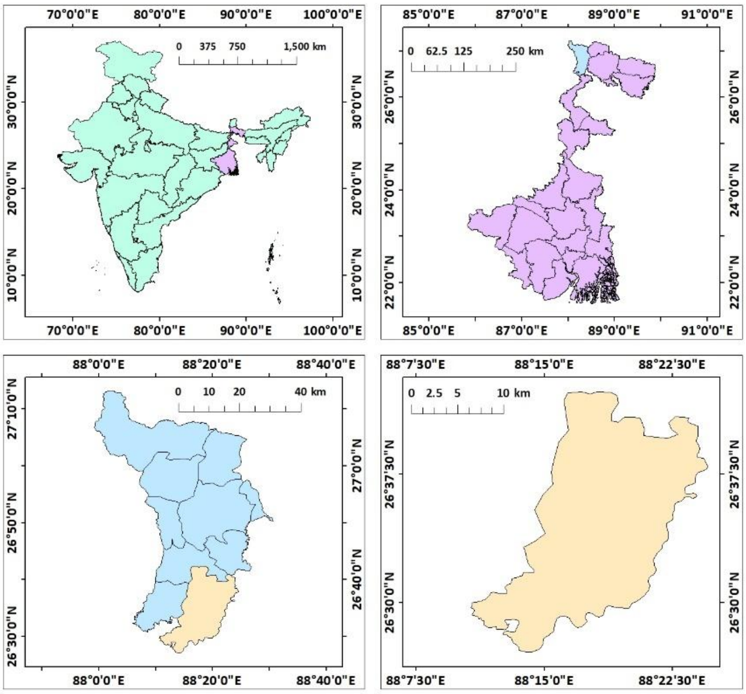


Figure 15: Location map of Phansidewa

The topography of the area plays a crucial role in groundwater potential. The steep slopes and deep valleys limit infiltration and promote surface runoff, reducing groundwater recharge. However, the presence of valley fills and alluvial deposits in river valleys can act as potential groundwater reservoirs. The dense vegetation cover in the region also influences groundwater recharge by intercepting rainfall and reducing soil erosion.

The land use and land cover patterns in the Phansidewa block are diverse, including forest, agriculture, and settlements. Forest areas, with their high infiltration rates and deep root systems, contribute significantly to groundwater recharge. Agricultural practices, such as terrace cultivation and irrigation, can impact groundwater levels and quality. Urban and rural settlements, with their impervious surfaces and wastewater discharge, can pose threats to groundwater quality.

Understanding the complex interplay of these factors is essential for assessing the groundwater potential of the Phansidewa block. By integrating geological, hydrological, and topographical information, it is possible to identify areas with favourable conditions for groundwater occurrence and extraction.

3. DATA SOURCES AND METHODOLOGY:

DATA SOURCES:

- **Soil Texture**:
  - Source: National Bureau of Soil Survey and Land Use Planning (NBSSLUP).
  - Details: Provides soil texture classification, essential for assessing erodibility.
- **Geomorphology**:
  - Source: Bhukosh portal, Geological Survey of India (GSI).
  - Details: Provides geomorphological maps for classifying terrain features.
- **Geology**:
  - Source: Bhukosh portal, Geological Survey of India (GSI).
  - Details: Provides geological maps for classifying terrain features.
- **Slope**:
  - Derived from: Digital Elevation Model (DEM) obtained from the USGS Earth Explorer.
  - Details: Calculated using Slope feature in Spatial analyst tool in ArcGIS
- **Rainfall Intensity**:
  - Source: Climate Hazards Group InfraRed Precipitation with Station Data (CHRS data portal).
  - Details: Provides NETCDF data.
- **Land Use and Land Cover (LULC)**:
  - Source: ESRI Sentinel-2 Land Cover Explorer.
  - Details: Classified LULC data for assessing human and natural influences on soil erosion.
- **Drainage Density**:
  - Source: Downloaded spatial datasets from BB Bike.
  - Derived from: Processed in ArcGIS using line density feature.
- **Distances from Road, River, and Settlement**:
  - Source: Downloaded spatial datasets from BB Bike.

- Derives from: Processed in ArcGIS to calculate Euclidean distances for each feature.

- **Lineament density**:
  - Derived from: Digital Elevation Model (DEM) obtained from the USGS Earth Explorer.
  - Details: Processed in ArcGIS using line density feature.
- **Waterbody density**:
  - Source: Downloaded spatial datasets from BB Bike.
  - Derives from: Processed in ArcGIS using point density feature.

These diverse data sources and derived parameters provide a comprehensive framework for analysing ground water potential zones using geospatial tools.

METHODOLOGY:

The Pairwise Comparison Matrix (PCM) is a valuable tool for decision-making processes, especially in multi-criteria decision analysis (MCDA). It allows for a structured and systematic comparison of multiple criteria or alternatives. First, relevant criteria are identified to evaluate the alternatives. A square matrix is then created, with each row and column representing a criterion. Each criterion is compared pairwise with every other criterion, assigning a numerical value to each comparison on a scale of 1 to 9. The pairwise comparison matrix is used to calculate the relative weights of each criterion, often using techniques like the Analytic Hierarchy Process (AHP). Once the weights are assigned, each alternative is evaluated against each criterion. The scores for each alternative are then multiplied by the corresponding weights, and the results are summed to obtain an overall score.

The methodology for this study incorporated a pairwise comparison matrix to evaluate and weight multiple criteria influencing groundwater recharge potential. The matrix, structured using the Analytical Hierarchy Process (AHP), compared criteria such as soil texture, geomorphology, geology, slope, rainfall intensity, land use and land cover (LULC), drainage density, proximity to roads, rivers, settlements, lineament density, and water body density. Each criterion was assigned relative importance based on expert judgment and hydrological principles, forming a weighted framework for subsequent spatial analysis.

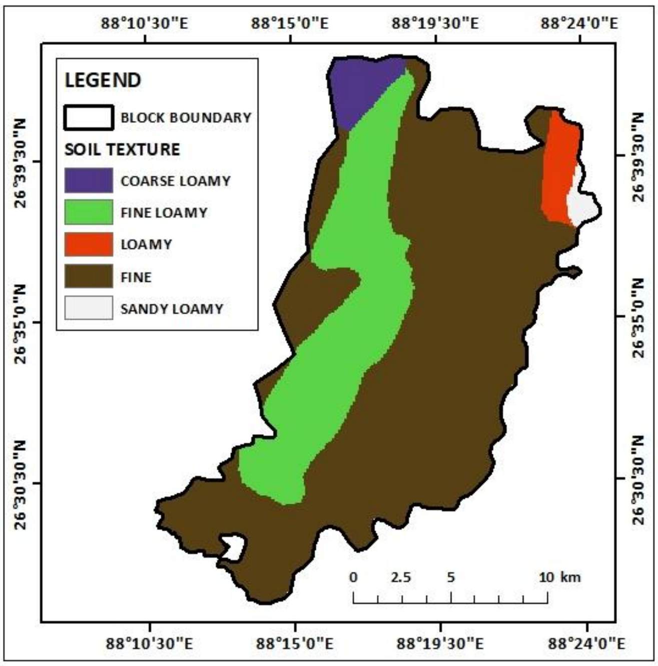
Criteria	Soil Texture	Geomorphology	Geology	Slope	Rainfall	LULC	Drainage Density	Dist.Roads	Dist.River	Dist.Settlement	Lineament Density	Wterbody Density
Soil Texture	1	3	5	7	3	3	5	3	3	5	3	5
Geomorphology	0.333333333	1	3	5	3	2	3	2	3	3	2	3
Geology	0.2	0.333333333	1	3	2	2	3	2	2	3	2	2
Slope	0.142857143	0.2	0.333333	1	2	0.5	1	1	0.5	2	1	1
Rainfall Intensity	0.333333333	0.333333333	0.5	0.5	1	2	3	3	2	3	3	3
LULC	0.333333333	0.5	0.5	2	0.5	1	2	2	2	2	2	2
Drainage Density	0.2	0.333333333	0.333333	1	0.33333	0.5	1	2	1	3	2	3
Dist.Roads	0.333333333	0.5	0.5	1	0.33333	0.5	0.5	1	0.5	1	1	1
Dist.River	0.333333333	0.333333333	0.5	2	0.5	0.5	1	2	1	2	2	3
Dist.Settlement	0.2	0.333333333	0.333333	0.5	0.33333	0.50	0.333333333	1	0.5	1	1	1
Lineament Density	0.333333333	0.5	0.5	1	0.33333	0.5	0.333333333	1	0.5	1	1	2
Wterbody Density	0.2	0.333333333	0.5	1	0.33333	0.5	0.333333333	1	0.3333333	1	0.5	1
SUM	3.942857143	7.7	13	25	13.6667	13.5	20.5	21	16.333333	27	20.5	27

Criteria	Soil Texture	Geomorphology	Geology	Slope	Rainfall	LULC	Drainage Density	Dist.Roads	Dist.River	Dist.Settlement	Lineament Density	Wterbody Density	SUM	AVG
Soil Texture	0.254	0.390	0.385	0.280	0.220	0.222	0.164	0.143	0.184	0.185	0.146	0.185	2.757	0.230
Geomorphology	0.085	0.130	0.231	0.200	0.220	0.148	0.098	0.095	0.184	0.111	0.098	0.111	1.710	0.142
Geology	0.051	0.043	0.077	0.120	0.146	0.148	0.098	0.095	0.122	0.111	0.098	0.074	4.467	0.372
Slope	0.036	0.026	0.026	0.040	0.146	0.037	0.033	0.048	0.031	0.074	0.049	0.037	0.582	0.049
Rainfall Intensity	0.085	0.043	0.038	0.020	0.073	0.148	0.098	0.143	0.122	0.111	0.146	0.111	1.140	0.095
LULC	0.085	0.065	0.038	0.080	0.037	0.074	0.066	0.095	0.122	0.074	0.098	0.074	1.722	0.143
Drainage Density	0.051	0.043	0.026	0.040	0.024	0.037	0.033	0.095	0.061	0.111	0.098	0.111	0.730	0.061
Dist.Roads	0.085	0.065	0.038	0.040	0.024	0.037	0.016	0.048	0.031	0.037	0.049	0.037	0.507	0.042
Dist.River	0.085	0.043	0.038	0.080	0.037	0.037	0.033	0.095	0.061	0.074	0.098	0.111	1.237	0.103
Dist.Settlement	0.051	0.043	0.026	0.020	0.024	0.037	0.011	0.048	0.031	0.037	0.049	0.037	0.413	0.034
Lineament Density	0.085	0.065	0.038	0.040	0.024	0.037	0.011	0.048	0.031	0.037	0.049	0.074	0.538	0.045
Wterbody Density	0.051	0.043	0.038	0.040	0.024	0.037	0.011	0.048	0.020	0.037	0.024	0.037	0.952	0.079

The weighted values derived from the pairwise comparison matrix was integrated into ArcGIS. Each criterion was represented as a thematic map, standardized and reclassified to align with the analysis framework. The reclassification ensured consistency across datasets and facilitated the use of weighted overlay techniques. Using these weighted overlays, a composite map was generated to illustrate zones of varying groundwater recharge potential, reflecting the combined influence of all criteria.

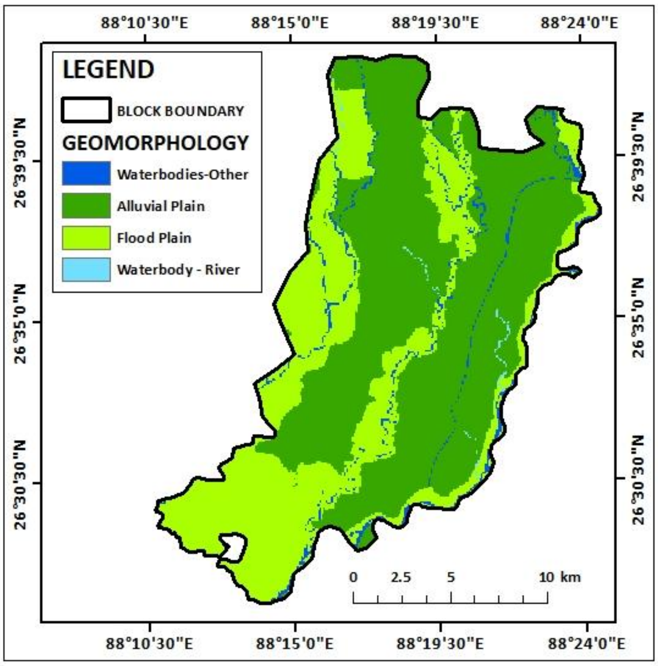
4. RESULTS AND DISCISSION:

The analysis of various thematic maps, including soil texture, geomorphology, geology, slope, rainfall intensity, land use/land cover, drainage density, distance from road, river, and settlement, lineament density, and waterbody density, provides valuable insights into the groundwater potential of Phansidewa block. By integrating these factors using a multi-criteria evaluation approach, we have delineated groundwater potential zones across the study area.



Soil texture map:

The provided map illustrates the spatial distribution of soil texture within the Phansidewa block of Darjeeling district, West Bengal. The map employs a color-coded legend to differentiate between various soil textures, including coarse loamy, coarse sandy, fine loamy, fine sandy, and sandy loamy. The map reveals a diverse range of soil textures across the study area, with finer-textured soils predominantly occupying the central and eastern portions, while coarser-textured soils are more prevalent in the western and northern regions. The spatial pattern of soil texture is likely influenced by factors such as parent material, topography, and climatic conditions.



Geomorphology map:

The geomorphological map of Phansidewa block, Darjeeling district, West Bengal, delineates four primary landforms: water bodies-other, alluvial plains, floodplains, and water bodies-river. Alluvial plains dominate the central and eastern parts, offering potential for groundwater accumulation. Floodplains, located along river courses, have high recharge potential but may face contamination risks. Water bodies, including rivers, lakes, and ponds, regulate water flow and contribute to groundwater recharge. Understanding these geomorphological features is crucial for assessing groundwater potential and implementing sustainable management strategies.

Geology map:

The map depicts the spatial distribution of the dominant soil type in Phansidewa block, Darjeeling district, West Bengal. The predominant soil type in the area is a mixture of sand, silt, and clay. This soil composition suggests a moderate level of permeability and water-holding capacity. While the specific proportions of sand, silt, and clay can vary within the block, this general soil type has implications for groundwater potential. The presence of sand and silt allows for reasonable

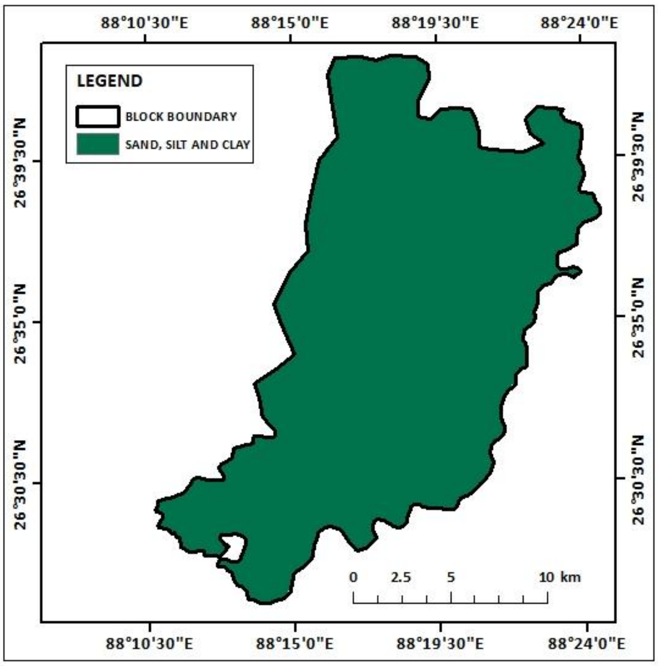


Figure 18: Geology map

infiltration of rainwater, contributing to groundwater recharge. However, the presence of clay can also restrict infiltration to some extent.

Slope map:

The slope map of Phansidewa block, Darjeeling district, West Bengal, illustrates the spatial distribution of slope gradients across the area. The map categorizes the landscape into five slope classes, ranging from gentle slopes to very steep slopes. The map reveals a diverse range of slope gradients, with gentle slopes dominating the central and eastern portions of the block. These areas are characterized by relatively flat terrain and are often associated with alluvial plains and valleys. In contrast, the western and northern parts of the block exhibit steeper slopes, particularly in the hilly regions. Understanding the spatial distribution of slope gradients is crucial for assessing groundwater potential and implementing sustainable water management practices. Gentle slopes generally have higher infiltration rates and promote groundwater recharge, while steeper slopes can lead to increased surface runoff and reduced infiltration.

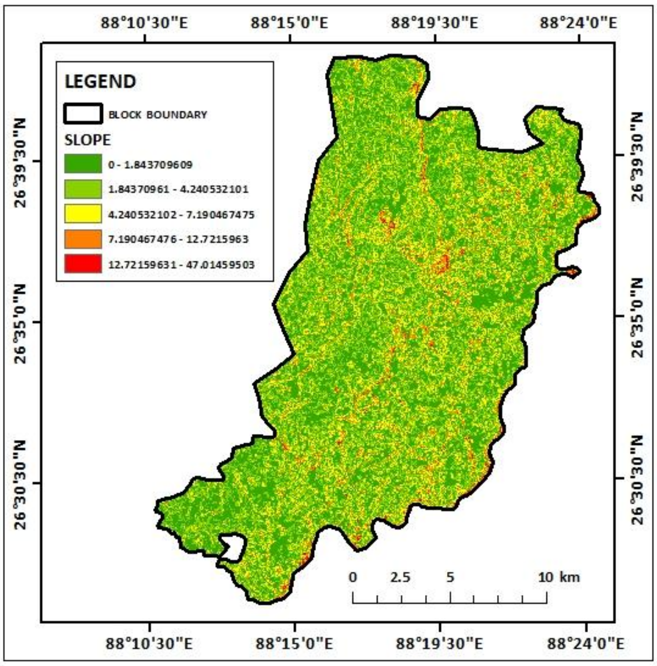


Figure 19: Slope map

Rainfall intensity map:

The rainfall intensity map of Phansidewa block, Darjeeling district, West Bengal, depicts the spatial distribution of rainfall intensity across the area. The map categorizes rainfall intensity into four classes: very low, low, moderate, and high. The map reveals that the majority of the block experiences moderate to high rainfall intensity, particularly in the central and eastern regions. These areas are influenced by the monsoon winds and receive abundant rainfall, which is crucial for groundwater recharge. In contrast, the western and northern parts of the block, which are more

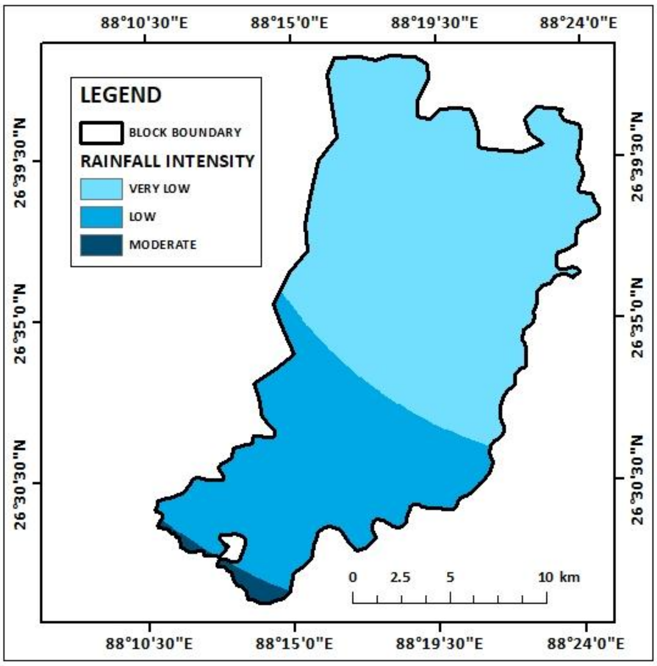


Figure 20: Rainfall intensity map

mountainous and receive less rainfall, exhibit lower rainfall intensities.

Drainage density map:

The drainage density map of Phansidewa block, Darjeeling district, West Bengal, illustrates the spatial distribution of drainage density across the area. Drainage density refers to the total length of stream channels per unit area and provides insights into the hydrological characteristics of a region. The map categorizes drainage density into five classes, ranging from low to very high. The map reveals a diverse range of drainage densities across the block. Higher drainage densities are observed in the central and eastern regions, where the river network is more developed. In contrast, lower drainage densities are found in the western and northern parts, which are characterized by steeper slopes and fewer stream channels.

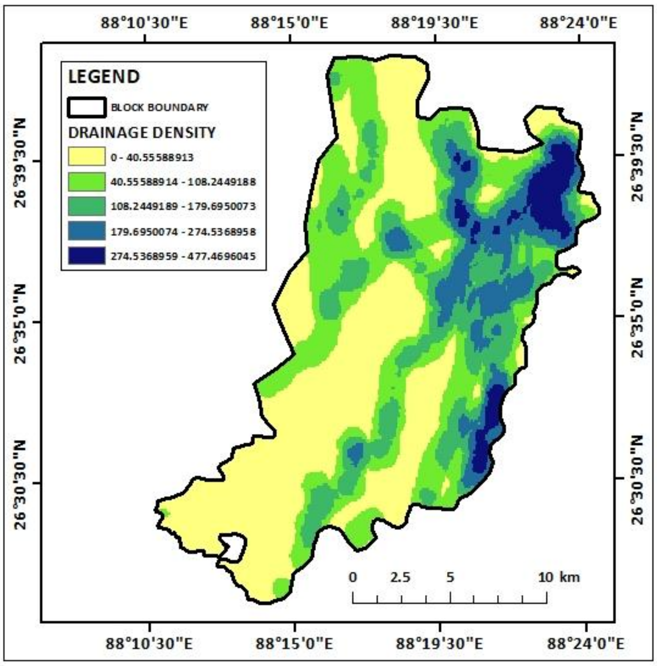


Figure 21: Drainage density map

Distance from road map:

The distance from road map of Phansidewa block, Darjeeling district, West Bengal, illustrates the spatial distribution of distances to the nearest road. The map categorizes distances into five classes, ranging from very close to very far. The map reveals that areas closer to roads are concentrated in the central and eastern parts of the block, particularly along major roads and highways. In contrast, areas farther from roads are more prevalent in the western and northern regions, which are characterized by hilly terrain and scattered settlements. Understanding the spatial distribution of distances from roads is crucial for assessing accessibility, transportation, and potential impacts on groundwater resources. Areas closer to roads may be more

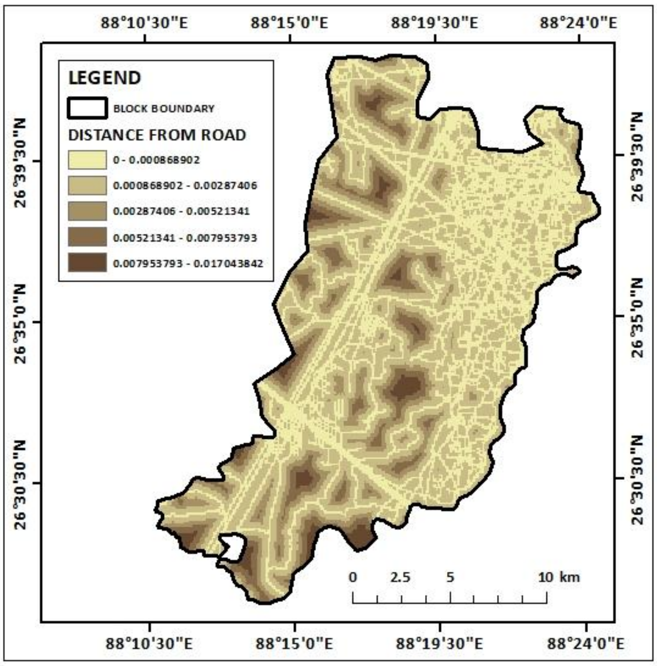


Figure 22: Distance from road map

susceptible to pollution from vehicular emissions, road runoff, and other anthropogenic activities.

Distance from river map:

The distance from river map of Phansidewa block, Darjeeling district, West Bengal, illustrates the spatial distribution of distances to the nearest river. The map categorizes distances into five classes, ranging from very close to very far. The map reveals that areas closer to rivers are concentrated along the river valleys and floodplains, particularly in the central and eastern parts of the block. In contrast, areas farther from rivers are more prevalent in the hilly regions of the western and northern parts. Understanding the spatial distribution of distances from rivers is crucial for assessing groundwater potential, flood risk, and ecological impact. Areas closer to rivers may have higher groundwater recharge due to infiltration from river water, but they may also be at risk of flooding and contamination from river water.

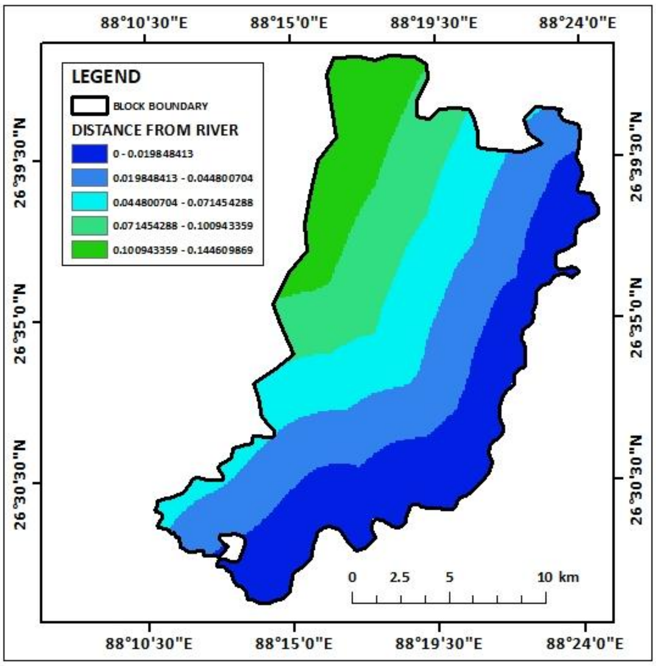


Figure 23: Distance from river map

Distance from settlement map:

The distance from settlement map of Phansidewa block, Darjeeling district, West Bengal, illustrates the spatial distribution of distances to the nearest settlement. The map categorizes distances into five classes, ranging from very close to very far. The map reveals that areas closer to settlements are concentrated in the central and eastern parts of the block, particularly along major roads and highways. In contrast, areas farther from settlements are more prevalent in the western and northern regions, which are characterized by hilly terrain and scattered settlements. Understanding the spatial distribution

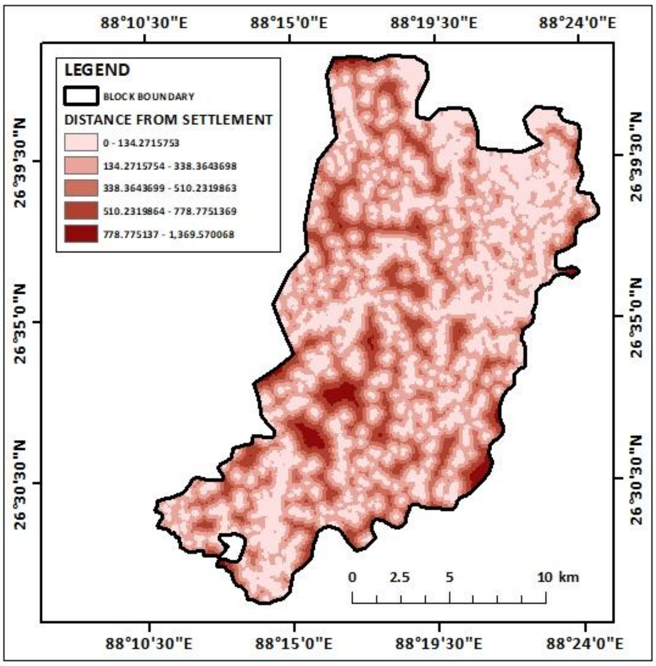


Figure 24: Distance from settlement map

of distances from settlements is crucial for assessing accessibility, transportation, and potential impacts on groundwater resources. Additionally, proximity to settlements can increase demand for water resources, potentially leading to overexploitation of groundwater.

Lineament density map:

The lineament density map of Phansidewa block, Darjeeling district, West Bengal, illustrates the spatial distribution of lineament density across the area. Lineaments are linear features, such as faults and fractures, that can affect groundwater flow and storage. The map categorizes lineament density into five classes, ranging from low to very high. The map reveals a diverse range of lineament densities across the block. Higher lineament densities are observed in the central and eastern regions, where the geological formations are more complex and fractured. In contrast, lower lineament densities are found in the western and northern parts, which are characterized by less fractured rocks. Understanding the spatial distribution of lineament density is crucial for assessing groundwater potential and implementing sustainable water management strategies.

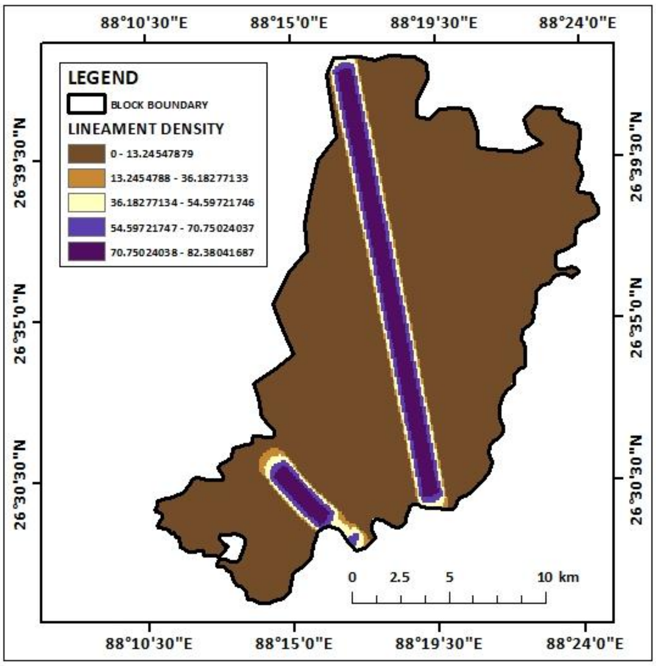


Figure 25: Lineament density map

Waterbody density map:

The waterbody density map of Phansidewa block, Darjeeling district, West Bengal, illustrates the spatial distribution of water bodies per unit area. The map categorizes waterbody density into five classes, ranging from low to very high. The map reveals a diverse range of waterbody densities across the block. Higher waterbody densities are observed in the central and eastern regions, where the river network is more developed and there are numerous lakes and ponds. In contrast, lower waterbody densities are

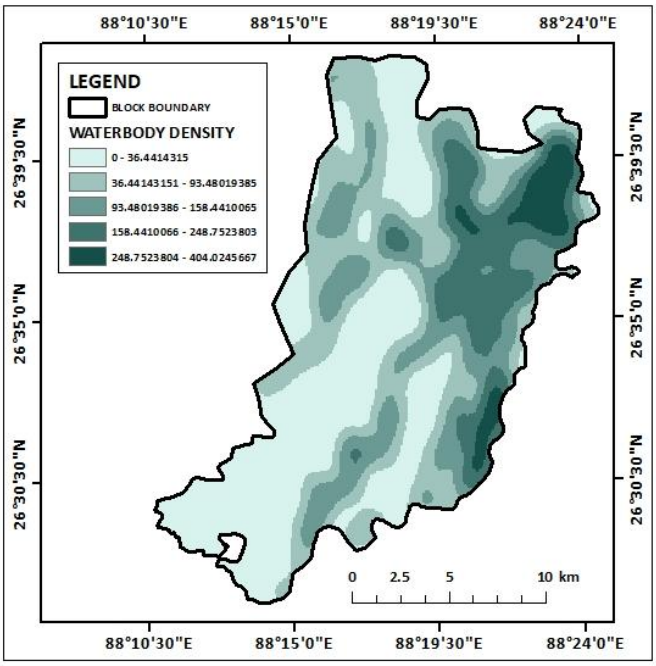


Figure 26: Waterbody density map

found in the western and northern parts, which are characterized by hilly terrain and fewer water bodies. Understanding the spatial distribution of waterbody density is crucial for assessing groundwater potential, flood risk, and ecological impact.

Ground water potential zones map:

The groundwater potential zone map of Phansidewa block, Darjeeling district, West Bengal, delineates five distinct zones based on the integrated analysis of multiple factors influencing groundwater occurrence and availability.

Very High Groundwater Potential Zones: These zones, predominantly located in the central and eastern parts of the block, exhibit favorable geological, hydrological, and geomorphological conditions. The presence of permeable rocks, moderate to high rainfall, and gentle to moderate slopes contributes to higher groundwater recharge and storage potential.

High Groundwater Potential Zones: These zones are characterized by moderate to high rainfall, moderate slopes, and a mix of permeable and less permeable rock formations. While the groundwater potential is not as high as in the very high potential zones, these areas still offer significant opportunities for groundwater development.

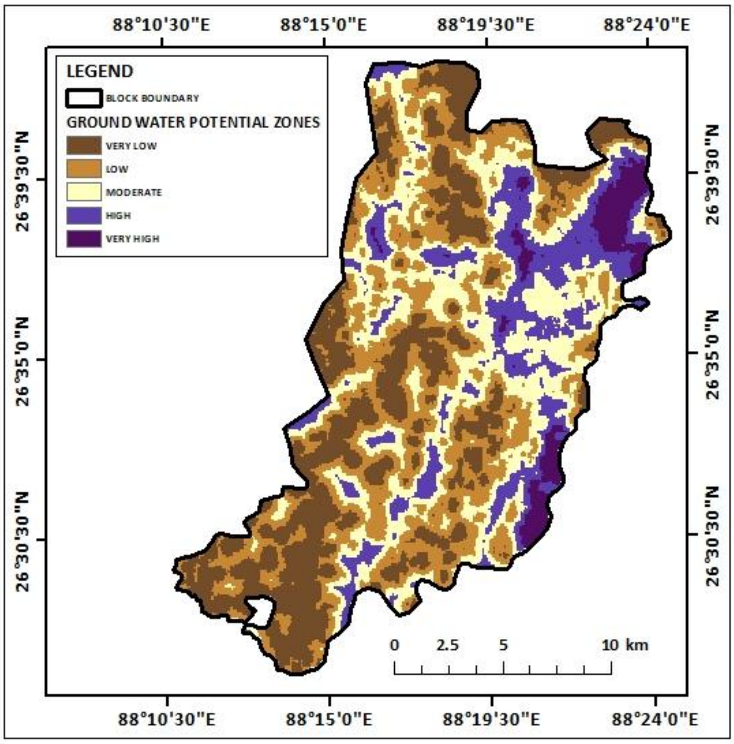


Figure 27: Ground water potential zones map

Moderate Groundwater Potential Zones: These zones are characterized by a combination of factors, including moderate to low rainfall, moderate to steep slopes, and a mix of permeable and less permeable rock formations. Groundwater potential in these zones is influenced by variations in rainfall, topography, and geology.

Low Groundwater Potential Zones: These zones are primarily located in the hilly and mountainous regions of the block, characterized by steep slopes, low rainfall, and less permeable rock formations. The combination of these factors limits groundwater recharge and storage potential.

Very Low Groundwater Potential Zones: These zones are characterized by very steep slopes, low rainfall, and highly impermeable rock formations. The limited infiltration and high runoff rates in these areas significantly reduce groundwater recharge potential.

Overall, the groundwater potential zone map provides valuable insights into the spatial distribution of groundwater resources within Phansidewa block. This information is crucial for sustainable groundwater management, including groundwater exploration, development, and conservation.

5. CONCLUSION:

The present study aimed to assess the groundwater potential of Phansidewa block, Darjeeling district, West Bengal, by integrating multiple factors using a GIS-based approach. Various thematic maps, including soil texture, geomorphology, geology, slope, rainfall intensity, land use/land cover, drainage density, distance from road, river, and settlement, lineament density, and waterbody density, were prepared and analysed. The analysis revealed significant spatial variability in groundwater potential across the study area. The central and eastern parts of the block, characterized by gentle slopes, moderate to high rainfall, and permeable geological formations, exhibited higher groundwater potential. In contrast, the western and northern regions, dominated by steep slopes, low rainfall, and less permeable rocks, exhibited lower groundwater potential.

The integration of these factors using a weighted overlay analysis enabled the delineation of five groundwater potential zones: very high, high, moderate, low, and very low. The very high and high potential zones were identified in the central and eastern parts of the block, indicating areas with favorable conditions for groundwater occurrence and extraction. The low and very low potential zones were primarily located in the hilly and mountainous regions, where groundwater potential is limited due to unfavorable geological and hydrological conditions. The findings of this study have important implications for sustainable groundwater management in Phansidewa block. The identification of groundwater potential zones can help in prioritizing groundwater exploration and development efforts, optimizing water resource allocation, and implementing appropriate water conservation measures.

It is recommended that further detailed investigations, such as hydrogeological surveys and groundwater modeling, be conducted to refine the groundwater potential assessment and provide more accurate information for decision-making. Additionally, it is essential to implement sustainable groundwater management practices, including rainwater harvesting, efficient irrigation techniques, and pollution control measures, to ensure the long-term sustainability of groundwater resources in the region. By integrating scientific knowledge, technological advancements, and participatory approaches, it is possible to achieve sustainable groundwater management and secure water resources for future generations.

FLOOD POTENTIAL ZONE MAPPING USING MCDM TECHNIQUE: A CASE STUDY ON KHANAKUL, GOGHAT AND ARAMBAG SUB-DIVISION OF HOOGHLY, WEST BENGAL

1. INTRODUCTION:

Flooding is a recurring natural disaster that poses significant risks to human life, property, and ecosystems. Understanding flood potential zones is crucial for effective disaster management, urban planning, and environmental conservation. The delineation of such zones requires a comprehensive approach that incorporates a variety of geospatial and environmental parameters, ensuring a holistic understanding of the factors contributing to flood susceptibility.

Soil texture plays a critical role in determining water infiltration rates and runoff potential. Coarser soils, like sandy textures, allow water to infiltrate quickly, reducing surface runoff, whereas clayey soils with lower permeability tend to increase the likelihood of flooding. Similarly, the Stream Power Index (SPI), which measures the erosive power of flowing water, is an essential parameter for understanding how stream dynamics influence flood risks, particularly in steep and narrow channels.

The Topographic Wetness Index (TWI) offers insight into areas likely to retain water due to their landscape position, reflecting the convergence of surface flow and soil moisture accumulation. Alongside this, geomorphology provides an understanding of landform features and their susceptibility to flooding, considering factors like valley configurations and plains that act as natural water pathways. Slope steepness and slope length are also critical as they determine the velocity and volume of surface runoff, with steeper slopes often leading to faster and more concentrated flows.

Rainfall intensity is a direct driver of floods, as high-intensity rainfall can overwhelm natural and manmade drainage systems. Land use and land cover (LULC) analysis further reveals human impacts on flood dynamics, with urbanized areas and impervious surfaces reducing infiltration and increasing runoff, while vegetated regions may mitigate flooding. Drainage density, defined as the length of streams per unit area, is another critical factor, reflecting the efficiency of surface water evacuation and its implications for flood spread.

Proximity to key features such as roads, rivers, and settlements adds a layer of societal and infrastructural vulnerability. Areas closer to rivers and settlements face heightened flood risks due to their exposure to overflow and drainage congestion. Distances from roads also play a role, as roads can act as barriers or conduits for water flow, exacerbating localized flooding.

By integrating these diverse criteria using geospatial technologies, flood potential zones can be identified with greater accuracy, enabling data-driven decisions for risk reduction and sustainable development planning.

2. STUDY AREA:

The study area encompasses the subdivisions of Khanakul I, Khanakul II, Goghat I, Goghat II, and Arambagh in the Hooghly district of West Bengal, India. Located in the lower Gangetic plain, this region is prone to seasonal flooding due to its geographical and hydrological characteristics. Geographically, the area lies approximately between latitudes **22°49'N to 23°6'N** and longitudes **87°40'E to 88°10'E**, covering a predominantly rural landscape.

The Hooghly district, known for its rich agricultural heritage, is influenced by the network of rivers and tributaries in the lower reaches of the Ganges. Major rivers, such as the Damodar and Mundeswari, flow through or near this area, contributing to its susceptibility to inundation during the monsoon season. The region's terrain is characterized by a combination of alluvial plains, low-lying flood-prone areas, and gently undulating landscapes, which significantly influence water movement and accumulation patterns.

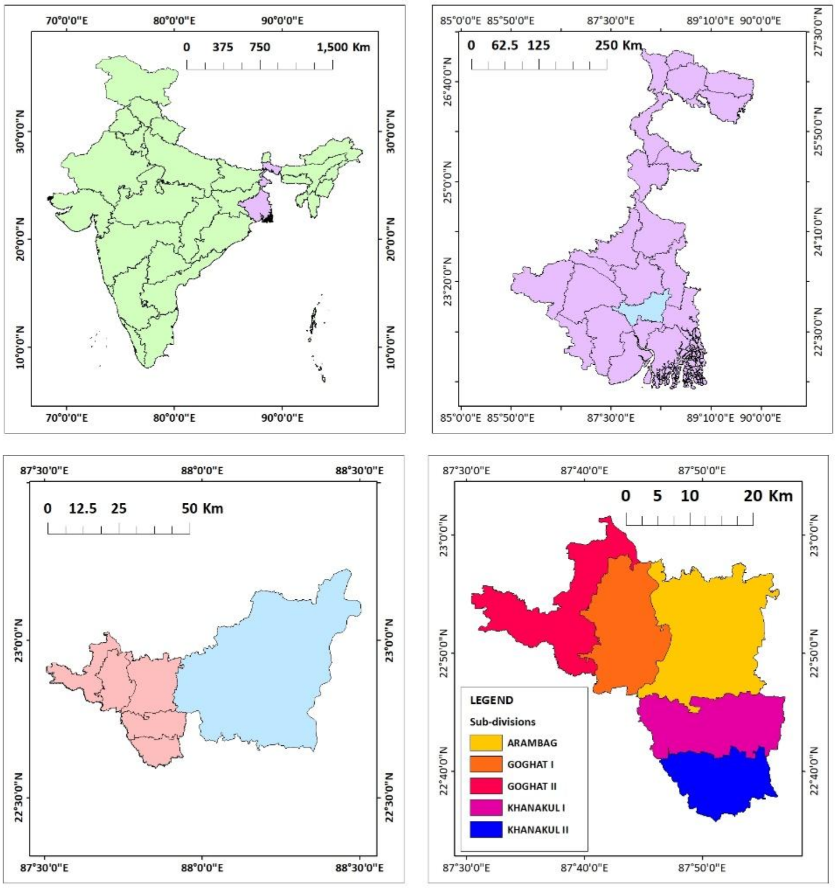


Figure 28: Location map of the sub-divisions of Hooghly district

Agriculture forms the backbone of the local economy, with crops like paddy and jute dominating the landscape. However, recurring floods disrupt agricultural activities and impact livelihoods. The settlements are interspersed with urban clusters like Arambagh town, which serve as administrative and commercial hubs. The growing population and expanding infrastructure further increase the vulnerability to flood hazards.

This area experiences a tropical monsoon climate, with an average annual rainfall of approximately 1,200–1,400 mm, most of which occurs between June and September. Intense rainfall during this period often exceeds the drainage capacity, leading to waterlogging and floods. The interaction of natural factors like soil texture, drainage patterns, and geomorphology, combined with anthropogenic pressures such as deforestation, unplanned urbanization, and improper land use, exacerbates the region's flood risks. Analyzing these factors in this specific context will provide valuable insights into flood potential zones and help in devising effective management strategies.

3. DATA SOURCES AND METHODOLOGY:

DATA SOURCES:

- **Soil Texture**:
  - Source: National Bureau of Soil Survey and Land Use Planning (NBSSLUP).
  - Details: Provides soil texture classification, essential for assessing erodibility.
- **Stream Power Index (SPI)**:
  - Derived from: DEM (USGS).
  - Calculated using the formula:

$$SPI = A \cdot \tan(\beta)$$

  - where:
    - A: Upslope contributing area (m^2).
    - β: Slope gradient in radians.
- **Topographic Wetness Index (TWI):**
  - Derived from: DEM (USGS).
  - Calculated using the formula:

$$\mathrm{TWI} = \ln\left(\frac{a}{\tan\beta}\right)$$

  - where:
    - α: The specific catchment area (upstream contributing area per unit width, typically in m^2/m).
    - β: The slope angle in radians (or the tangent of the slope).
- **Geomorphology**:
  - Source: Bhukosh portal, Geological Survey of India (GSI).
  - Details: Provides geomorphological maps for classifying terrain features.
- **Slope Length and Steepness Factor (LS)**:
  - Derived from: Digital Elevation Model (DEM) obtained from the USGS Earth Explorer.
  - Calculated using ArcGIS with the formula:

$$LS = \left(\frac{\lambda}{22.13}\right)^m \cdot (\sin\theta \cdot 0.0896)^n$$

- where:
  - λ: Slope length (m).
  - θ: Slope angle in radians.
  - m and n: Empirical constants (typically m=0.5, n=1.3m)

- **Rainfall Intensity**:
  - Source: Climate Hazards Group InfraRed Precipitation with Station Data (CHRS data portal).
  - Details: Provides NETCDF data.
- **Land Use and Land Cover (LULC)**:
  - Source: ESRI Sentinel-2 Land Cover Explorer.
  - Details: Classified LULC data for assessing human and natural influences on soil erosion.
- **Drainage Density**:
  - Source: Downloaded spatial datasets from BB Bike.
  - Derived from: Processed in ArcGIS using line density feature.
- **Distances from Road, River, and Settlement**:
  - Source: Downloaded spatial datasets from BB Bike.
  - Derives from: Processed in ArcGIS to calculate Euclidean distances for each feature.

These diverse data sources and derived parameters provide a comprehensive framework for analysing flood susceptibility using geospatial tools.

METHODOLOGY:

Dataset which are captured during the report preparation, are subjected to detail analysis work. District Survey Report involves the analytical implication of the captured dataset to prepare relevant maps. Methodology adopted for prepare implication of maps is explained below.

The methodology for delineating flood potential zones involved a systematic approach using geospatial analysis in ArcGIS, combining multiple environmental and anthropogenic parameters. The process began with the collection and preprocessing of spatial data for the chosen criteria: soil texture, Stream Power Index (SPI), Topographic Wetness Index (TWI), geomorphology, slope steepness and length, rainfall intensity, land use and land cover

(LULC), drainage density, and distances from roads, rivers, and settlements. Each parameter was converted into a raster format and prepared for further analysis by ensuring consistency in resolution and projection.

To standardize the data for analysis, all parameters were reclassified into specific classes based on their influence on flood susceptibility. For instance, areas with higher TWI or closer proximity to rivers were assigned higher flood susceptibility values, whereas areas with favourable geomorphological features were assigned lower values. These reclassified raster layers were then weighted according to their relative importance in contributing to flood risks. The weights were determined based on expert judgment and literature review, ensuring that the contribution of each parameter to flooding was appropriately represented.

The weighted raster layers were combined using the weighted overlay tool in ArcGIS, resulting in a composite flood potential map. This map effectively highlighted areas with varying levels of flood susceptibility, ranging from low to high potential zones.

To enhance spatial analysis, a fishnet grid was created over the study area, dividing it into uniform cells. The fishnet points, representing the centre of each cell, were clipped to the boundaries of the study area. Using these points, an Inverse Distance Weighted (IDW) interpolation was performed to visualize spatial variations in flood potential across the entire region. The IDW technique allowed for the generation of a smooth gradient map, capturing localized variations and further refining the delineation of flood-prone zones. This methodology ensured a comprehensive, data-driven approach to identifying and analysing flood risks in the study area.

4. RESULTS AND DISCUSSION:

The results section presents the spatial distribution of various factors influencing flood potential, including soil texture, stream power index, topographic wetness index, geomorphology, slope steepness and length, rainfall intensity, land use/land cover, drainage density, distance from roads, rivers, and settlements, and landslide susceptibility. The weighted sum method was employed to integrate these factors and generate a flood potential map. The discussion section delves into the interpretation of these maps, highlighting areas of high and low flood potential. It analyses the factors contributing to flood occurrences and discusses the implications of the findings for flood risk assessment and mitigation planning.

The study successfully assessed the spatial distribution of flood potential in the study area using a combination of geospatial techniques and multiple factors. The analysis revealed a significant variation in flood potential across the study area, with certain regions exhibiting higher risk due to factors like low-lying areas, high rainfall intensity, impervious surfaces, and proximity to rivers. The weighted sum method proved effective in integrating these factors and generating a comprehensive flood potential map.

The results of this study have important implications for flood risk management and disaster preparedness in the region. By identifying areas of high flood potential, policymakers and stakeholders can prioritize mitigation measures, such as flood control structures, early warning systems, and land use planning. Additionally, the study can inform decision-making regarding infrastructure development, urban planning, and disaster response. However, it is important to note that this study has certain limitations, including the availability and accuracy of input data. Future research can further refine the flood potential assessment by incorporating additional factors, such as climate change impacts and socioeconomic factors, and by using more advanced modelling techniques.

Soil texture map:

The map depicts the spatial distribution of soil texture within the study area. Soil texture, characterized by the relative proportions of sand, silt, and clay, significantly influences infiltration capacity and runoff generation. The study area exhibits a diverse range of soil textures, including fine loamy, fine, fine-fine loamy, very fine-fine loamy, fine loamy-fine, coarse loamy-fine, coarse loamy-fine loamy, and very fine soils. Fine-textured soils, such as clay and silt, have low infiltration capacity and high runoff potential, making them more susceptible to flooding. Conversely, coarse-textured soils, such as sand, have higher infiltration capacity and lower runoff potential. Understanding the spatial distribution of soil texture is crucial for assessing flood potential and implementing appropriate flood mitigation measures. By identifying areas with low infiltration capacity, targeted interventions like drainage improvement and flood control structures can be implemented to reduce flood risk.

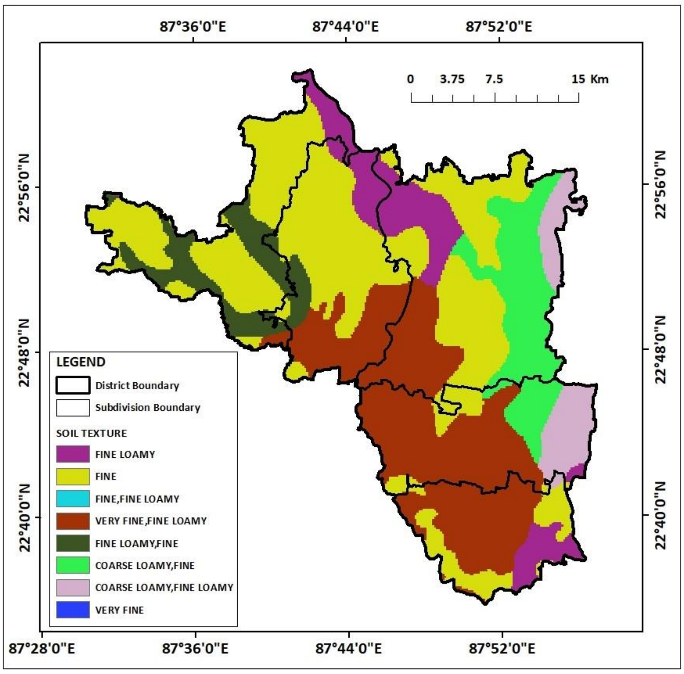


Figure 29: Soil texture map

Stream Power Index (SPI) map:

The map depicts the spatial distribution of Stream Power Index (SPI) values within the study area. SPI is a geomorphic index that quantifies the erosive power of flowing water.

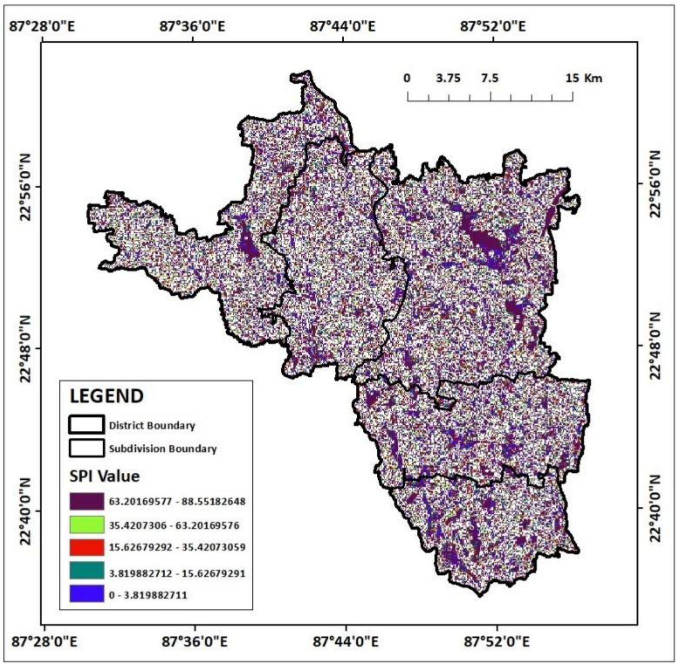


Figure 30: SPI map

Higher SPI values indicate areas with greater erosive potential, while lower values indicate areas with lower erosive potential. The study area exhibits a significant variation in SPI values. Areas with higher SPI values, often associated with steeper slopes and higher flow velocities, are more prone to flooding and erosion. Conversely, areas with lower SPI values, typically characterized by gentle slopes and lower flow velocities, are less susceptible to flooding and erosion. Understanding the spatial distribution of SPI is crucial for assessing flood potential and implementing appropriate flood mitigation measures. By identifying areas with high SPI values, targeted interventions like channel improvement, flood embankments, and early warning systems can be implemented to reduce flood risk.

Topographic Wetness Index (TWI) map:

The map depicts the spatial distribution of Topographic Wetness Index (TWI) values within the study area. TWI is a geomorphic index that quantifies the potential for water accumulation on a slope. Higher TWI values indicate areas with higher water accumulation potential, while lower values indicate areas with lower water accumulation potential. The study area exhibits a significant variation in TWI values. Areas with higher TWI values are more prone to waterlogging and flooding, especially during heavy rainfall events. Conversely, areas with lower TWI values are less susceptible to waterlogging and flooding. Understanding the spatial distribution of TWI is crucial for assessing flood potential and implementing appropriate flood mitigation measures. By identifying areas with high TWI values, targeted interventions like drainage improvement, flood embankments, and early warning systems can be implemented to reduce flood risk.

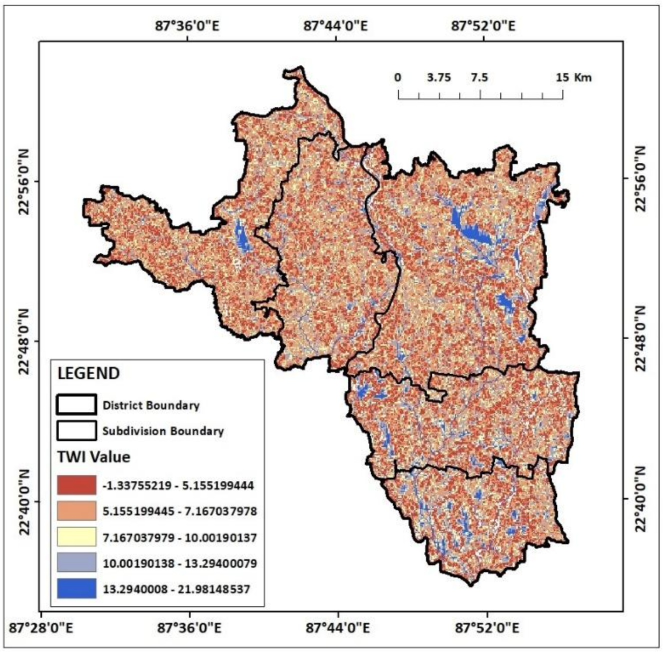


Figure 31: TWI map

Geomorphology map:

The map depicts the geomorphological characteristics of the study area. Geomorphology plays a crucial role in determining flood potential as different landforms have varying susceptibility to flooding. The study area exhibits a diverse range of geomorphological features, including older floodplains, waterbodies, older alluvial plains, rivers, ponds, active floodplains, and older deltaic plains. Older floodplains and older alluvial plains, characterized by gentle slopes and low elevation, are more prone to

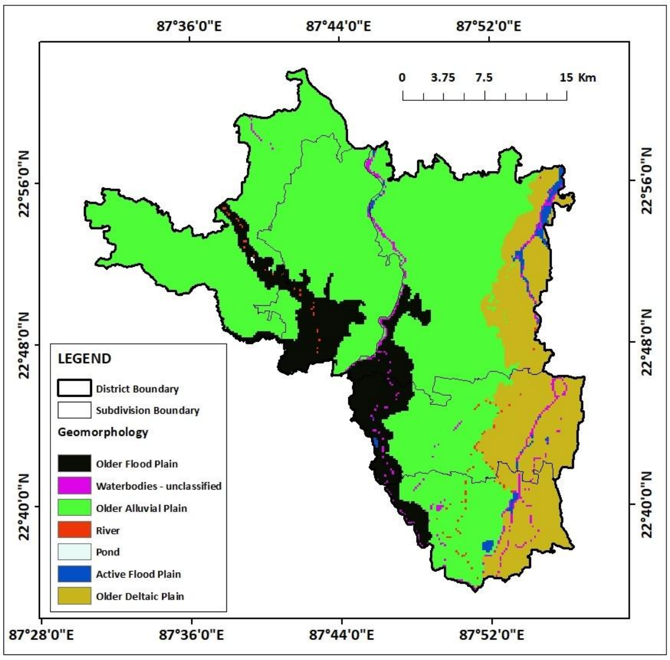


Figure 32: Geomorphology map

flooding, especially during heavy rainfall events. In contrast, areas with higher elevations, such as hills and ridges, are less susceptible to flooding. Understanding the spatial distribution of geomorphological features is crucial for assessing flood potential and implementing appropriate flood mitigation measures. By identifying areas with high flood potential, targeted interventions like channel improvement, flood embankments, and early warning systems can be implemented to reduce flood risk.

LS map:

The map depicts the spatial distribution of slope steepness and length within the study area. Slope steepness and length are crucial factors influencing flood potential as they determine the potential for surface runoff and soil erosion. Steeper slopes and longer slopes are more prone to soil erosion, which can lead to increased sediment load in rivers and streams, potentially contributing to flooding. The study area exhibits a significant variation in slope steepness and length. Areas with steeper slopes and longer slopes are more susceptible to flooding, especially during intense rainfall events. Conversely, areas with gentle slopes and shorter slopes are less prone to flooding. Understanding the spatial distribution of slope steepness and length is crucial for assessing flood potential and implementing appropriate flood mitigation measures. By identifying areas with steeper slopes and longer slopes, targeted interventions like slope stabilization, reforestation, and soil conservation practices can be implemented to reduce the risk of flooding and sediment-related problems.

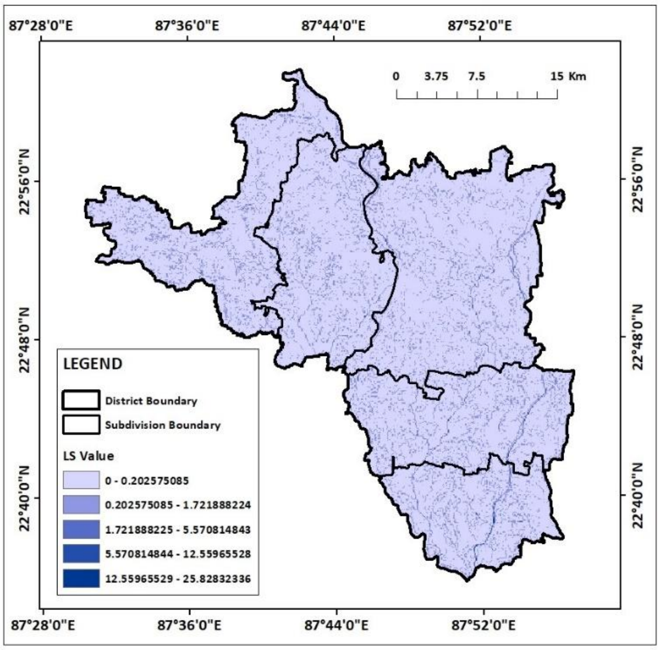


Figure 33: LS map

Rainfall intensity map:

The map depicts the spatial distribution of rainfall intensity within the study area. Rainfall intensity is a crucial factor influencing flood potential as it determines the volume and rate of surface runoff. Higher rainfall intensity can lead to increased

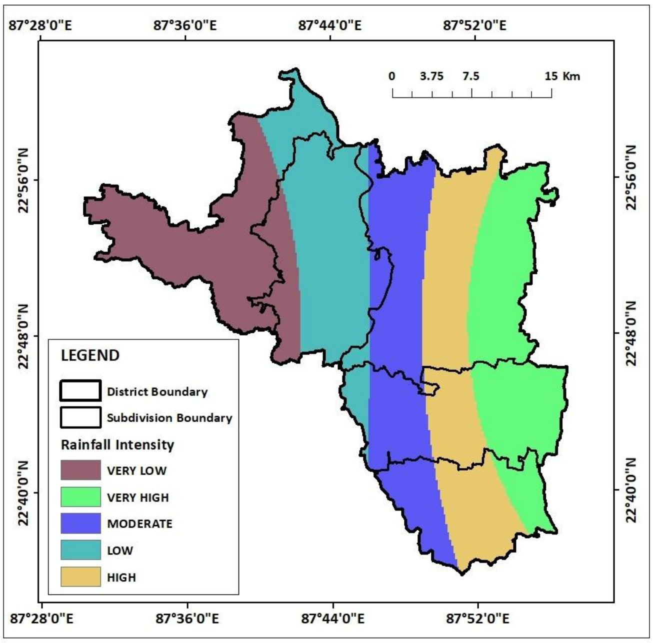


Figure 34: Rainfall intensity map

surface runoff and higher flood risk. The study area exhibits a significant variation in rainfall intensity. Areas with higher rainfall intensity are more prone to flooding, especially during intense rainfall events. Conversely, areas with lower rainfall intensity are less susceptible to flooding. Understanding the spatial distribution of rainfall intensity is crucial for assessing flood potential and implementing appropriate flood mitigation measures. By identifying areas with high rainfall intensity, targeted interventions like channel improvement, flood embankments, and early warning systems can be implemented to reduce flood risk.

LULC map:

The map depicts the spatial distribution of land use and land cover (LULC) classes within the study area. LULC plays a crucial role in determining flood potential as different land cover types have varying infiltration capacities and runoff generation rates. The study area exhibits a diverse range of LULC classes, including water bodies, vegetation, flooded vegetation, crops, built-up areas, and bare ground. Permeable surfaces like vegetation and crops generally have higher infiltration rates, resulting in lower runoff. Conversely, impervious surfaces like built-up areas and bare ground have lower infiltration rates, leading to higher runoff and increased flood risk. Understanding the spatial distribution of LULC classes is crucial for assessing flood potential and implementing appropriate flood mitigation measures. By identifying areas with high runoff potential, targeted interventions like afforestation, contour bunding, and terrace farming can be implemented to reduce runoff and mitigate flood risk.

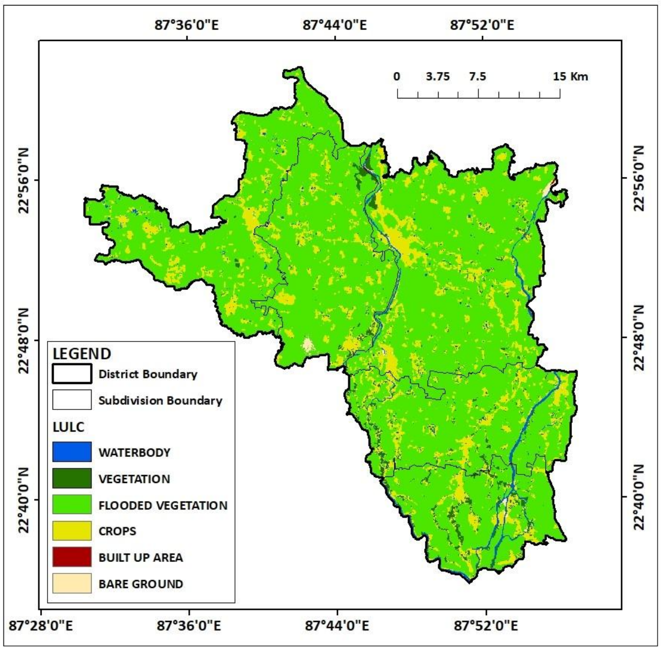


Figure 35: LULC map

Drainage density map:

The map depicts the spatial distribution of drainage density within the study area. Drainage density is a measure of the total length of streams in a given area, and it significantly influences flood potential. Higher drainage density can lead to increased surface runoff and higher flood risk, especially in areas with steep slopes and impervious surfaces. The study area exhibits a significant variation in drainage density. Areas with higher drainage density are more prone to

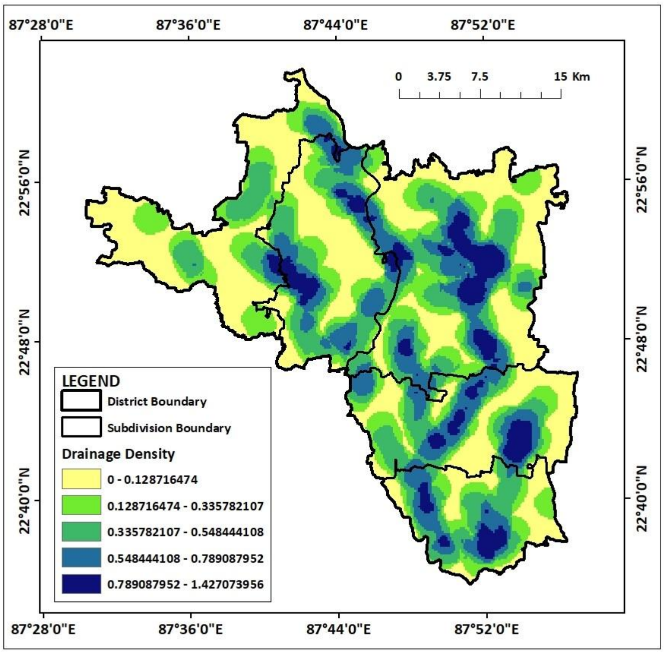


Figure 36: Drainage density map

flooding, while areas with lower drainage density are less susceptible. Understanding the spatial distribution of drainage density is crucial for assessing flood potential and implementing appropriate flood mitigation measures. By identifying areas with high drainage density, targeted interventions like channel improvement, flood embankments, and early warning systems can be implemented to reduce flood risk.

Distance from road map:

The map depicts the spatial distribution of distances from roads within the study area. Distance from roads is a crucial factor influencing flood potential as it affects accessibility for flood mitigation measures and evacuation routes. Areas closer to roads may have better access to emergency services and infrastructure, reducing flood risk. However, proximity to roads can also increase the risk of flooding due to factors like road construction and traffic-related activities. The study area exhibits a significant variation in distances from roads. Areas closer to roads are generally less prone to flooding, while areas farther from roads may be more susceptible. Understanding the spatial distribution of distances from roads is crucial for assessing flood potential and implementing appropriate flood mitigation measures. By identifying areas with limited road access, targeted interventions like the construction of flood embankments and evacuation routes can be prioritized.

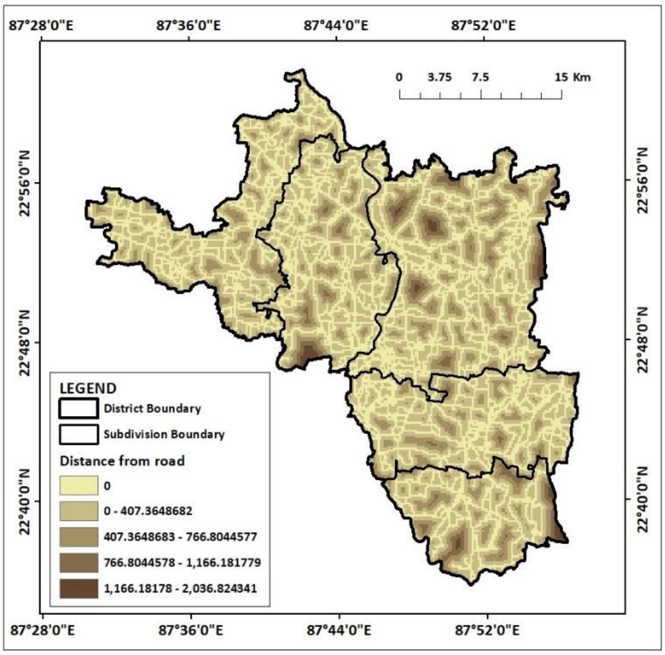


Figure 37: Distance from road map

Distance from rivers map:

The map depicts the spatial distribution of distances from rivers within the study area. Distance from rivers is a crucial factor influencing flood potential as it affects the availability of water for vegetation, the intensity of surface runoff, and the potential for riverbank erosion. Areas closer to rivers may have higher soil moisture content, reduced surface runoff, and increased vegetation cover, which can help mitigate flood risk. However, proximity to rivers can also increase the risk of riverbank erosion and flooding, which can contribute to flood

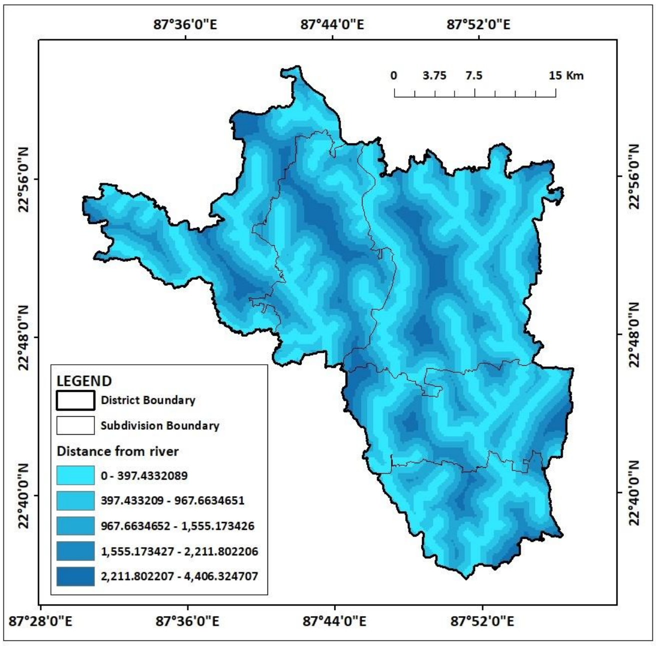


Figure 38: Distance from river map

damage. The study area exhibits a significant variation in distances from rivers. Areas closer to rivers are generally more influenced by riverine processes and may have lower flood risk due to increased infiltration and reduced surface runoff. Conversely, areas farther from rivers may be more susceptible to flooding due to factors like lower soil moisture content and increased surface runoff.

Distance from settlements map:

The map depicts the spatial distribution of distances from settlements within the study area. Distance from settlements is a crucial factor influencing flood potential as it affects the intensity of human activities and land use practices. Areas closer to settlements may experience higher levels of disturbance, such as deforestation, agriculture, and construction, which can increase flood risk. Conversely, areas farther from settlements may be less impacted by human activities and retain more natural vegetation cover, which can help mitigate flood risk. The study area exhibits a significant variation in distances from settlements. Areas closer to settlements are generally more prone to flooding due to increased impervious surfaces and altered drainage patterns. Conversely, areas farther from settlements may be less susceptible to flooding due to lower levels of disturbance and more natural drainage patterns. Understanding the spatial distribution of distances from settlements is crucial for assessing flood potential and implementing appropriate flood mitigation measures. By identifying areas closer to settlements, targeted interventions like flood zoning, urban planning, and drainage improvement can be implemented to reduce flood risk.

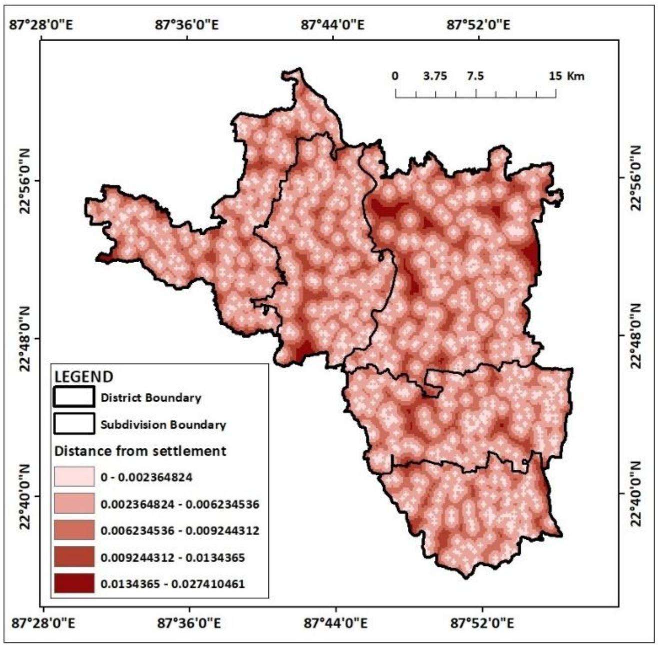


Figure 39: Settlement map

Weighted Sum map:

The map depicts the spatial distribution of flood susceptibility within the study area. Flood susceptibility is calculated using a weighted sum index, which combines multiple factors such as elevation, slope, soil type, land use/land cover, drainage density, and distance from rivers. Higher values indicate areas with higher flood susceptibility, while lower values indicate areas with lower susceptibility. The study area exhibits a significant variation in flood

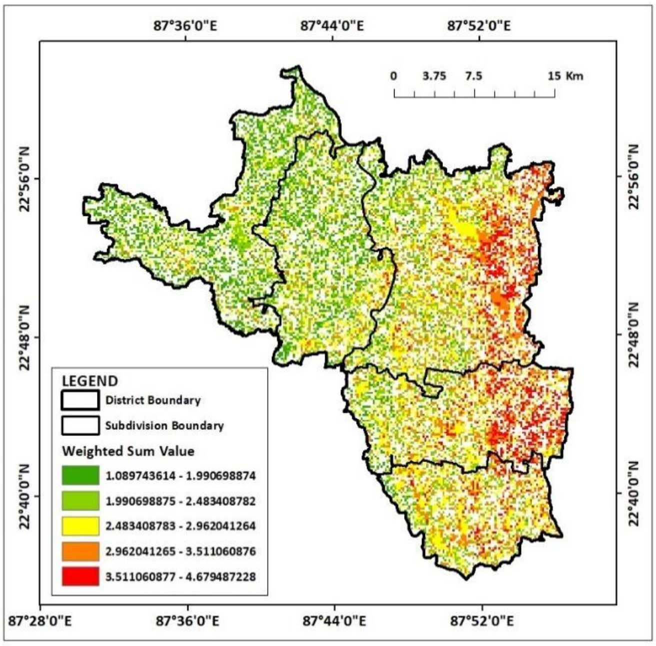


Figure 40: Weighted Sum map

susceptibility. Areas with higher values are more prone to flooding, especially during heavy rainfall events. Conversely, areas with lower values are less susceptible to flooding. Understanding the spatial distribution of flood susceptibility is crucial for implementing effective flood mitigation measures. By identifying high-risk areas, targeted interventions like flood zoning, urban planning, and drainage improvement can be prioritized to reduce flood risk and minimize damage to infrastructure and property.

IDW map based on flood susceptible zones:

From the above maps we can easily conclude that high flood potential zones are predominantly concentrated in low-lying regions, areas with clayey soils, and locations near major rivers such as the Damodar and Mundeswari. These areas are further exacerbated by high rainfall intensity and dense drainage networks, which, although efficient in evacuating water, are prone to overflow during peak rainfall events. The geomorphology, dominated by floodplains near river systems, aligns closely with the high susceptibility zones, highlighting the natural vulnerability of these areas.

Moderate flood potential zones represent a mix of contributing and mitigating factors. For instance, areas with moderate slopes and vegetative cover help reduce runoff but may still face localized flooding due to poor drainage or proximity to infrastructure like roads and settlements. The integration of proximity factors, such as distance from rivers, roads, and settlements, emphasizes the role of human-induced vulnerabilities, where unplanned development and impermeable surfaces compound the flood risks.

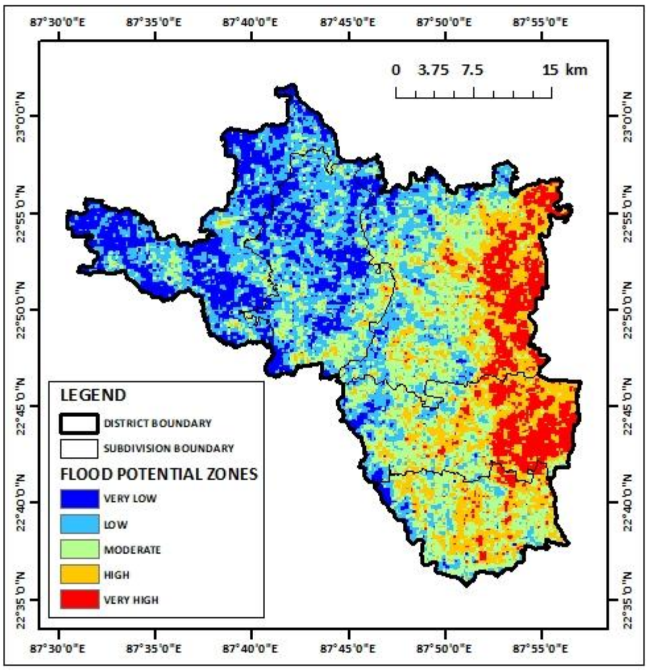


Figure 41: IDW map based on flood susceptibility zones

Low flood potential zones are primarily located in uplands or areas with favorable soil and drainage characteristics. These findings stress the need for targeted flood management strategies, prioritizing high-risk areas for mitigation measures like embankments, afforestation, and improved drainage systems while protecting low-risk zones from future development pressures.

5. CONSLUSION:

his study provides a comprehensive assessment of flood potentiality in Khanakul (1 and 2), Goghat (1 and 2), and Arambag subdivisions of the Hooghly district, West Bengal, through the integration of eleven critical environmental and anthropogenic factors using geospatial techniques. The use of ArcGIS tools to generate thematic maps and perform a weighted overlay analysis has enabled the identification of zones with varying flood susceptibility, ranging from low to high risk. The results highlight the interplay of natural factors such as soil texture, slope, geomorphology, and rainfall intensity with human-induced factors like land use changes, road proximity, and settlement expansion in shaping the flood risk patterns.

The weighted flood susceptibility map effectively delineates high-risk zones concentrated near river systems, low-lying floodplains, and areas with poor drainage infrastructure, underscoring their vulnerability to inundation. Moderate-risk areas, influenced by a combination of topographic and anthropogenic factors, present opportunities for targeted mitigation efforts. Low-risk zones, primarily located in elevated terrains, offer natural resilience but require sustainable management to prevent future risks.

This study underscores the importance of integrating geospatial analysis and multi-criteria evaluation for flood risk assessment. The findings provide valuable insights for disaster management, land use planning, and the implementation of mitigation measures to reduce flood impacts and enhance community resilience.

LAND SLIDE POTENTIAL ZONE MAPPING USING MCDM TECHNIQUE: A CASE STUDY ON KALIMPONG, WEST BENGAL

1. INTRODUCTION:

Landslides are a significant natural hazard that cause substantial economic, environmental, and human losses worldwide. The increasing frequency and intensity of landslide events are often attributed to complex interactions between natural processes and anthropogenic activities. In this context, identifying potential landslide zones has become a critical task for effective disaster risk management and mitigation. The integration of multiple geospatial and environmental criteria is essential for accurately delineating these zones, allowing for proactive planning and safeguarding vulnerable communities.

Among the key factors influencing landslide occurrences, soil texture plays a pivotal role in determining the stability of slopes. Loose or sandy soils are more prone to erosion and sliding compared to compact clayey soils. Similarly, hydrological indices such as the Stream Power Index (SPI) and Topographic Wetness Index (TWI) are instrumental in assessing water flow and accumulation, which can weaken slope stability. Geomorphological features, which reflect the terrain's structural and historical evolution, also significantly contribute to landslide susceptibility. The steepness and length of slopes further amplify the gravitational forces acting on soil and rock materials, making steep terrains particularly vulnerable.

Rainfall intensity is another critical driver of landslides, as it triggers soil saturation, reduces cohesion, and increases pore-water pressure. The role of land use and land cover (LULC) cannot be overlooked, as vegetation cover tends to stabilize slopes while urban or deforested areas are often associated with heightened susceptibility. Drainage density, representing the extent of surface water channels, highlights areas prone to rapid water accumulation that can destabilize slopes. Proximity to roads, rivers, and settlements introduces anthropogenic pressures that may exacerbate natural vulnerabilities by altering drainage patterns or adding load to slopes.

Topographical parameters such as slope aspect and curvature provide additional insights into microclimatic and structural influences on landslide susceptibility. North-facing slopes may retain moisture longer, whereas convex or concave curvatures influence the accumulation and flow of water, affecting slope stability. Analysing these factors in a comprehensive manner using advanced geospatial techniques and multi-criteria decision-making frameworks offers a systematic approach to mapping landslide-prone areas.

This assignment aims to employ a multidisciplinary perspective to delineate landslide potential zones by integrating diverse criteria. Leveraging tools such as Geographic Information Systems (GIS) and remote sensing, the analysis seeks to provide an accurate and practical understanding of landslide risks. This effort not only contributes to disaster risk reduction but also promotes sustainable land management practices in vulnerable regions.

2. STUDY AREA:

Kalimpong district in West Bengal, located in the eastern Himalayas, is an area highly susceptible to landslides due to its unique topographical, geological, and climatic characteristics. The district, spanning rugged terrains with steep slopes, is part of the Darjeeling Himalayas and is known for its fragile geology. The elevation varies from 125 meters to over 3,000 meters, with steep gradients that make the region prone to slope instability. The combination of high rainfall intensity during the monsoon season and anthropogenic activities like deforestation, road construction, and urban expansion exacerbates the landslide potential in this region.

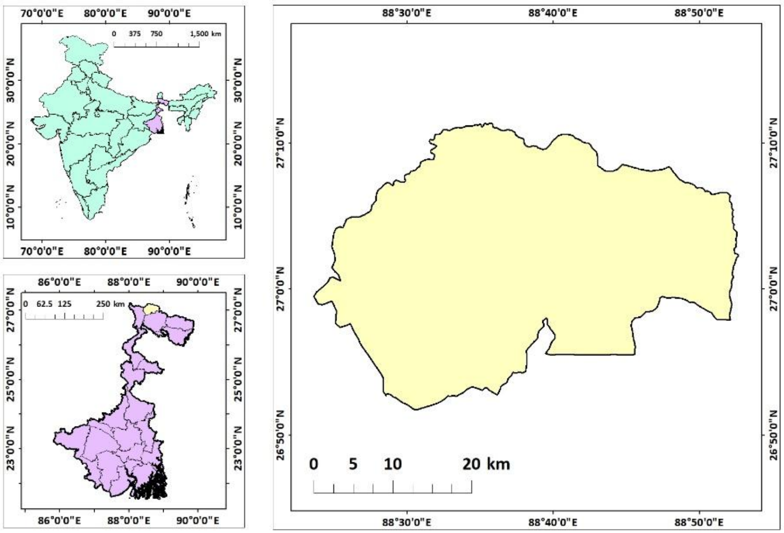


Figure 42: Location map of Kalimpong district

The soil texture in Kalimpong plays a significant role in determining the susceptibility to landslides. Sandy loam and loose soils, coupled with heavy rainfall, can lead to soil saturation and increased pore pressure, reducing slope stability. Stream Power Index (SPI) is another critical factor, as the area features numerous small rivers and streams with high erosive power, carving out steep and unstable slopes. The Topographic Wetness Index (TWI), representing the soil moisture distribution, highlights zones where water accumulates, further weakening the terrain and triggering slope failures during heavy rainfall events.

Geomorphology is another significant determinant in Kalimpong's landslide potential. The region exhibits a variety of geomorphological features, including ridges, escarpments, and river valleys, each contributing differently to slope stability. Slope steepness and length are critical parameters, as areas with long, steep slopes are more prone to landslides due to the gravitational pull and accumulation of surface runoff. Rainfall intensity, a major landslide-triggering factor, is particularly high during the monsoon season, often exceeding the soil's absorption capacity, leading to runoff and erosion. The district also features diverse Land Use and Land Cover (LULC), where urbanized and deforested areas are more vulnerable to slope instability compared to forested regions that provide natural reinforcement to the soil.

Drainage density, another key factor, influences the concentration of runoff and the potential for erosion. Proximity to roads, rivers, and settlements further compounds the landslide risk, as these areas often involve human activities that disturb the natural equilibrium of slopes. The slope aspect and curvature also affect the potential for landslides; for instance, south-facing slopes in this region, which receive more direct sunlight and rainfall, are more prone to weathering and instability. Convex slopes often act as detachment zones for debris, while concave slopes serve as accumulation zones, both contributing to the overall landslide dynamics.

3. DATA SOURCES AND METHODOLOGY:

DATA SOURCES:

- **Soil Texture**:
  - Source: National Bureau of Soil Survey and Land Use Planning (NBSSLUP).
  - Details: Provides soil texture classification, essential for assessing erodibility.
- **Stream Power Index (SPI)**:
  - Derived from: DEM (USGS).
  - Calculated using the formula:

$$SPI = A \cdot \tan(\beta)$$

  - where:
    - A: Upslope contributing area (m^2).
    - β: Slope gradient in radians.
- **Topographic Wetness Index (TWI):**
  - Derived from: DEM (USGS).
  - Calculated using the formula:

$$\mathrm{TWI} = \ln\left(\frac{a}{\tan\beta}\right)$$

  - where:
    - α: The specific catchment area (upstream contributing area per unit width, typically in m^2/m).
    - β: The slope angle in radians (or the tangent of the slope).
- **Geomorphology**:
  - Source: Bhukosh portal, Geological Survey of India (GSI).
  - Details: Provides geomorphological maps for classifying terrain features.
- **Slope Length and Steepness Factor (LS)**:
  - Derived from: Digital Elevation Model (DEM) obtained from the USGS Earth Explorer.
  - Calculated using ArcGIS with the formula:

$$LS = \left(\frac{\lambda}{22.13}\right)^m \cdot (\sin\theta \cdot 0.0896)^n$$

- where:
    - λ: Slope length (m).
    - θ: Slope angle in radians.
    - m and n: Empirical constants (typically m=0.5, n=1.3m)

- **Rainfall Intensity**:
  - Source: Climate Hazards Group InfraRed Precipitation with Station Data (CHRS data portal).
  - Details: Provides NETCDF data.
- **Land Use and Land Cover (LULC)**:
  - Source: ESRI Sentinel-2 Land Cover Explorer.
  - Details: Classified LULC data for assessing human and natural influences on soil erosion.
- **Drainage Density**:
  - Source: Downloaded spatial datasets from BB Bike.
  - Derived from: Processed in ArcGIS using line density feature.
- **Distances from Road, River, and Settlement**:
  - Source: Downloaded spatial datasets from BB Bike.
  - Derives from: Processed in ArcGIS to calculate Euclidean distances for each feature.
- **Slope**:
  - Derived from: Digital Elevation Model (DEM) obtained from the USGS Earth Explorer.
  - Details: Calculated using Slope feature in Spatial analyst tool in ArcGIS
- **Aspect**:
  - Derived from: Digital Elevation Model (DEM) obtained from the USGS Earth Explorer.
  - Details: Calculated using Aspect feature in Spatial analyst tool in ArcGIS
- **Curvature**:
  - Derived from: Digital Elevation Model (DEM) obtained from the USGS Earth Explorer.
  - Details: Calculated using Curvature feature in Spatial analyst tool in ArcGIS

These diverse data sources and derived parameters provide a comprehensive framework for analysing land slide potential using geospatial tools.

METHODOLOGY:

The methodology for this landslide susceptibility analysis in the Kalimpong district involves several stages, including data preparation, reclassification of criteria, spatial analysis, and the application of geospatial techniques in ArcGIS to generate a final landslide potential map.

Initially, we gathered raster datasets for a range of criteria influencing landslide susceptibility, including soil texture, Stream Power Index (SPI), Topographic Wetness Index (TWI), geomorphology, slope steepness and length, rainfall intensity, Land Use/Land Cover (LULC), drainage density, distances from roads, rivers, and settlements, slope aspect, and curvature. These datasets were sourced from reliable remote sensing platforms and local government records.

The first step in the analysis was to reclassify each raster dataset based on predefined classes that represent different levels of susceptibility to landslides. Each criterion was assigned specific values or classes reflecting its impact on landslide occurrence, such as low, medium, or high susceptibility. For instance, areas with steep slopes were assigned higher values, while regions with dense vegetation were classified as having lower susceptibility.

To assess the relative importance of each criterion, we used the Frequency Ratio (FR) method. This involved calculating the frequency of landslides in each reclassified class of each criterion and determining the ratio between the frequency of landslides in a particular class and the total number of landslides across all classes. This method helped assign

TABLE FOR FREQUENCY RATIO FOR LAND SLIDE POTENTIAL						
rk	**n**	**rj**	**n-rj+1**	**n-rk+1**	**wij**	**Percentage**
1	RAINFALL	2	13	14	0.12381	12.38095238
2	LULC	9	6	13	0.057143	5.714285714
3	GEOMORPHOLOGY	3	12	12	0.114286	11.42857143
4	SOIL	1	14	11	0.133333	13.33333333
5	DRAINAGE DENSITY	7	8	10	0.07619	7.619047619
6	SLPOE STEEPNESS & LENGTH	8	7	9	0.066667	6.666666667
7	ASPECT	6	9	8	0.085714	8.571428571
8	CURVATURE	5	10	7	0.095238	9.523809524
9	TWI	10	5	6	0.047619	4.761904762
10	SPI	11	4	5	0.038095	3.80952381
11	DISTANCE FROM ROAD	12	3	4	0.028571	2.857142857
12	DISTANCE FROM RIVER	14	1	3	0.009524	0.952380952
13	DISTANCE FROM SETTLEMENT	13	2	2	0.019048	1.904761905
14	SLOPE	4	11	1	0.104762	10.47619048
N=14				**105**	**1**	**100**

weights to the different criteria based on their correlation with landslide occurrence.

Once the Frequency Ratio values were calculated, we employed the Weighted Sum Method (WSM) in ArcGIS. This method involved combining the weighted raster layers to produce a composite landslide susceptibility map. The weights assigned to each criterion were based on their significance in influencing landslide risk, as derived from the Frequency Ratio.

To enhance the spatial accuracy and resolution of the final map, we created a fishnet grid over the entire study area. The fishnet points were then clipped to the area of interest and used as input for the Inverse Distance Weighting (IDW) interpolation technique. IDW was applied to estimate landslide susceptibility at unsampled locations, providing a continuous surface representing potential landslide zones across the district.

This methodology combines multiple spatial analysis techniques, including raster reclassification, frequency ratio calculation, weighted sum overlay, and interpolation, to create a robust model for assessing landslide susceptibility in the Kalimpong district. The final result offers a detailed, spatially explicit understanding of landslide risk, which can inform decision-making in land use planning and disaster mitigation.

4. RESULTS AND DISCUSSION:

The subsequent maps created from these parameters, including the individual criterion maps and the weighted sum model, provide a visual representation of landslide susceptibility across the region. These results are critical for understanding how each factor interacts with others to influence the stability of the terrain. The discussion delves into the significance of each map, highlighting key areas at risk, and provides insights into the factors that contribute to landslide vulnerability in the study area. By interpreting these maps, we aim to offer targeted recommendations for landslide risk mitigation and management strategies, particularly in regions with high human activity or vulnerable topographic features. The findings of this analysis contribute to a better understanding of the complex interplay between natural and human-induced factors in landslide occurrences, providing valuable information for disaster preparedness and land-use planning.

Soil texture map:

The soil texture map provides valuable insights into the distribution of soil types across the study area. It indicates varying levels of susceptibility to landslides, with areas of sandy or loamy soil types showing higher vulnerability due to their lower cohesiveness and greater erosion potential. Conversely, clayey soils, which are more cohesive, exhibit lower landslide susceptibility. The regions with fine-textured soils are typically located in flatter areas or valleys, while coarser soils are found in steeper terrains. This

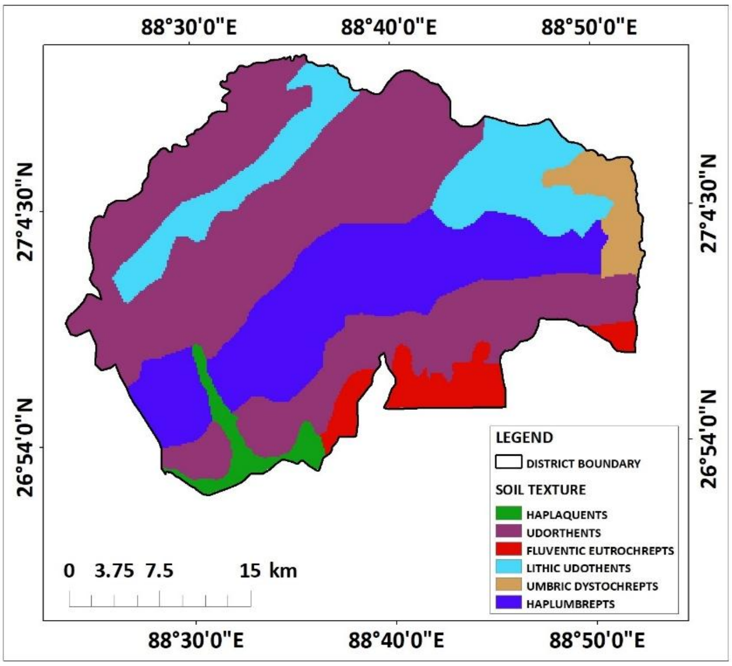


Figure 43: Soil texture map

distribution of soil textures plays a critical role in determining the overall stability of slopes in the district.

Stream Power Index (SPI) map:

The SPI map reflects the potential energy of water flow across the landscape. Areas with high SPI values, typically found along rivers and steep terrains, indicate strong water flow capable of eroding and destabilizing slopes. These areas are more prone to landslides, especially during periods of intense rainfall. Conversely, low SPI values occur in less active or flatter regions, where water flow is less forceful. The map highlights the critical areas where water erosion, coupled with steep slopes, could significantly contribute to landslide initiation, making these zones high-risk for landslides, particularly in the rainy season.

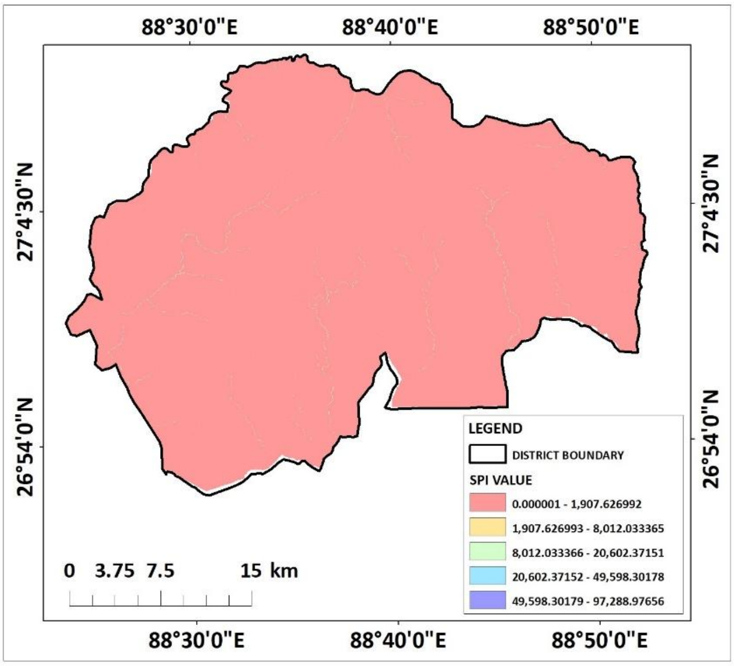


Figure 44: SPI map

Topographic Wetness Index (TWI) map:

The TWI map is used to assess the distribution of moisture in the landscape, based on topographic features such as slope and catchment area. Areas with higher TWI values are characterized by increased moisture accumulation, which can weaken the soil and trigger landslides. The TWI map indicates that steep slopes combined with poor drainage conditions are more susceptible to landslide events. Regions with lower TWI values, typically found in well-drained areas, show lower susceptibility to landslides. This map is crucial for understanding how water

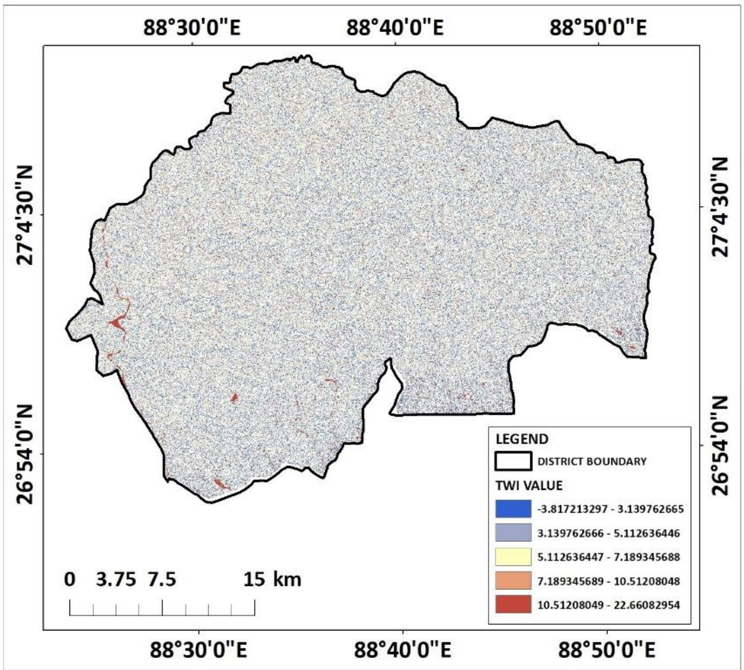


Figure 45: TWI map

accumulates in different parts of the landscape and contributes to slope instability.

Geomorphology map:

The geomorphology map classifies the landscape into distinct geomorphic units based on landforms and terrain characteristics.

Geomorphological features such as hills, valleys, floodplains, and slopes directly affect the stability of the land. Steep hill slopes, mountain ridges, and active river valleys are more prone to landslides due to their susceptibility to erosion and the influence of gravitational forces. Conversely, flat terrain or areas with depositional features, such as floodplains, tend to be more stable. This map provides valuable insights into the landform types, helping to identify regions where geomorphic processes like erosion or deposition might contribute to landslide susceptibility. For instance, areas on the edge of valleys or near active river beds are identified as high-risk zones.

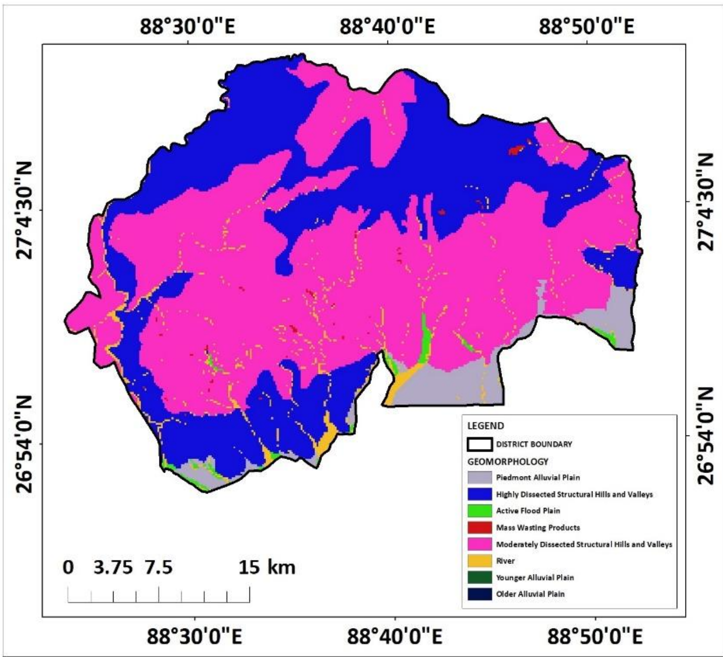


Figure 46: Geomorphology map

Slope steepness and length map:

The slope steepness and length map is a crucial in identifying areas vulnerable to landslides, as the steepness of slopes directly influences gravitational forces acting on the soil. Steep slopes (indicated by higher values on the map) are more prone to instability, especially in areas where the terrain is prolonged or lengthy. The combination of slope steepness and length results in a higher likelihood of soil movement, particularly when combined with heavy rainfall or rapid water flow. The map clearly highlights regions with steep and

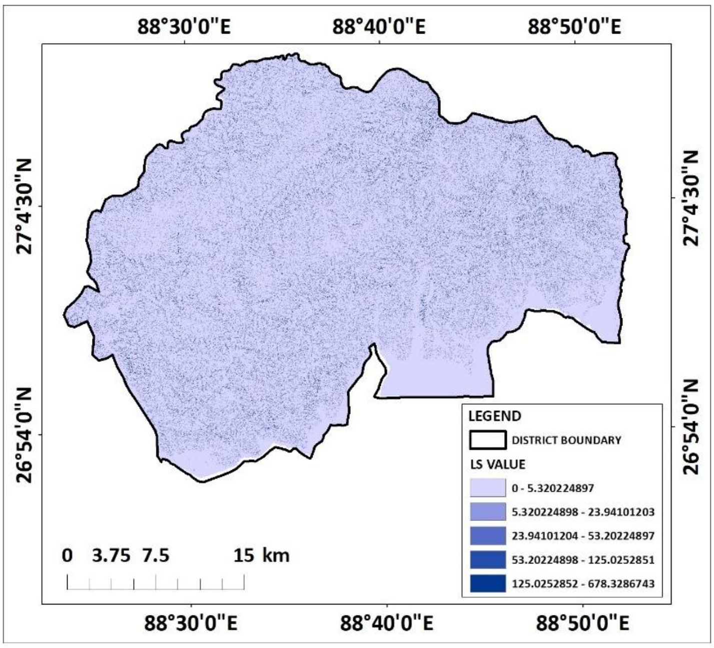


Figure 47: LS map

extended slopes, typically located on ridges or escarpments, as critical landslide-prone zones. These regions are particularly vulnerable to both shallow and deep-seated landslides.

Rainfall intensity map:

The rainfall intensity map indicates the spatial distribution of precipitation within the study area, which is a critical factor in landslide initiation. The areas with high rainfall intensity (represented by darker shades on the map) experience greater soil saturation, which can lead to increased pore pressure, making slopes more prone to failure. These regions coincide with areas of steep terrain, where rainfall rapidly accumulates and cannot be absorbed effectively by the soil. Conversely, regions with lower rainfall intensity (indicated by lighter shades) are less susceptible to landslides due to lower water accumulation. The map emphasizes the importance of rainfall patterns in determining landslide risk, particularly in the monsoon season.

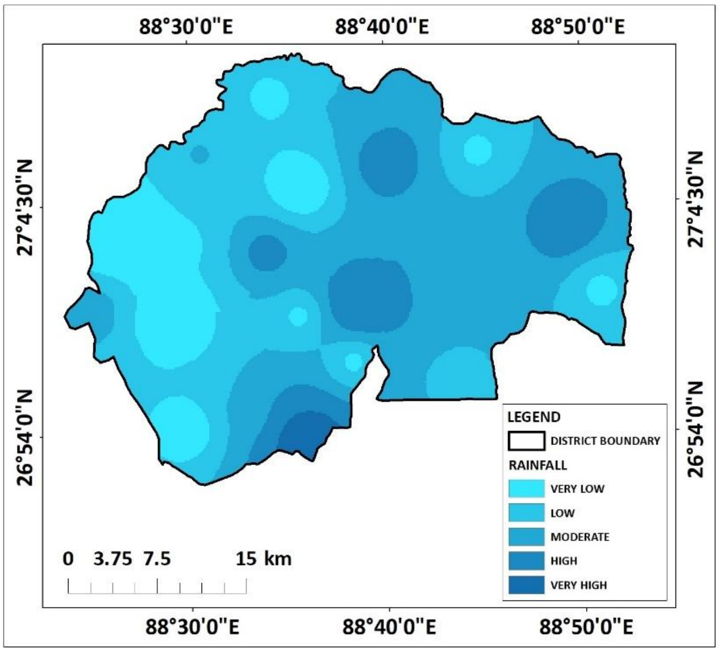


Figure 58: Rainfall map

LULC map:

The LULC map highlights the distribution of different land uses and covers across the study area, which directly impacts the stability of slopes. Deforestation, agricultural activities, and urban expansion (shown by areas with reduced vegetation cover) can contribute to soil degradation and increased vulnerability to landslides. In contrast, forested areas (represented by green shades) are generally more stable due to their soil-binding vegetation. The map suggests that human activities, such as construction or farming on steep slopes, exacerbate the risk of landslides, especially in areas

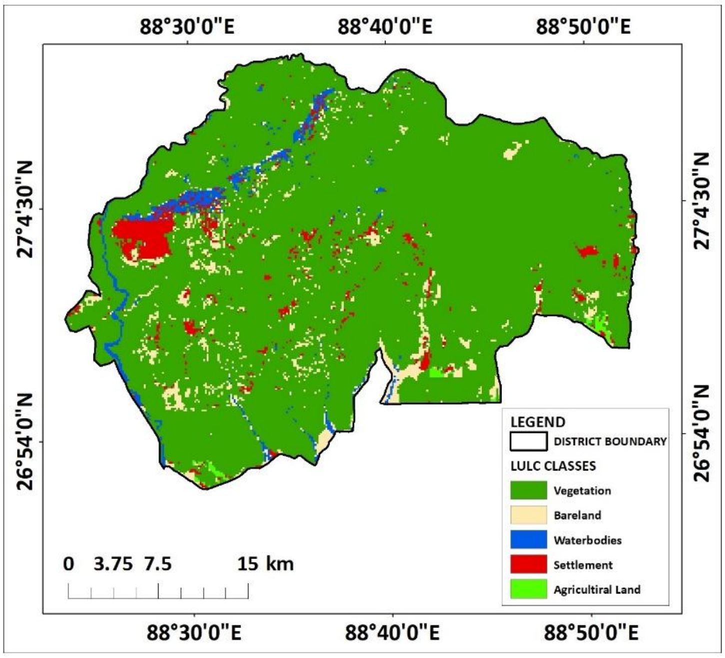


Figure 59: LULC map

with poor soil structure or improper land management practices. This map provides valuable information for land-use planning and disaster risk reduction efforts.

Drainage density map:

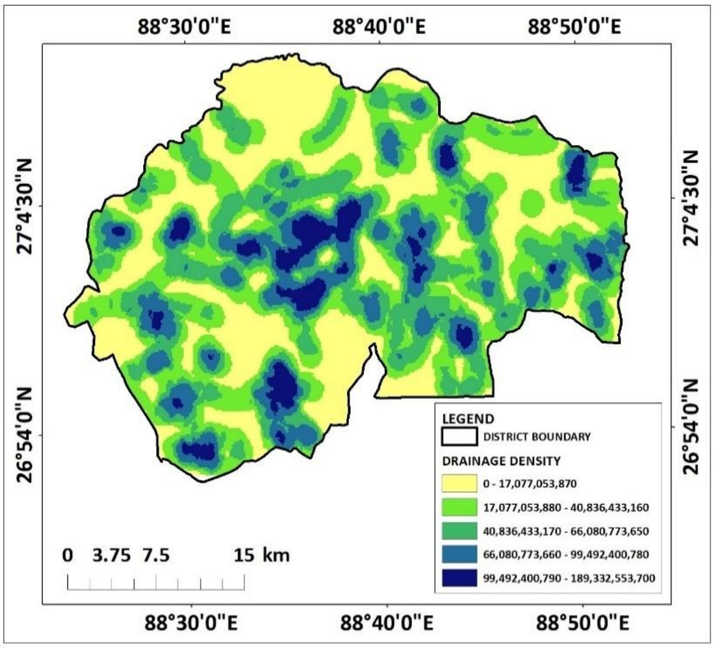


Figure 60: Drainage density map

The drainage density map provides important insights into the network of rivers and streams within the study area. Drainage density refers to the total length of the drainage network (rivers, streams, etc.) per unit area, with higher values indicating a denser network. Areas with high drainage density typically experience quicker water runoff, which can lead to increased soil erosion and slope instability, especially in regions with steep terrain. This map highlights areas that are more likely to experience concentrated water flow, exacerbating landslide susceptibility. Lower drainage density areas, on the other hand, have slower water movement, contributing to better soil stability. The drainage density map, therefore, is a crucial factor in identifying zones prone to water-induced landslides.

Distance from roads map:

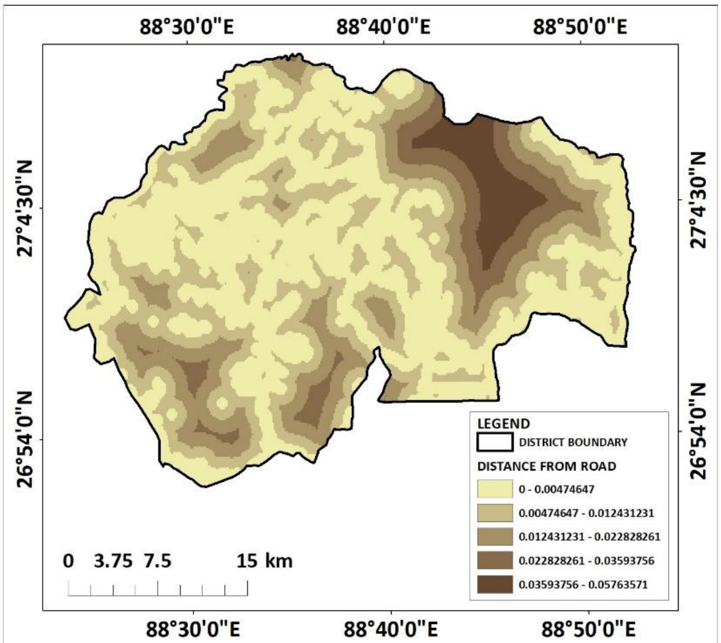


Figure 61: Distance from road map

The distance from roads map shows the spatial relationship between transportation infrastructure and landslide-prone areas. Roads, especially those constructed on or near steep slopes, can destabilize the surrounding land by altering natural drainage patterns and disturbing the soil. Additionally, areas closer to roads may have increased human activity and development, further contributing to landslide risk due to excavation, deforestation, and construction. This map identifies regions near roadways as potentially more vulnerable to landslides, making it an important layer for disaster management planning. Conversely, areas farther from roads tend to be less disturbed and have a lower likelihood of landslide incidents, though they can still be at risk due to natural topographic and climatic factors.

Distance from river map:

The distance from rivers map represents the proximity of different areas to riverbanks, which can influence the likelihood of landslides due to erosion. Areas that are in close proximity to rivers, especially those located along steep slopes, are more susceptible to river-induced landslides. Erosion caused by riverbank undercutting can lead to the destabilization of nearby soil, resulting in slope failure. The map highlights regions near rivers as high-risk zones, particularly where the riverbanks are not stabilized or protected. In contrast, areas further away from rivers are less exposed to river-induced erosion and are generally at a lower risk of landslides related to water flow from rivers.

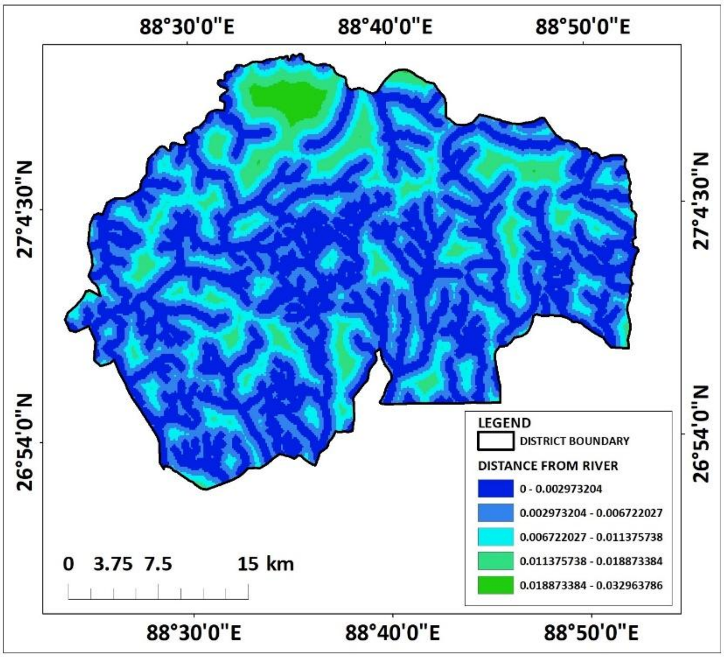


Figure 62: Distance from river map

Distance from settlement map:

The distance from settlement map indicates how proximity to human settlements influences landslide susceptibility. Areas closer to settlements are often more disturbed due to human activities such as construction, road building, and land modification, which can alter natural drainage patterns and destabilize the soil. Furthermore, human settlements near steep slopes are at higher risk, as these areas are more likely to experience both direct impacts from landslides and secondary hazards like debris flows or road blockages. Conversely, areas farther from settlements, often

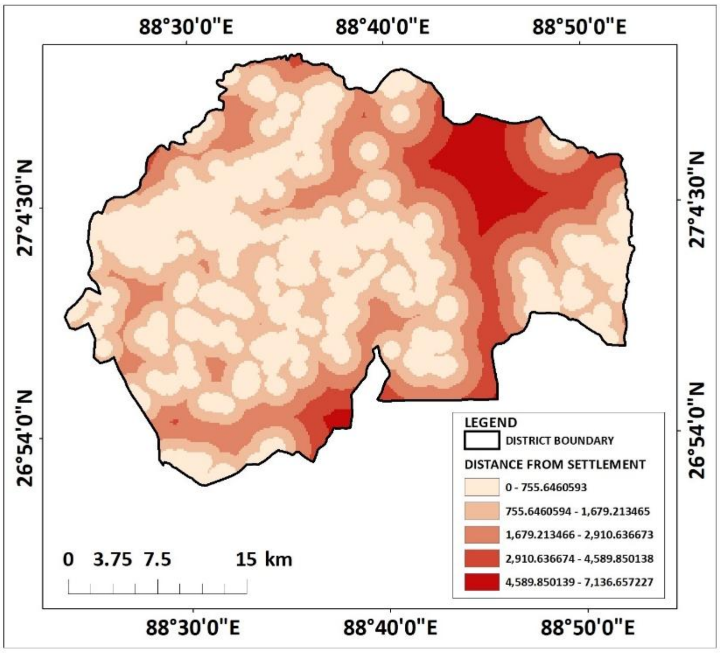


Figure 63: Distance from settlements map

characterized by more undisturbed land, show lower susceptibility to landslides. This map highlights regions where infrastructure development needs to consider landslide risks and where disaster mitigation efforts should focus.

Slope map:

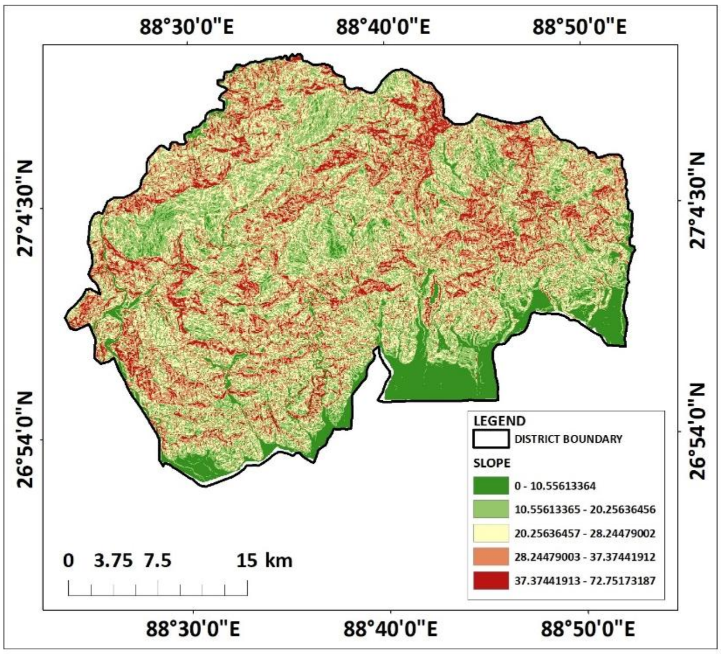


Figure 64: Slope map

The slope map represents the steepness or degree of incline of the terrain in the study area. Slope plays a crucial role in landslide susceptibility, as steeper slopes are generally more prone to soil movement due to gravitational forces. Areas with high slope values are more vulnerable to landslides, especially under conditions of heavy rainfall or seismic activity, as the soil becomes unstable when it is saturated with water. In the Kalimpong district, steep areas, such as those along ridgelines or mountain slopes, are at higher risk for mass wasting events, including landslides, because of the increased pressure exerted on the soil and bedrock. On the other hand, flatter areas, such as valleys or plateaus, experience less gravitational stress and are less likely to experience landslides, although other factors such as soil type or water drainage might still contribute to instability.

Aspect map:

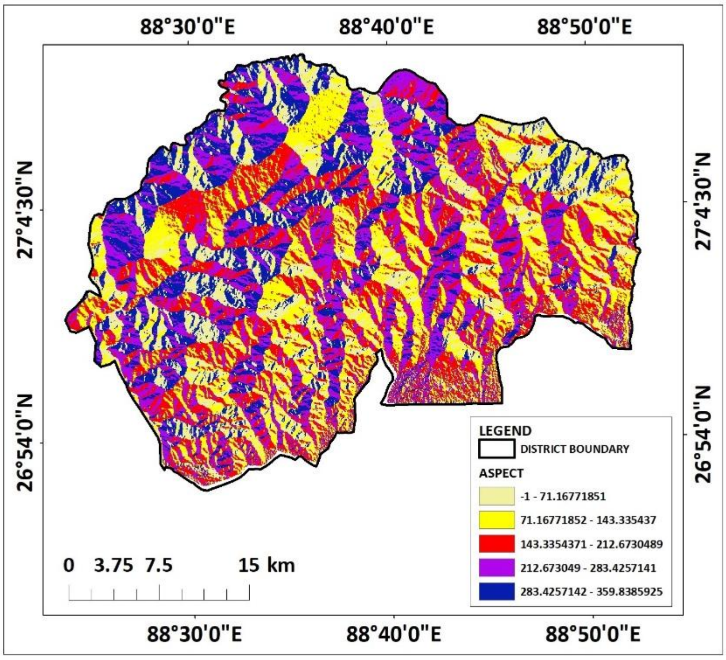
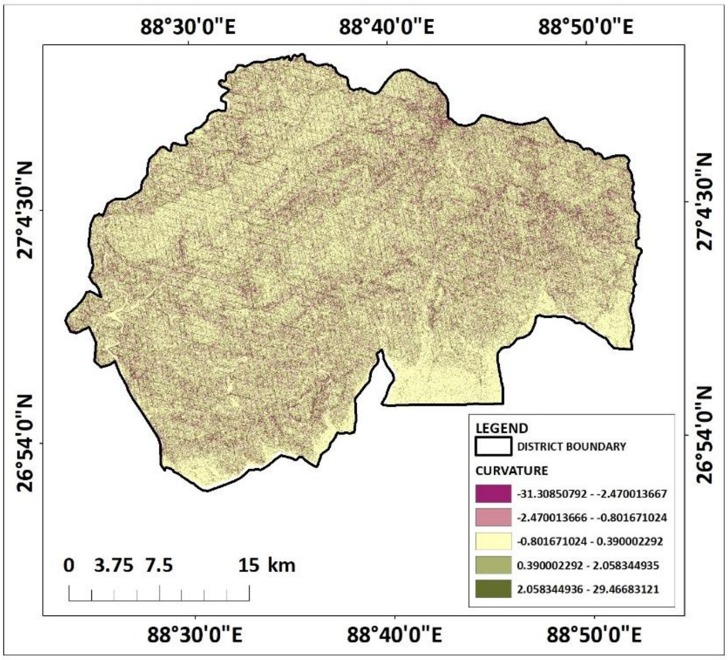


Figure 65: Aspect map

The aspect map indicates the direction each slope faces in relation to the cardinal directions (north, south, east, or west). This factor significantly affects landslide susceptibility because it influences several environmental conditions such as sunlight, temperature, and moisture content. Slopes that face the sun (typically south or west-facing slopes in the Northern Hemisphere) tend to have drier soils due to higher evaporation and greater exposure to sunlight, which can reduce vegetation cover and lead to soil weakening. As a result, these slopes are more prone to dry landslides, especially during the dry season when moisture is low. Conversely, north-facing or east-facing slopes tend to receive less direct sunlight, leading to cooler conditions and higher moisture retention. These slopes are more prone to wet landslides, particularly during the monsoon season, as the moisture saturation weakens the soil, making it more likely to fail. The slope aspect map helps in identifying areas where particular weather conditions—such as prolonged rain or dry spells—can increase the likelihood of landslides, depending on the slope orientation and its relationship with the surrounding environment. Therefore, the slope aspect map is an essential tool for assessing the risk of both dry and wet landslides in different topographic settings.



Curvature map:

The curvature map represents the convexity or concavity of the terrain and is used to assess how the landscape shape influences water flow and soil stability. Convex areas tend to promote water runoff, which can lead to erosion and slope instability, while concave areas often trap water, creating conditions for increased moisture accumulation and soil weakening. Steep concave slopes are more prone to landslides, especially when combined with heavy rainfall, as they have a higher likelihood of saturation. The curvature map helps to identify areas where water is likely to accumulate or run off, providing insights into the distribution of landslide susceptibility across different topographic features.

Weighted Sum map:

The weighted sum map combines all the individual parameter maps using a weighted sum method to produce a comprehensive landslide susceptibility map. Each criterion, such as soil texture, slope steepness, SPI, TWI, and others, is assigned a weight based on its contribution to landslide potential in the study area. The weighted sum map synthesizes these individual layers to provide a final landslide susceptibility classification for the entire study area. This map identifies high-risk zones where multiple factors,

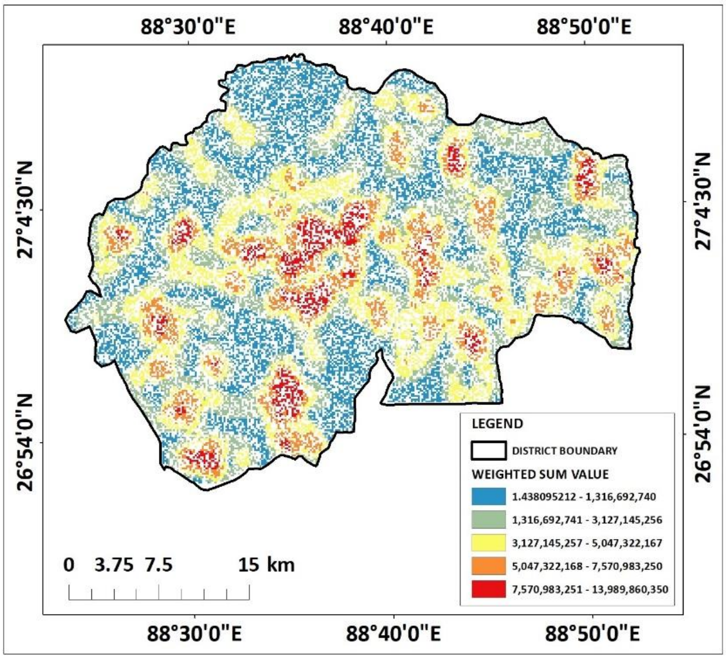


Figure 67: Weighted Sum map

such as steep slopes, high SPI, and poor soil stability, converge. The weighted sum map is an essential tool for land-use planning and risk mitigation, as it visually represents areas that require immediate attention and intervention to prevent landslide disasters.

IDW map based on landslide potential zones:

The landslide susceptibility map generated through the integration of multiple criteria reveals a complex spatial pattern of vulnerability within the Kalimpong district. The map effectively delineates areas with varying degrees of landslide potential, ranging from very low to very high. Notably, the northern and eastern regions of the district exhibit higher susceptibility, with the presence of steep slopes, dense drainage networks, and areas with high rainfall intensity. These factors, coupled with the influence of human activities like road construction and urbanization, contribute to the increased risk of landslides in these zones.

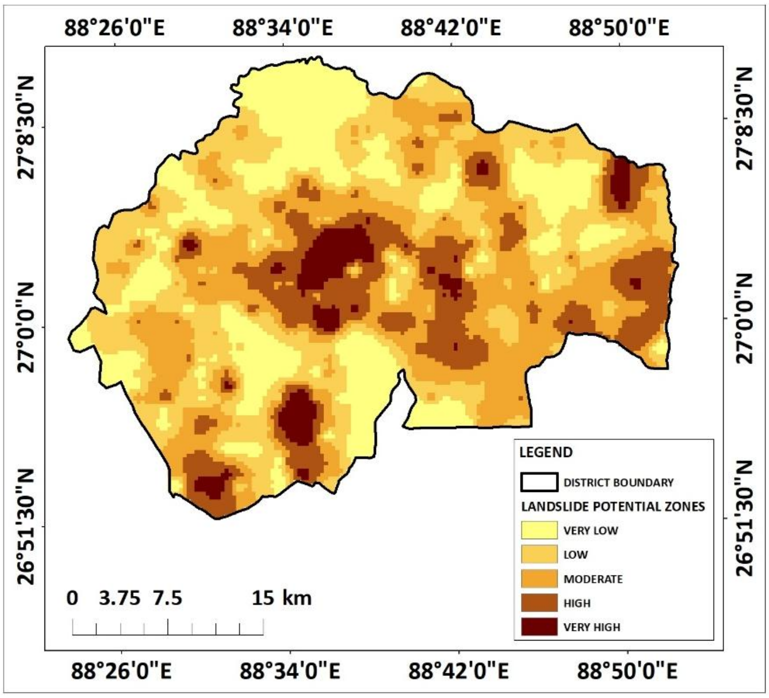


Figure 68: IDW map based on landslide potential zones

The analysis highlights the importance of geomorphic factors, such as slope steepness and aspect, in controlling landslide occurrence. Areas with steeper slopes and those facing the southwest direction, which are more exposed to rainfall, are identified as particularly vulnerable. Additionally, the influence of soil texture and topography on water infiltration and runoff generation is evident. Areas with finer-grained soils and higher topographic wetness indices are more prone to landslides, as they retain more water and are susceptible to saturation.

The integration of anthropogenic factors, such as land use and land cover (LULC), road density, and proximity to settlements, further enhances the understanding of landslide susceptibility. Areas with higher road densities and settlements are more likely to experience landslides, as these activities often involve slope cutting, excavation, and other disturbances that can destabilize slopes. Furthermore, the presence of existing landslide scars serves as a valuable indicator of past landslide activity and potential future events.

From the above information we can easily conclude that, the landslide susceptibility map provides a valuable tool for identifying areas at risk and guiding future land use planning and disaster management strategies in the Kalimpong district. By understanding the complex interplay of natural and human-induced factors, it is possible to implement measures to mitigate landslide risks and protect vulnerable communities.

5. CONSLUSION:

The landslide susceptibility map generated through the integration of multiple criteria reveals a complex spatial pattern of vulnerability within the Kalimpong district. The map effectively delineates areas with varying degrees of landslide potential, ranging from very low to very high. Notably, the northern and eastern regions of the district exhibit higher susceptibility, with the presence of steep slopes, dense drainage networks, and areas with high rainfall intensity. These factors, coupled with the influence of human activities like road construction and urbanization, contribute to the increased risk of landslides in these zones.

The analysis highlights the importance of geomorphic factors, such as slope steepness and aspect, in controlling landslide occurrence. Areas with steeper slopes and those facing the southwest direction, which are more exposed to rainfall, are identified as particularly vulnerable. Additionally, the influence of soil texture and topography on water infiltration and runoff generation is evident. Areas with finer-grained soils and higher topographic wetness indices are more prone to landslides, as they retain more water and are susceptible to saturation.

The integration of anthropogenic factors, such as land use and land cover (LULC), road density, and proximity to settlements, further enhances the understanding of landslide susceptibility. Areas with higher road densities and settlements are more likely to experience landslides, as these activities often involve slope cutting, excavation, and other disturbances that can destabilize slopes. Furthermore, the presence of existing landslide scars serves as a valuable indicator of past landslide activity and potential future events.

In conclusion, the landslide susceptibility map provides a valuable tool for identifying areas at risk and guiding future land use planning and disaster management strategies in the Kalimpong district. By understanding the complex interplay of natural and human-induced factors, it is possible to implement measures to mitigate landslide risks and protect vulnerable communities.

LANDSLIDE SUSCEPTIBLE ZONE MAPPING USING MCDM TECHNIQUE: A CASE STUDY ON DARJEELING AND KALIMPONG, WEST BENGAL

1. INTRODUCTION:

Landslides are one of the most devastating natural hazards, significantly impacting human life, infrastructure, and the environment, particularly in mountainous and hilly regions. The increasing frequency and intensity of landslides due to natural processes and human-induced activities necessitate a detailed and comprehensive understanding of the factors contributing to their occurrence. Landslides are influenced by a variety of geophysical, hydrological, and anthropogenic factors, making their susceptibility analysis a complex but crucial endeavor. To mitigate their impacts and ensure sustainable land-use planning, it is imperative to identify areas prone to landslides using a multi-criteria approach.

Soil texture plays a fundamental role in landslide susceptibility, as it determines the water retention capacity and shear strength of the soil. Fine-grained soils, such as clay, are more prone to saturation and slippage, whereas coarser soils like sand exhibit better drainage characteristics. The Stream Power Index (SPI) and Topographic Wetness Index (TWI) are valuable hydrological indicators for assessing the potential for surface runoff and saturation, which can trigger slope instability. These indices, derived from digital elevation models, help in quantifying the effects of water flow and accumulation on slope stability, particularly in areas with intense rainfall.

Geomorphological features and slope characteristics, including slope steepness, slope length, and slope aspect, are critical in determining landslide susceptibility. Steeper slopes are inherently more prone to instability due to gravitational forces, while the aspect influences the amount of solar radiation, temperature, and vegetation cover, all of which affect soil moisture dynamics. Rainfall intensity acts as a direct trigger for many landslides, as it increases pore water pressure and reduces the cohesive strength of the soil. Similarly, curvature, which represents the shape of the terrain, can indicate zones of convergence or divergence of surface flow, further influencing stability.

Land Use and Land Cover (LULC) patterns, drainage density, and proximity to anthropogenic features such as roads, rivers, and settlements also play pivotal roles in determining landslide susceptibility. Urbanization, deforestation, and agricultural expansion often disrupt the natural equilibrium of the landscape, exacerbating the risk of landslides. Roads and settlements not only impose direct physical alterations to the terrain but also increase the vulnerability of nearby areas due to construction activities, excavation, and the concentration of water runoff. Drainage density reflects the intensity of channel development in a region, which directly correlates with erosion potential and slope failure.

This study aims to comprehensively assess landslide susceptibility by integrating diverse criteria using geospatial tools. The inclusion of multiple parameters ensures a holistic understanding of the factors contributing to landslides, enabling effective risk management and disaster preparedness.

2. STUDY AREA:

The Darjeeling and Kalimpong districts of West Bengal, India, located in the eastern Himalayan region, are renowned for their scenic landscapes, unique biodiversity, and tea plantations. Geographically, the area lies between latitudes 26°27′ N to 27°13′ N and longitudes 87°59′ E to 88°53′ E, covering a region characterized by steep slopes, rugged terrain, and fragile geological formations. These districts are part of the Eastern Himalayan range, which is inherently prone to landslides due to its active tectonics, high rainfall, and complex topography.

The climate of the region plays a significant role in shaping its susceptibility to landslides. Darjeeling and Kalimpong receive heavy monsoonal rainfall, with an annual average of 2000–4000 mm. This intense precipitation often leads to saturated soils, increased pore water pressure, and erosion, triggering landslides across the districts. Furthermore, the area's diverse land-use patterns, ranging from dense forests to sprawling tea gardens and urban settlements, contribute to the variability in slope stability. Deforestation, agricultural expansion, and infrastructure development have further exacerbated the risks by altering the natural drainage and destabilizing slopes.

The geomorphology of the region is dominated by steep hills, deep valleys, and narrow ridges, with elevations ranging from about 150 meters in the foothills to over 3600 meters at Sandakphu, the highest peak in West Bengal. The steep slopes combined with loose and weathered soil make the terrain particularly vulnerable to landslides. The region's geology consists predominantly of metamorphic rocks such as gneisses, schists, and phyllites, which are prone to weathering and erosion, further increasing the instability of slopes.

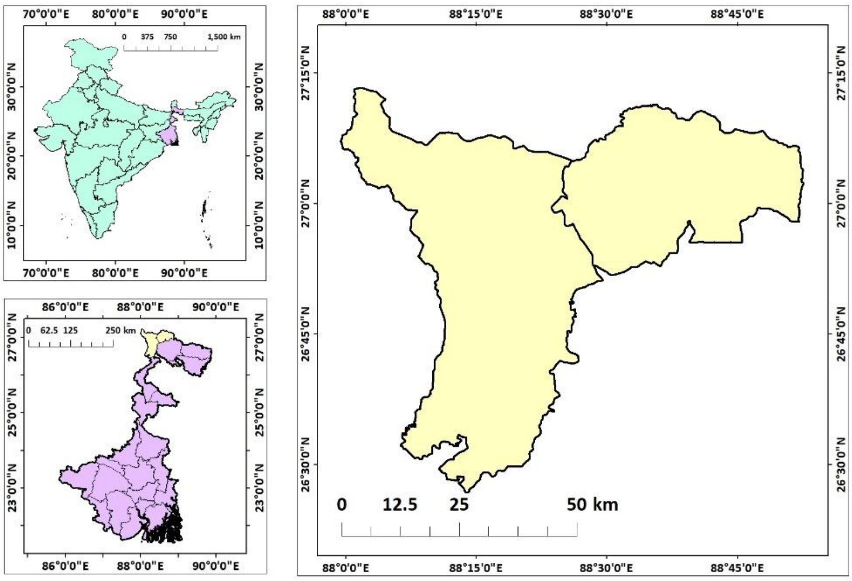


Figure 69: Location map of Darjeeling and Kalimpong districts

The population in the districts is distributed across urban centers like Darjeeling, Kalimpong, and Kurseong, as well as numerous rural villages. Rapid urbanization, coupled with unplanned development and poor drainage management, has heightened the risk of landslides in densely populated areas. Additionally, critical infrastructure such as roads, railway tracks, and hydropower projects traverses the region, making it essential to assess landslide susceptibility for the safety of these structures and the communities they serve.

Given the socioeconomic importance of these districts, a detailed landslide susceptibility analysis is crucial. The findings will aid in planning and implementing effective disaster risk reduction measures, safeguarding lives, livelihoods, and the region's natural heritage.

3. DATA SOURCES AND METHODOLOGY:

DATA SOURCES:

- **Soil Texture**:
  - Source: National Bureau of Soil Survey and Land Use Planning (NBSSLUP).
  - Details: Provides soil texture classification, essential for assessing erodibility.
- **Stream Power Index (SPI)**:
  - Derived from: DEM (USGS).
  - Calculated using the formula: $SPI = A \cdot \tan(\beta)$
  - where:
    - A: Upslope contributing area (m^2).
    - β: Slope gradient in radians.
- **Topographic Wetness Index (TWI):**
  - Derived from: DEM (USGS).
  - Calculated using the $\mathrm{TWI} = \ln\left(\frac{a}{\tan\beta}\right)$ formula:
  - where:
    - α: The specific catchment area (upstream contributing area per unit width, typically in m^2/m).
    - β: The slope angle in radians (or the tangent of the slope).
- **Geomorphology**:
  - Source: Bhukosh portal, Geological Survey of India (GSI).
  - Details: Provides geomorphological maps for classifying terrain features.
- **Slope Length and Steepness Factor (LS)**:
  - Derived from: Digital Elevation Model (DEM) obtained from the USGS Earth Explorer.
  - Calculated using ArcGIS with the $LS = \left(\frac{\lambda}{22.13}\right)^m \cdot (\sin\theta \cdot 0.0896)^n$ formula:
  - where:
    - λ: Slope length (m).
    - θ: Slope angle in radians.
    - m and n: Empirical constants (typically m=0.5, n=1.3m)

- **Rainfall Intensity**:
  - Source: Climate Hazards Group InfraRed Precipitation with Station Data (CHRS data portal).
  - Details: Provides NETCDF data.
- **Land Use and Land Cover (LULC)**:
  - Source: ESRI Sentinel-2 Land Cover Explorer.
  - Details: Classified LULC data for assessing human and natural influences on soil erosion.
- **Drainage Density**:
  - Source: Downloaded spatial datasets from BB Bike.
  - Derived from: Processed in ArcGIS using line density feature.
- **Distances from Road, River, and Settlement**:
  - Source: Downloaded spatial datasets from BB Bike.
  - Derives from: Processed in ArcGIS to calculate Euclidean distances for each feature.
- **Slope**:
  - Derived from: Digital Elevation Model (DEM) obtained from the USGS Earth Explorer.
  - Details: Calculated using Slope feature in Spatial analyst tool in ArcGIS
- **Aspect**:
  - Derived from: Digital Elevation Model (DEM) obtained from the USGS Earth Explorer.
  - Details: Calculated using Aspect feature in Spatial analyst tool in ArcGIS
- **Curvature**:
  - Derived from: Digital Elevation Model (DEM) obtained from the USGS Earth Explorer.
  - Details: Calculated using Curvature feature in Spatial analyst tool in ArcGIS
- **Existing landslide scars**:
  - Derived from: Digital Elevation Model (DEM) obtained from the USGS Earth Explorer.
  - Details: Processed in ArcGIS.

These diverse data sources and derived parameters provide a comprehensive framework for analysing landslide susceptible zones using geospatial tools.

METHODOLOGY:

The methodology for assessing landslide susceptibility in the Darjeeling and Kalimpong districts was designed to integrate multiple criteria using frequency ratio and weighted sum methods, implemented in a GIS environment. This comprehensive approach utilized various geophysical, hydrological, and anthropogenic parameters to ensure robust spatial modeling of landslide-prone zones.

Frequency ratio analysis was applied to evaluate the relationship between landslide occurrence and individual causative factors, such as soil texture, slope, rainfall intensity, geomorphology, and proximity to features like roads and rivers. Data for each parameter was categorized into classes, and the spatial distribution of past landslides was used to calculate the frequency ratio for each class. This ratio quantifies the degree of association between landslide occurrences and specific factor classes, which were subsequently normalized and utilized in the weighted sum analysis. The formula is:

$$FR = \frac{\text{Area of the factor class with hazard occurrence}}{\text{Total area of the factor class}} \div \frac{\text{Total area of hazard occurrence}}{\text{Total study area}}$$

Alternatively, this can be expressed as:

$$FR = \frac{\frac{\text{Number of hazard events in a factor class}}{\text{Total hazard events}}}{\frac{\text{Area of the factor class}}{\text{Total study area}}}$$

Where:

- **Factor class**: A specific category or range of a causative factor (e.g., slope range, land use type, etc.).
- **Hazard occurrence**: Locations where the hazard has been observed (e.g., landslide scars or flood-affected areas).
- **Study area**: The entire geographic region being analyzed.

Interpretation:

- **FR > 1**: The factor class has a higher influence on hazard occurrence.
- **FR = 1**: The factor class has a neutral influence.
- **FR < 1**: The factor class has less influence on hazard occurrence.

The weighted index overlay analysis (WIJ) was conducted to assign relative weights to each parameter based on its significance in influencing landslides. The weights were derived through expert judgment and validated using statistical measures. Parameters with higher weights exerted greater influence in the final susceptibility model. All spatial data layers were prepared and standardized in ArcGIS to ensure compatibility, and rasterized datasets were used for overlay analysis. This process culminated in the generation of a comprehensive landslide susceptibility map highlighting high-risk zones.

TABLE FOR FREQUENCY RATIO

SL NO	Criteria	Sub class	Landslide zone pixel (count)	Landslide zone total pixel (count)	sub class pixel (count)	sub class total pixel (count)	FR
1	DD	1	50890	236353	685846	3497550	1.098015962
		2	60254		826502		1.078809561
		3	53172		796823		0.987470084
		4	42969		677209		0.938934589
		5	29010		511170		0.83981801
SL NO	Criteria	Sub class	Landslide zone pixel (count)	Landslide zone total pixel (count)	sub class pixel (count)	sub class total pixel (count)	FR
2	ASPECT	1	51869	236353	713510	3621903	1.113995681
		2	47559		690932		1.054807243
		3	43838		769897		0.872556988
		4	44134		722111		0.936580316
		5	48817		725453		1.031187172
SL NO	Criteria	Sub class	Landslide zone pixel (count)	Landslide zone total pixel (count)	sub class pixel (count)	sub class total pixel (count)	FR
3	CURVETURE	1	71365	236353	804561	3621902	1.359257911
		2	41267		850744		0.743326437
		3	33106		819580		0.619000407
		4	42398		621517		1.04536495
		5	48081		525500		1.402091471
SL NO	Criteria	Sub class	Landslide zone pixel (count)	Landslide zone total pixel (count)	sub class pixel (count)	sub class total pixel (count)	FR
4	RAINFALL	1	20940	236353	694596	3497550	0.446115396
		2	94629		706046		1.983325823
		3	32568		678955		0.709827648
		4	20160		707304		0.421781208
		5	67998		710649		1.415936583
SL NO	Criteria	Sub class	Landslide zone pixel (count)	Landslide zone total pixel (count)	sub class pixel (count)	sub class total pixel (count)	FR
5	TWI	1	17920	236353	342260	1862066	0.774981
		2	24395		415983		0.86803
		3	25649		380446		0.9979
		4	26291		372950		1.043436
		5	23493		350427		0.992317
SL NO	Criteria	Sub class	Landslide zone pixel (count)	Landslide zone total pixel (count)	sub class pixel (count)	sub class total pixel (count)	FR
6	SLOPE	1	1915	236353	734197	3622734	0.038607
		2	32100		722476		0.657644
		3	65119		722541		1.333996
		4	68218		722723		1.397129
		5	68904		720797		1.414949
SL NO	Criteria	Sub class	Landslide zone pixel (count)	Landslide zone total pixel (count)	sub class pixel (count)	sub class total pixel (count)	FR
7	GEOMORPHOLOGY	Waterbodies-Other	844	236353	53136	3464590	0.232832884
		Alluvial Plain	835		615739		0.01987836
		Flood plain	0		372245		0
		Waterbody - River	7172		60171		1.747204663
		Moderately Dissected Hills and Valleys	164739		1322251		1.826304754
		Highly Dissected Hills and Valleys	60884		1037250		0.860419931
		Mass Wasting Products	827		3798		3.19184121
SL NO	Criteria	Sub class	Landslide zone pixel (count)	Landslide zone total pixel (count)	sub class pixel (count)	sub class total pixel (count)	FR
8	SOIL	LITHIC UDORTHENTS	3406	236353	268717	3453054	0.185179008
		TYPIC UDORTHENTS	131484		1507539		1.274226411
		UMBRIC DYSTROCHREPTS	7305		393044		0.271532176
		TYPIC HAPLUMBREPTS	79177		498029		2.322665457
		UFLUVENTIC EUTROCHREPTS	9948		225006		0.645927728
		AQUIC UDIFLUVENTS	0		560006		0
		TYPIC FLUVAQUENTS	0		428		0
		TYPIC USTOCHREPTS	0		285		0
SL NO	Criteria	Sub class	Landslide zone pixel (count)	Landslide zone total pixel (count)	sub class pixel (count)	sub class total pixel (count)	FR
9	LULC	WATER	2189	236353	201338	31478058	1.447994204
		TREES	203445		22253111		1.21759568
		CROP LAND	16		3496825		0.000609387
		BUILT UP AREA	16328		3319447		0.655109753
		BARE LAND	1106		298124		0.494088789
		CLOUDS	16		4663		0.456984405
		RANGELAND	13230		1897104		0.928785692
		FLOOD VEGETATION	0		7446		0
SL NO	Criteria	Sub class	Landslide zone pixel (count)	Landslide zone total pixel (count)	sub class pixel (count)	sub class total pixel (count)	FR
10	SPI	1	27046	236353	520124	1789218	1.279505
		2	83883		1175143		0.086298
		3	3793		57002		0.032456
		4	1431		21667		0.805328
		5	921		15282		1.388509
SL NO	Criteria	Sub class	Landslide zone pixel (count)	Landslide zone total pixel (count)	sub class pixel (count)	sub class total pixel (count)	FR
11	DISTANCE FROM ROAD	1	18125	236353	300638	3497550	0.955978
		2	59754		1059554		0.869946
		3	43843		740756		0.903854
		4	53053		716121		1.093435
		5	61520		680481		1.251097
SL NO	Criteria	Sub class	Landslide zone pixel (count)	Landslide zone total pixel (count)	sub class pixel (count)	sub class total pixel (count)	FR
12	DISTANCE FROM SETTLEMENT	1	74922	236353	691526	3497550	1.603258907
		2	55735		714709		1.153988729
		3	105606		701310		2.22833958
		4	32		695687		0.000680674
		5	0		694318		0
SL NO	Criteria	Sub class	Landslide zone pixel (count)	Landslide zone total pixel (count)	sub class pixel (count)	sub class total pixel (count)	FR
13	DISTANCE FROM RIVER	1	40534	236353	675291	3497550	0.92855
		2	45363		739260		0.862857
		3	49910		705236		1.034088
		4	54739		696755		1.161008
		5	45749		681008		1.019692
SL NO	Criteria	Sub class	Landslide zone pixel (count)	Landslide zone total pixel (count)	sub class pixel (count)	sub class total pixel (count)	FR
14	LS	1	55659	236353	1131081	3622734	0.728368
		2	160474		2217660		1.071073
		3	12801		172938		1.095627
		4	4433		61195		1.0772238
		5	2889		39860		1.072802

TABLE FOR Wij CALCULATION					
rk	Name	rj	n-rj+1	n-rk+1	wij
1	TWI	6	10	15	0.08333
2	SPI	5	11	14	0.09167
3	RAINFALL	1	15	13	0.125
4	DD	7	9	12	0.075
5	SETTLEMENT	13	3	11	0.025
6	ROAD	12	4	10	0.03333
7	RIVER	14	2	9	0.01667
8	LS	8	8	8	0.06667
9	CURVATURE	3	13	7	0.10833
10	ASPECT	4	12	6	0.1
11	SLOPE	2	14	5	0.11667
12	LULC	10	6	4	0.05
13	SOIL	11	5	3	0.04167
14	GEOMORPHOLOGY	9	7	2	0.05833
15	Landlside Scurs	15	1	1	0.00833
15				120	1

To enhance spatial precision, a fishnet grid was created over the study area, generating uniformly distributed points. These points were then clipped to the boundaries of the study region to focus the analysis on the relevant spatial extent. The spatial integration of the fishnet points and the weighted layers enabled localized analysis and facilitated the interpolation of landslide susceptibility across the entire area.

Finally, the Inverse Distance Weighting (IDW) method was employed to interpolate the susceptibility values and produce a continuous susceptibility surface. IDW allowed for the prediction of susceptibility at unsampled locations based on the weighted influence of nearby points, resulting in a smooth and accurate spatial representation of landslide risk. This methodological framework provided a reliable and scientifically robust tool for identifying landslide-prone areas, supporting disaster risk management and sustainable planning efforts in the region.

4. RESULTS AND DISCUSSION:

The results section presents the spatial distribution of various factors influencing landslide susceptibility, including soil texture, SPI, TWI, geomorphology, slope characteristics, rainfall intensity, land use/land cover, drainage density, distance from roads, rivers, and settlements. The weighted sum method was employed to integrate these factors and generate a landslide susceptibility map. The discussion section delves into the interpretation of these maps, highlighting areas of high and low susceptibility. It analyses the factors contributing to landslide occurrences and discusses the implications of the findings for risk assessment and mitigation planning.

Soil texture map:

The soil texture map of the Darjeeling and Kalimpong districts reveals a complex mosaic of soil types that significantly influence the region's susceptibility to landslides. The map showcases a diverse distribution of coarse loamy, fine loamy, loamy skeletal, and rocky outcrop soils. Coarse-textured soils, primarily found in the southern parts of the districts, exhibit lower cohesion and are more prone to erosion, especially under heavy rainfall. Fine loamy soils, distributed across the mid-elevational ranges, retain moisture but are susceptible to saturation, leading to reduced slope stability. Loamy skeletal soils, observed in the hilly regions, are characterized by high drainage potential, which can reduce waterlogging but may increase erosion in steep areas. Rocky outcrop zones, generally found at higher elevations, display a lower risk of landslides due to their relatively stable structure but can act as triggering zones for debris flows when combined with loose surrounding materials. This spatial variability in soil texture forms a critical input for understanding landslide dynamics.

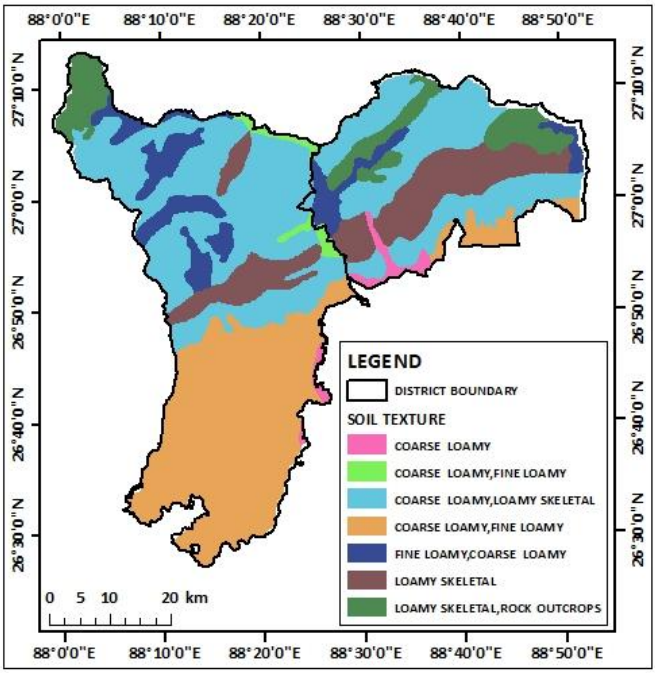


Figure 70: Soil texture map

Stream Power Index (SPI) map:

The Stream Power Index (SPI) map highlights the variation in runoff accumulation and the erosive potential of water across the study area. Higher SPI values are observed in steeply sloped regions and along major drainage channels, indicating zones where concentrated water flow exerts significant erosive forces. These areas, predominantly in the central and northern parts of the districts, are at greater risk of soil destabilization and gully formation, leading to potential landslide initiation points. Conversely, regions with lower SPI values, typically in flatter or gently sloped terrains, exhibit a reduced risk of erosion due to lower runoff velocity. The SPI map

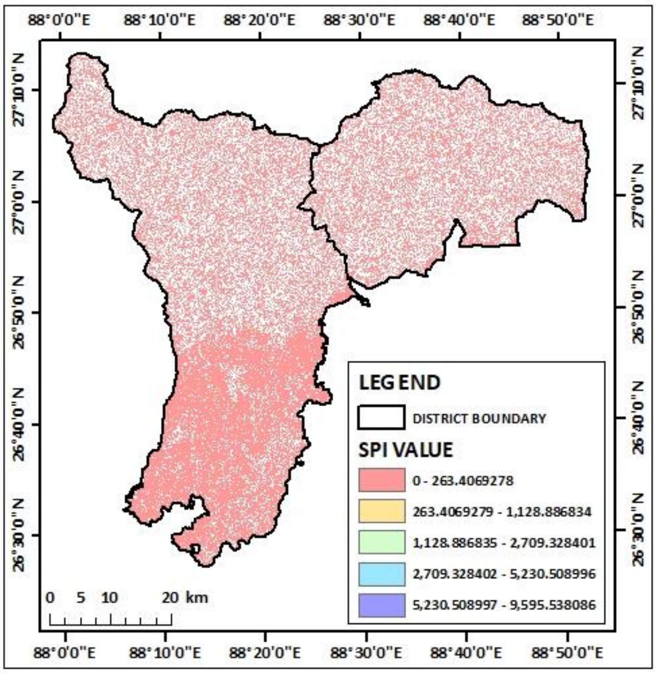


Figure 71: SPI map

provides a detailed representation of the hydrological energy distribution, allowing for targeted identification of high-risk zones. This parameter, when combined with other factors such as soil type and vegetation cover, serves as a critical determinant in assessing overall slope stability and prioritizing areas for mitigation measures.

Topographic Wetness Index (TWI) Map:

The Topographic Wetness Index (TWI) map illustrates the spatial variation in soil moisture potential and water accumulation within the study area. Higher TWI values, primarily observed in valley bottoms and concave slope regions, signify zones with greater water retention and saturation potential. These areas are prone to increased pore water pressure, which reduces shear strength and elevates the risk of landslides. Conversely, lower TWI values are associated with steep, convex slopes where water runoff is rapid, leading to drier conditions and reduced saturation risks. The distribution pattern reflects the interplay between slope gradient and topographic position, with flatter regions retaining more water and steep terrains facilitating quicker drainage. This map is particularly valuable for identifying waterlogging-prone areas, which can act as critical triggers for slope failures, especially during intense rainfall events. Integrating TWI with other parameters enhances the understanding of hydrological influences on landslide susceptibility.

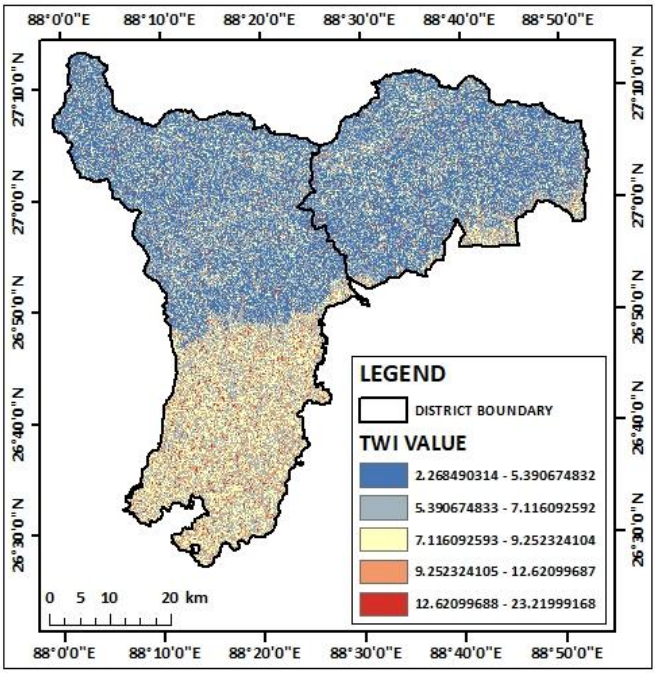


Figure 72: TWI map

Geomorphology map:

The map presents the geomorphological characteristics of the study area, highlighting diverse landforms that influence landslide susceptibility. The region is predominantly characterized by moderately to highly dissected structural hills and valleys, with pockets of piedmont alluvial plains and older alluvial plains. The presence of rivers

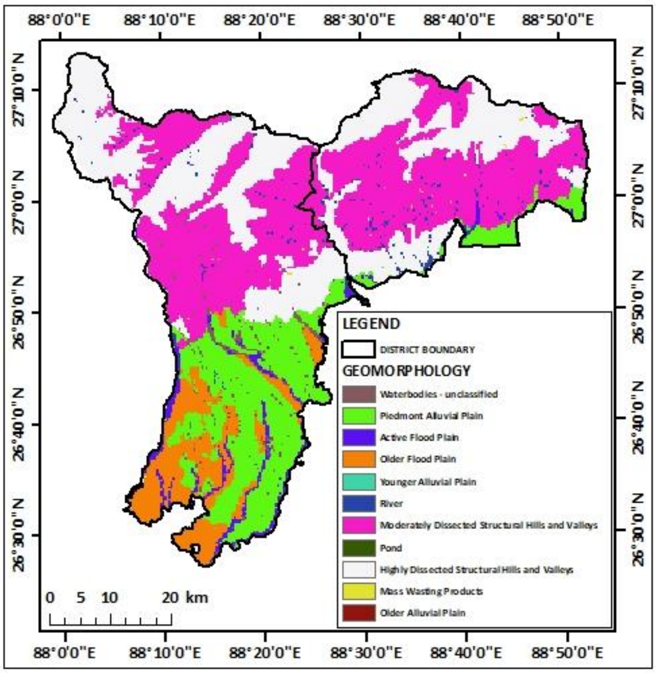


Figure73: Geomorphology map

and ponds further shapes the landscape. Active and older floodplains are also discernible, indicating areas prone to inundation and potential erosion. Mass wasting products are evident in certain regions, suggesting past landslides and ongoing slope instability. The combination of these geomorphological features, along with other factors like slope steepness, rainfall intensity, and land use/land cover, contributes to the varying degrees of landslide susceptibility across the study area.

Slope steepness and length map:

cThis map depicts the spatial distribution of landslide susceptibility (LS) values within the study area. The LS values are calculated using the weighted sum method, incorporating various factors like soil texture, stream power index, topographic wetness index, geomorphology, slope steepness and length, rainfall intensity, land use/land cover, drainage density, distance from roads, rivers, and settlements, slope aspect, curvature, and existing landslide scars. The map reveals that the study area exhibits a wide range of landslide susceptibility, with higher values concentrated in specific regions. These areas are likely characterized by steep slopes, erodible soils, and intense rainfall, making them more prone to landslides. Understanding this spatial variation in landslide susceptibility is crucial for effective risk assessment and mitigation planning.

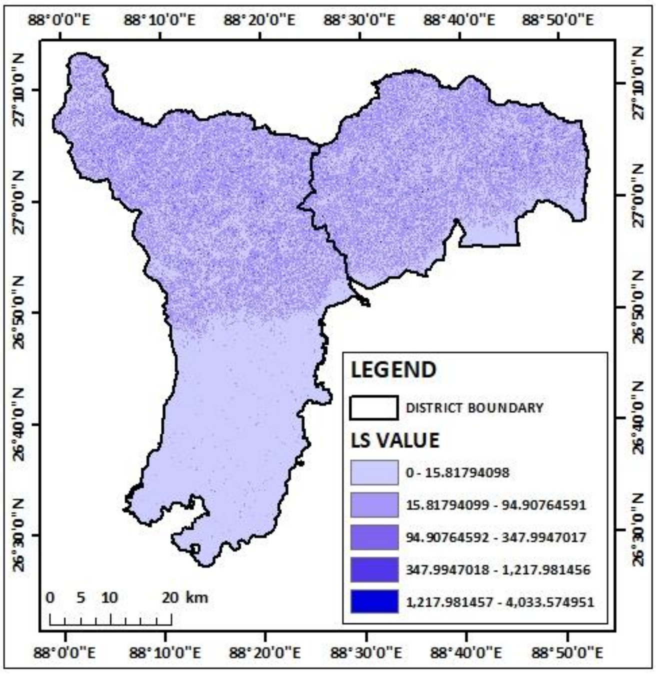


Figure 74: LS map

Rainfall intensity map:

This map illustrates the spatial distribution of rainfall intensity across the study area. The map reveals a significant variation in rainfall patterns, with areas of very high rainfall intensity concentrated in specific regions. These areas are likely characterized by higher elevation, proximity to moisture sources, and orographic effects, leading to increased precipitation. In contrast, lower rainfall intensity is observed in other parts of the study area, possibly due to lower elevations, rain shadow effects, or distance from moisture sources. Understanding this spatial

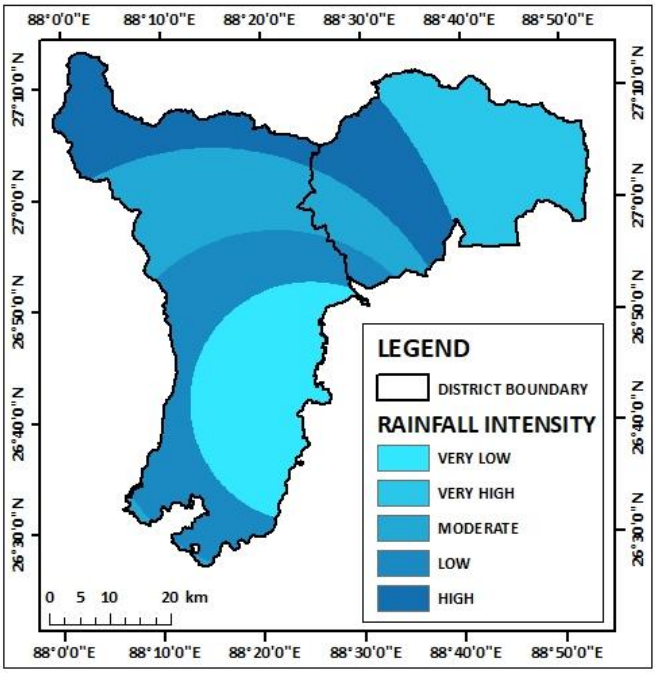


Figure 75: Rainfall intensity map

variability in rainfall intensity is crucial for assessing hydrological processes, soil erosion potential, and the overall impact of rainfall on the landscape.

LULC map:

This map presents the land use and land cover (LULC) classification of the study area. The LULC classes identified include water bodies, trees, flooded vegetation, crops, built-up areas, bare ground, clouds, and rangeland. The map reveals a diverse LULC pattern, with significant areas covered by trees and crops. Built-up areas are concentrated in specific regions, indicating urban and semi-urban development. Water bodies are present throughout the study area, likely contributing to the local hydrology and ecosystem. The presence of bare ground and rangeland highlights areas with limited vegetation cover, which may be susceptible to erosion and other environmental issues. Understanding the spatial distribution of LULC is essential for assessing land use change, ecosystem services, and the potential impacts of human activities on the environment.

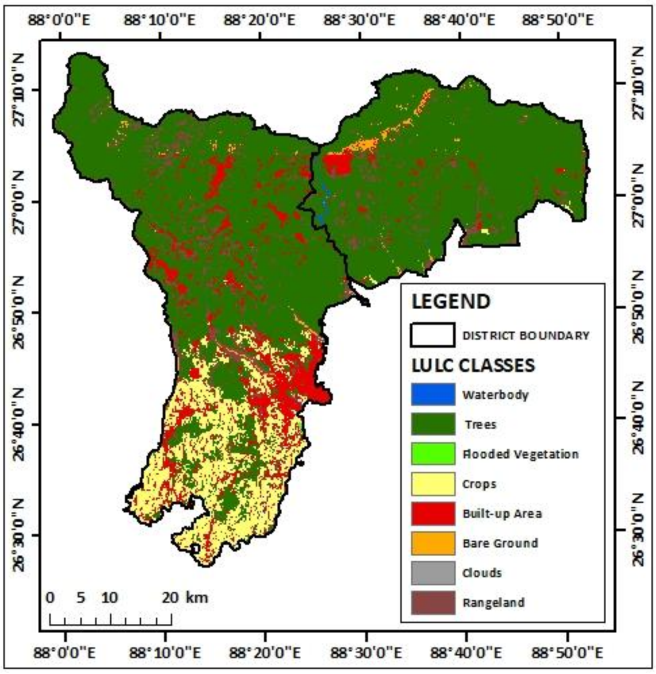


Figure 76: LULC map

Drainage density map:

This map illustrates the spatial distribution of drainage density across the study area. Drainage density is a measure of the total length of streams in a given area, and it provides insights into the hydrological processes and landscape characteristics. The map reveals a variation in drainage density, with higher values concentrated in certain regions. These areas likely have a denser network of streams and rivers, indicating a higher potential for runoff and erosion. In contrast, lower drainage density values are observed in other parts of the study area, suggesting a less developed drainage network and potentially lower runoff potential. Understanding the spatial pattern of drainage density is crucial

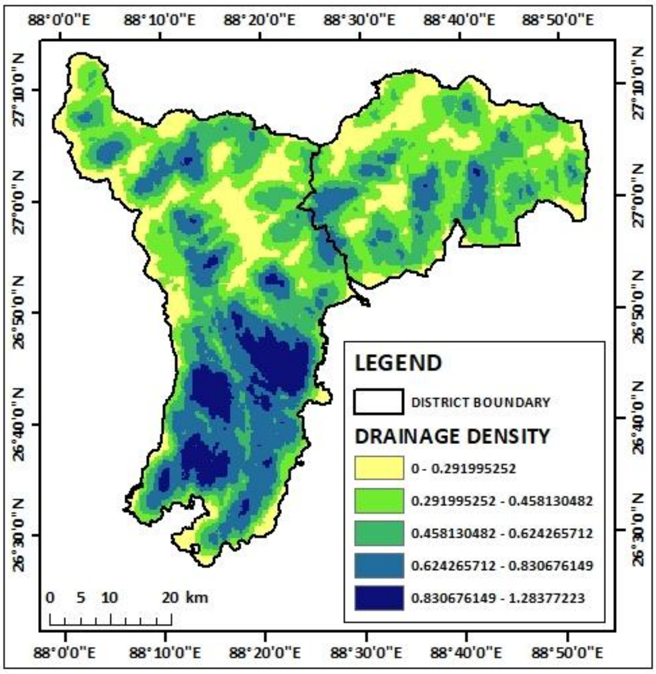


Figure 77: Drainage density map

for assessing hydrological processes, flood risk, and the overall impact of water on the landscape.

Distance from road map:

This map illustrates the spatial distribution of distances from roads within the study area. The map reveals a variation in distances from roads, with some areas closer to roads and others farther away. This information is crucial for understanding accessibility, transportation patterns, and the potential impact of human activities on the landscape. Areas closer to roads may experience greater human influence, such as development, pollution, and disturbance to natural habitats. In contrast, areas farther from roads may be more remote and less impacted by human activities. Understanding this spatial pattern of distances from roads is essential for land use planning, infrastructure development, and environmental conservation.

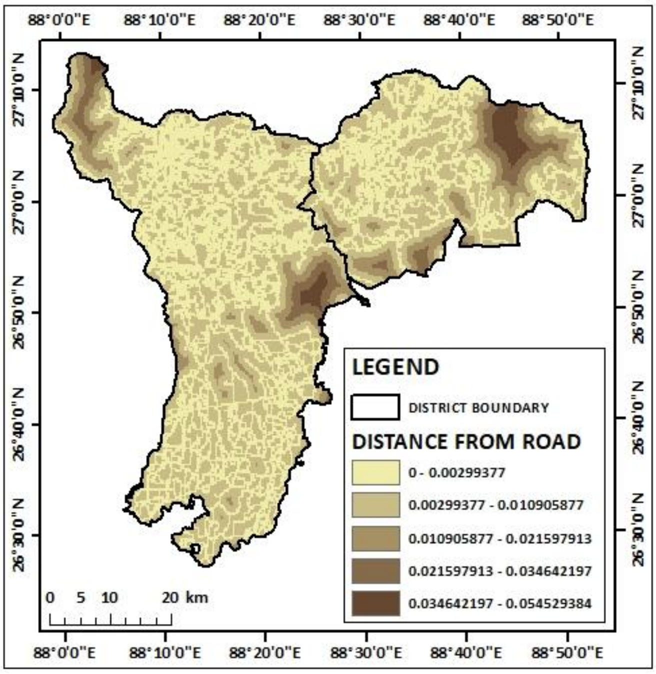


Figure 78: Distance from road map

Distance from river map:

This map illustrates the spatial distribution of distances from rivers within the study area. The map reveals a variation in distances from rivers, with some areas closer to rivers and others farther away. This information is crucial for understanding hydrological processes, flood risk, and the potential impact of rivers on the landscape. Areas closer to rivers may be more prone to flooding, erosion, and sedimentation, while areas farther from rivers may experience less hydrological influence. Understanding this spatial pattern of distances from rivers is essential for water resource management, flood control, and environmental protection.

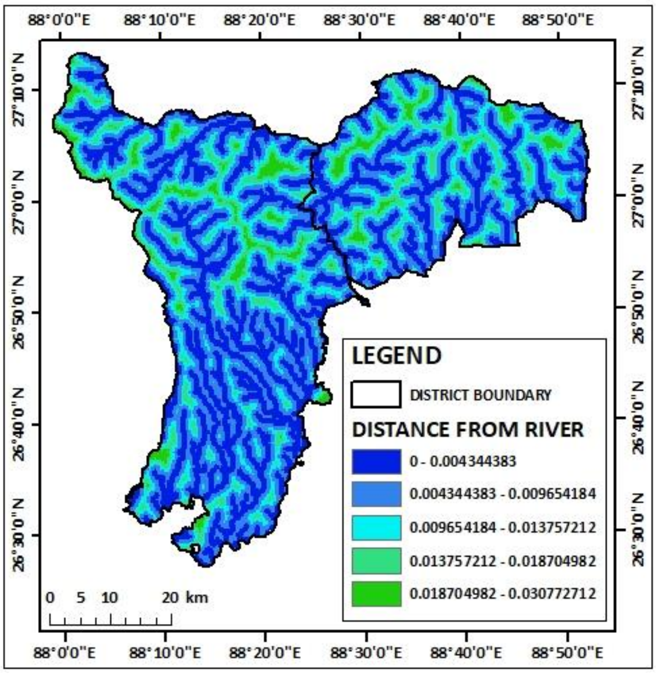


Figure 79: Distance from river map

Distance from settlement map:

This map illustrates the spatial distribution of distances from settlements within the study area. The map reveals a variation in distances from settlements, with some areas closer to settlements and others farther away. This information is crucial for understanding human impact on the landscape, land use patterns, and the potential for development and urbanization. Areas closer to settlements may experience greater human influence, such as development, pollution, and disturbance to natural habitats. In contrast, areas farther from settlements may be more remote and less impacted by human activities. Understanding this spatial pattern of distances from settlements is essential for land use planning, urban development, and environmental conservation.

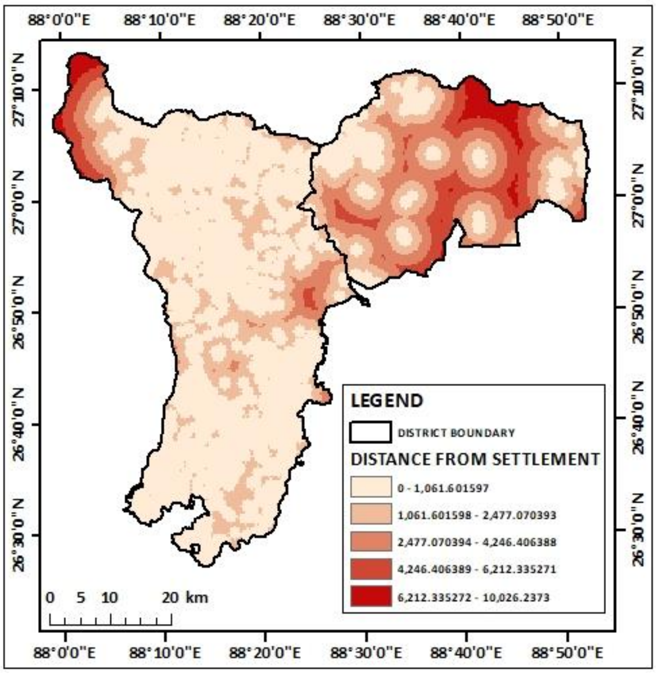


Figure 80: Distance from settlement map

Slope map:

This map illustrates the spatial distribution of slope steepness across the study area. Slope steepness is a critical factor influencing various geomorphological processes, including erosion, landslides, and surface runoff. The map reveals a variation in slope steepness, with steeper slopes concentrated in specific regions. These areas are more susceptible to erosion and landslides due to the increased gravitational force acting on the slope. In contrast, areas with gentler slopes are less prone to these processes. Understanding the spatial pattern of slope steepness is essential for assessing slope stability, erosion risk, and the overall impact of slope processes on the landscape.

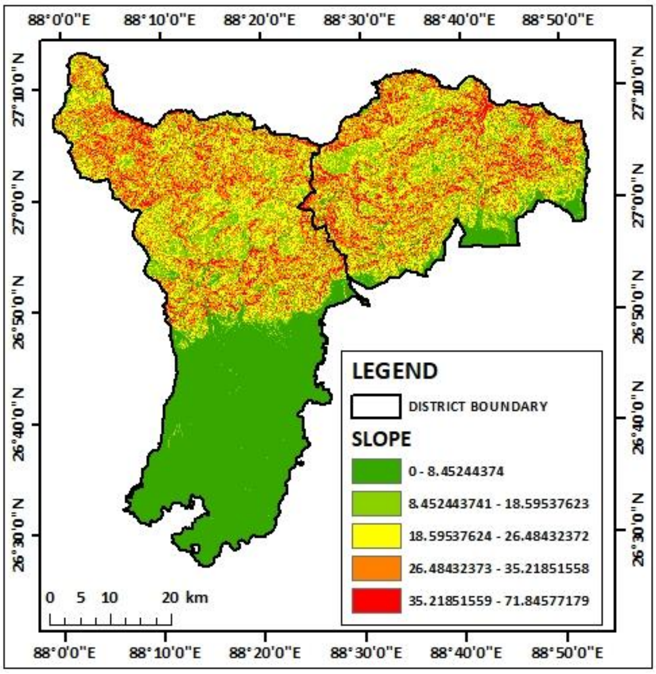


Figure 81: Slope map

Aspect map:

This map illustrates the spatial distribution of slope aspect across the study area. Slope aspect refers to the compass direction that a slope faces. The map reveals a variation in slope aspect, with different directions represented by different colors. This information is crucial for understanding solar radiation exposure, microclimate variations, and the potential impact of sunlight on various processes, such as erosion, vegetation growth, and snowmelt. For example, slopes facing south or west generally receive more sunlight, leading to higher temperatures and increased evaporation. Understanding the spatial pattern of slope aspect is essential for assessing microclimate conditions, vegetation distribution, and the overall impact of solar radiation on the landscape.

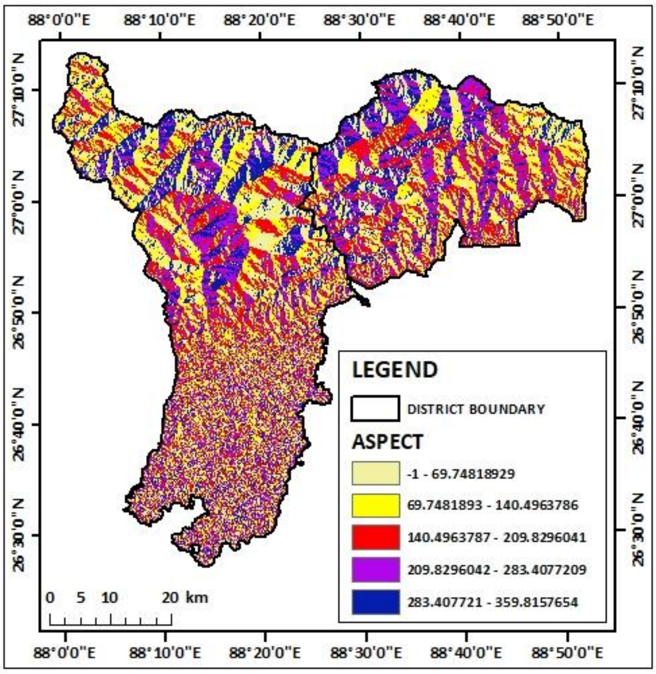


Figure 82: Aspect map

Curvature map:

This map illustrates the spatial distribution of topographic curvature across the study area. Topographic curvature is a measure of the rate of change in slope, and it provides insights into the shape of the land surface. The map reveals a variation in curvature, with different values indicating different landform types. Positive curvature values indicate convex slopes, while negative values indicate concave slopes. Understanding the spatial pattern of curvature is crucial for assessing landscape stability, water flow patterns, and the overall impact of topography on the environment.

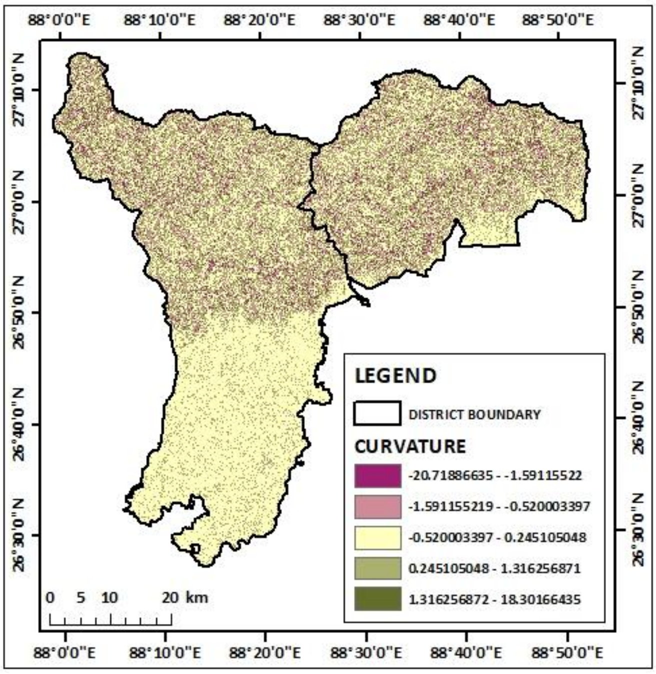


Figure 83: Curvature map

Existing landslide scur map:

This map illustrates the spatial distribution of landslide scars within the study area. Landslide scars are visible remnants of past landslides, and they provide valuable information about the areas that are prone to landslides. The map reveals that landslide scars are concentrated in specific regions, indicating areas with higher landslide susceptibility. These areas may be characterized by steep slopes, erodible soils, and intense rainfall, making them more prone to landslides. Understanding the spatial pattern of landslide scars is crucial for identifying areas at risk of future landslides and implementing appropriate mitigation measures.

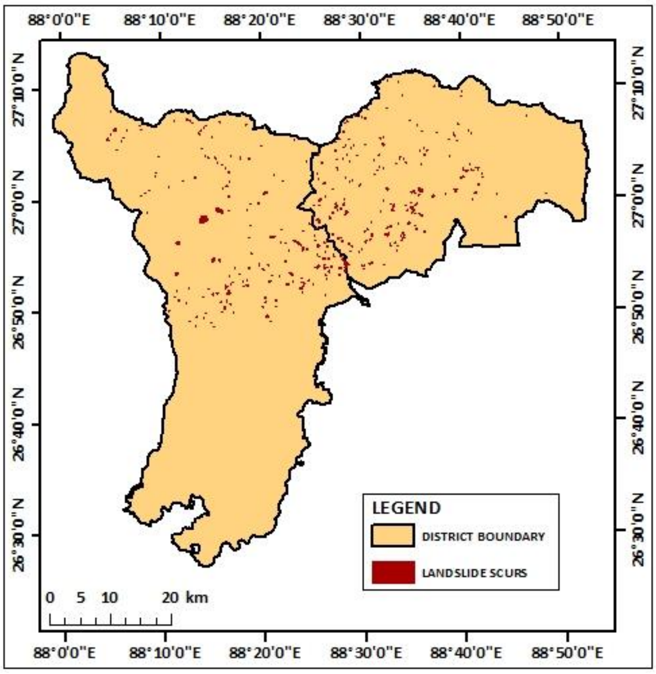


Figure 84: Existing landslide scurs map

Weighted Sum map:

This map illustrates the spatial distribution of landslide susceptibility based on the weighted sum method. The weighted sum method combines multiple factors like soil texture, stream power index, topographic wetness index, geomorphology, slope steepness and length, rainfall intensity, land use/land cover, drainage density, distance from roads, rivers, and settlements, slope aspect, curvature, and existing landslide scars. The map reveals a variation in landslide susceptibility, with higher values indicating areas with greater risk of landslides. These areas are likely characterized by a combination of factors that make them more prone to landslides, such as steep slopes, erodible soils, and intense rainfall. Understanding the spatial pattern of landslide susceptibility is crucial for identifying areas at risk of future landslides and implementing appropriate mitigation measures.

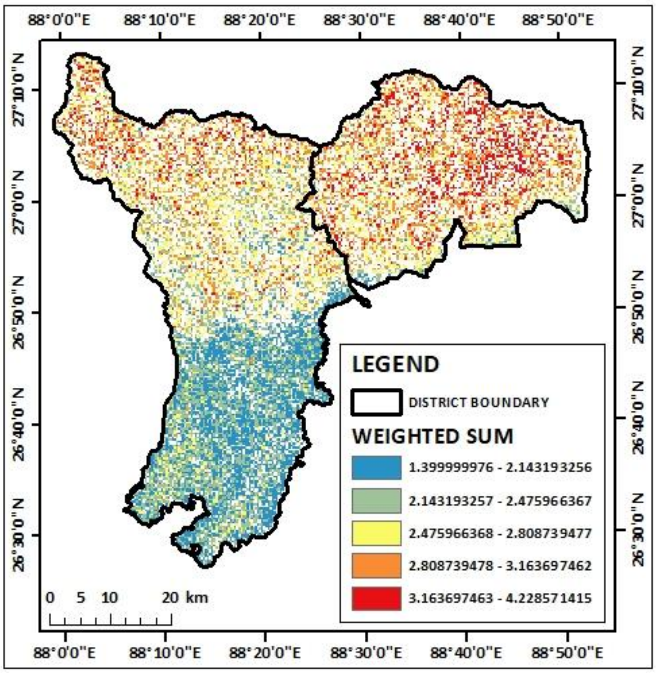


Figure 85: Weighted Sum map

IDW map based on landslide susceptible zones:

This map illustrates the spatial distribution of landslide susceptibility across the study area. The map reveals a variation in landslide susceptibility, with different levels of risk represented by different colors. High and very high landslide susceptibility zones are concentrated in specific regions, indicating areas that are more prone to landslides. These areas may be characterized by steep slopes, erodible soils, and intense rainfall, making them more vulnerable to landslides. Understanding the spatial pattern of landslide susceptibility is crucial for identifying areas at risk of future landslides and implementing appropriate mitigation measures.

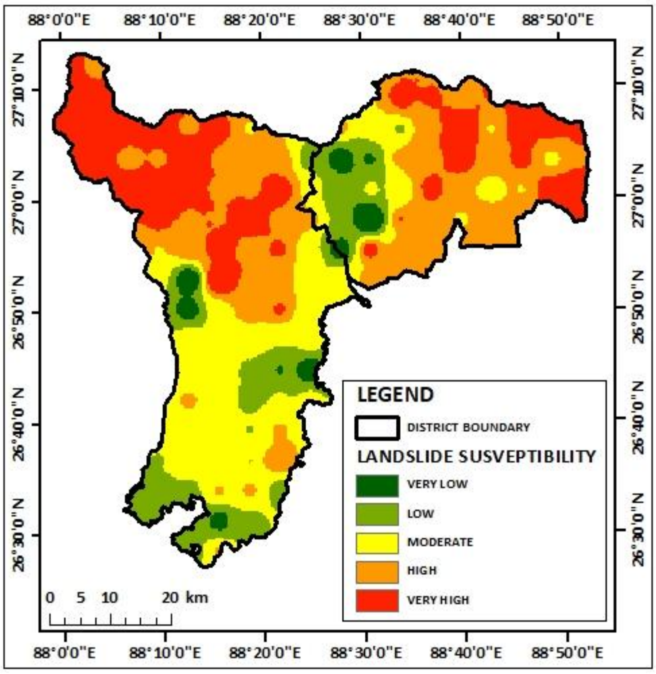


Figure 86: IDW map based on landslide susceptible zones

The results section presented the spatial distribution of various factors influencing landslide susceptibility, including geomorphology, slope characteristics, rainfall intensity, land use/land cover, drainage density, distance from roads, rivers, and settlements. The weighted sum method was employed to integrate these factors and generate a landslide susceptibility map. The discussion section delved into the interpretation of these maps, highlighting areas of high and low susceptibility. It analysed the factors contributing to landslide occurrences and discussed the implications of the findings for risk assessment and mitigation planning.

5. CONCLUSION:

The study successfully assessed the spatial distribution of landslide susceptibility in the Darjeeling and Kalimpong districts of West Bengal using a combination of geospatial techniques and multiple factors. The analysis revealed a significant variation in landslide susceptibility across the study area, with certain regions exhibiting higher risk due to factors like steep slopes, erodible soils, intense rainfall, and proximity to rivers and settlements. The weighted sum method proved effective in integrating these factors and generating a comprehensive landslide susceptibility map.

The results of this study have important implications for land use planning, disaster management, and sustainable development in the region. By identifying areas of high landslide susceptibility, policymakers and stakeholders can prioritize mitigation measures, such as slope stabilization, reforestation, and early warning systems. Additionally, this study provides a valuable baseline for future monitoring and assessment of landslide risk. However, it is important to note that this study has certain limitations, including the availability and accuracy of input data. Future research can further refine the landslide susceptibility assessment by incorporating additional factors, such as seismic activity and groundwater conditions, and by using more advanced modeling techniques.

THEMATIC REPRESENTATION OF POTENTIAL SITES FOR GROUND WATER RECHARGE USING MCDM TECHNIQUE: A CASE STUDY ON BANKURA SADAR, WEST BENGAL

1. INTRODUCTION:

Groundwater is a critical resource for sustaining agricultural, industrial, and domestic needs, especially in regions where surface water availability is limited. The increasing demand for groundwater, coupled with the impacts of climate change and human activities, has led to significant depletion in many areas. Identifying and prioritizing potential sites for groundwater recharge is essential to ensure sustainable water management. This process requires a detailed analysis of multiple environmental and anthropogenic factors that influence the infiltration and storage of water in aquifers.

One of the most significant factors in determining groundwater recharge potential is soil texture. Soils with high permeability, such as sandy soils, facilitate greater infiltration compared to clayey soils. Similarly, geomorphology and geology play a vital role in understanding the natural recharge zones. Features like alluvial plains, fractured rock formations, and porous geological strata are highly conducive to groundwater recharge. The slope of the terrain also influences the movement of surface runoff; flat or gentle slopes allow water to percolate into the ground, whereas steep slopes lead to rapid runoff, reducing recharge potential.

Rainfall intensity is another critical parameter, as areas with moderate and well-distributed rainfall are more favourable for recharge. Land use and land cover (LULC) patterns, including forests, agricultural fields, and urbanized areas, further dictate the recharge capacity. Vegetated lands, for instance, often enhance infiltration, while urban areas with impervious surfaces hinder it. Drainage density, which reflects the concentration of drainage channels, is also a key consideration. High drainage density often indicates areas with significant runoff and low infiltration capacity, while lower drainage density may indicate regions more suitable for recharge.

Proximity to roads, rivers, settlements, and other infrastructure also significantly impacts the selection of recharge sites. Areas close to rivers or water bodies are often more favourable due to the higher availability of water, while regions near roads and settlements may be less desirable due to pollution risks and reduced permeability. Lineament density, which represents the presence of fractures and faults, is a vital geological factor, as these features can enhance groundwater movement and storage.

By integrating these diverse criteria using geospatial technologies like remote sensing and GIS, a comprehensive analysis can be conducted to delineate suitable sites for groundwater recharge.

2. STUDY AREA:

The study area for the assessment of potential groundwater recharge sites is the Bankura Sadar subdivision of Bankura district, located in the state of West Bengal, India. This region lies between approximately 22°38'N to 23°38'N latitude and 86°36'E to 87°22'E longitude, covering a significant portion of the district's diverse geomorphological and hydrological landscape. Bankura Sadar is characterized by a combination of undulating terrain, plateaus, and plains, offering a unique setting for groundwater studies.

The region is predominantly rural, with agriculture serving as the primary occupation for the local population. Its climate is classified as tropical, marked by distinct wet and dry seasons. The average annual rainfall ranges from 1,200 mm to 1,400 mm, concentrated mainly during the monsoon months from June to September. This rainfall pattern makes the region highly dependent on effective water conservation and recharge practices to sustain water availability during the dry season.

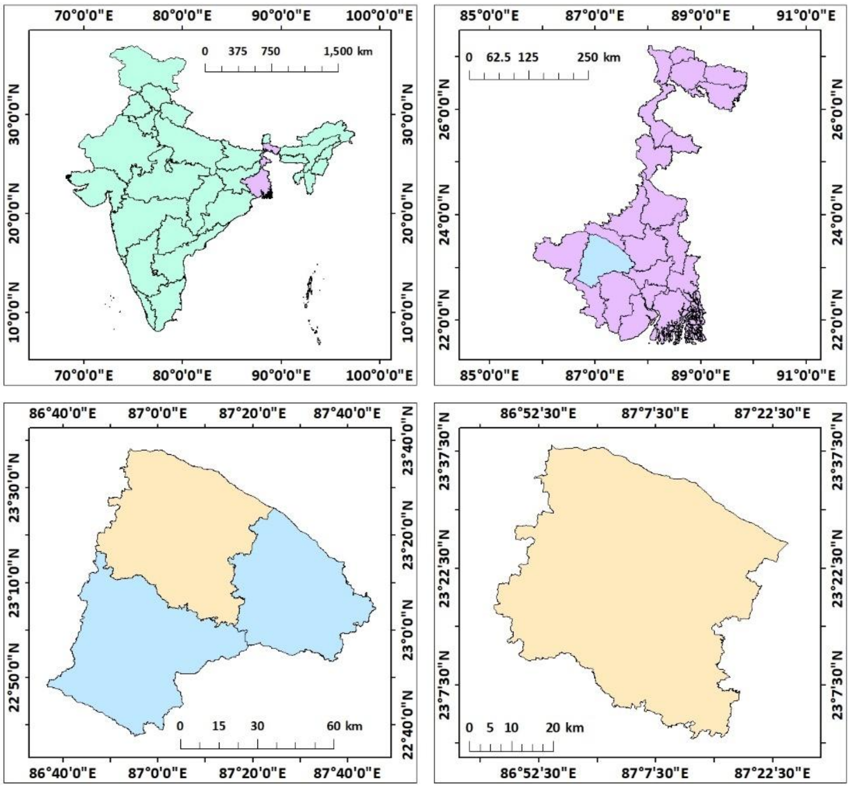


Figure 87: Location map of Bankura sadar

The geology of Bankura Sadar is a mix of hard rock formations, including granite and gneiss, interspersed with alluvial deposits. These features play a critical role in determining the area's groundwater potential. The soil types vary across the subdivision, with sandy loam and lateritic soils being dominant. Vegetation cover is sparse in certain parts, while forests and cultivated lands occupy other areas, adding to the spatial variability in land use and land cover (LULC).

Hydrologically, the area is drained by the major river systems of the Damodar and Dwarakeswar, along with their tributaries. Drainage density and the presence of small water bodies contribute to the recharge dynamics. The presence of settlements, roads, and other infrastructure adds anthropogenic complexity to the region's natural recharge potential. These diverse factors make Bankura Sadar a compelling case study for groundwater recharge site selection using geospatial techniques.

3. DATA SOURCES AND METHODOLOGY:

DATA SOURCES:

- **Soil Texture**:
    - Source: National Bureau of Soil Survey and Land Use Planning (NBSSLUP).
    - Details: Provides soil texture classification, essential for assessing erodibility.
- **Geomorphology**:
    - Source: Bhukosh portal, Geological Survey of India (GSI).
    - Details: Provides geomorphological maps for classifying terrain features.
- **Geology**:
    - Source: Bhukosh portal, Geological Survey of India (GSI).
    - Details: Provides geological maps for classifying terrain features.
- **Slope**:
    - Derived from: Digital Elevation Model (DEM) obtained from the USGS Earth Explorer.
    - Details: Calculated using Slope feature in Spatial analyst tool in ArcGIS
- **Rainfall Intensity**:
    - Source: Climate Hazards Group InfraRed Precipitation with Station Data (CHRS data portal).
    - Details: Provides NETCDF data.
- **Land Use and Land Cover (LULC)**:
    - Source: ESRI Sentinel-2 Land Cover Explorer.
    - Details: Classified LULC data for assessing human and natural influences on soil erosion.
- **Drainage Density**:
    - Source: Downloaded spatial datasets from BB Bike.
    - Derived from: Processed in ArcGIS using line density feature.
- **Distances from Road, River, and Settlement**:
    - Source: Downloaded spatial datasets from BB Bike.
    - Derives from: Processed in ArcGIS to calculate Euclidean distances for each feature.

- **Lineament density**:
    - Derived from: Digital Elevation Model (DEM) obtained from the USGS Earth Explorer.
    - Details: Processed in ArcGIS using line density feature.
- **Waterbody density**:
    - Source: Downloaded spatial datasets from BB Bike.
    - Derives from: Processed in ArcGIS using point density feature.

These diverse data sources and derived parameters provide a comprehensive framework for analysing ground water recharge potential zones using geospatial tools.

METHODOLOGY:

The Pairwise Comparison Matrix (PCM) is a valuable tool for decision-making processes, especially in multi-criteria decision analysis (MCDA). It allows for a structured and systematic comparison of multiple criteria or alternatives. First, relevant criteria are identified to evaluate the alternatives. A square matrix is then created, with each row and column representing a criterion. Each criterion is compared pairwise with every other criterion, assigning a numerical value to each comparison on a scale of 1 to 9. The pairwise comparison matrix is used to calculate the relative weights of each criterion, often using techniques like the Analytic Hierarchy Process (AHP). Once the weights are assigned, each alternative is evaluated against each criterion. The scores for each alternative are then multiplied by the corresponding weights, and the results are summed to obtain an overall score.

The methodology for this study incorporated a pairwise comparison matrix to evaluate and weight multiple criteria influencing groundwater recharge potential. The matrix, structured using the Analytical Hierarchy Process (AHP), compared criteria such as soil texture, geomorphology, geology, slope, rainfall intensity, land use and land cover (LULC), drainage density, proximity to roads, rivers, settlements, lineament density, and water body density. Each criterion was assigned relative importance based on expert judgment and hydrological principles, forming a weighted framework for subsequent spatial analysis.

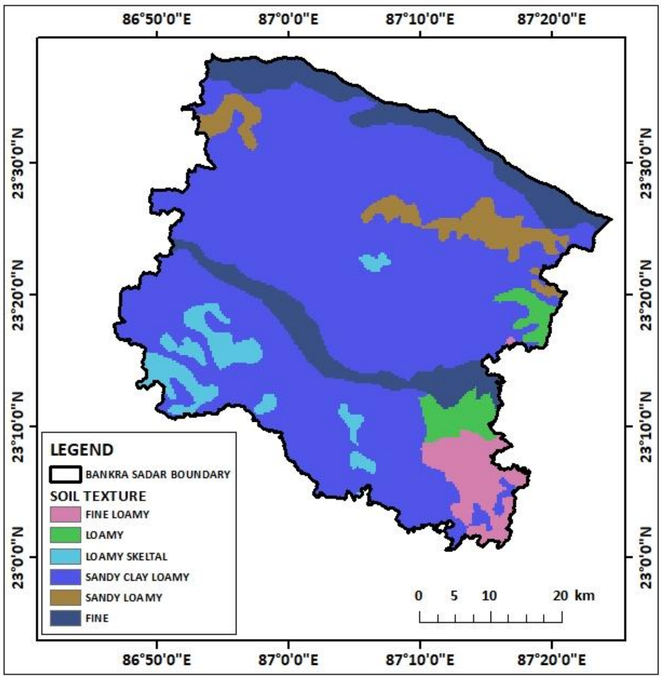
Criteria	Soil Texture	Geomorphology	Geology	Slope	Rainfall	LULC	Drainage Density	Dist.Roads	Dist.River	Dist.Settlement	Lineament Density	Wterbody Density
Soil Texture	1.000	3.000	5.000	7.000	5.000	3.000	3.000	5.000	7.000	5.000	7.000	5.000
Geomorphology	0.333	1.000	3.000	5.000	3.000	3.000	3.000	5.000	7.000	3.000	5.000	3.000
Geology	0.200	0.333	1.000	3.000	3.000	3.000	3.000	5.000	5.000	3.000	3.000	3.000
Slope	0.143	0.200	0.333	1.000	3.000	3.000	3.000	5.000	5.000	3.000	3.000	3.000
Rainfall Intensity	0.200	0.333	0.333	0.333	1.000	3.000	3.000	5.000	3.000	3.000	3.000	3.000
LULC	0.333	0.333	0.333	0.333	0.333	1.000	3.000	3.000	5.000	3.000	5.000	3.000
Drainage Density	0.333	0.333	0.333	0.333	0.333	0.333	1.000	3.000	3.000	3.000	3.000	3.000
Dist.Roads	0.200	0.200	0.200	0.200	0.200	0.333	0.333	1.000	3.000	3.000	3.000	3.000
Dist.River	0.143	0.143	0.200	0.200	0.333	0.200	0.333	0.333	1.000	3.000	3.000	3.000
Dist.Settlement	0.200	0.333	0.333	0.333	0.333	0.333	0.333	0.333	0.333	1.000	3.000	3.000
Lineament Density	0.143	0.200	0.333	0.333	0.333	0.200	0.333	0.333	0.333	0.333	1.000	3.000
Wterbody Density	0.200	0.333	0.333	0.333	0.333	0.333	0.333	0.333	0.333	0.333	0.333	1.000
SUM	3.429	6.743	11.733	18.400	17.200	17.733	20.667	33.333	40.000	30.667	39.333	36.000

Criteria	Soil Texture	Geomorphology	Geology	Slope	Rainfall	LULC	Drainage Density	Dist.Roads	Dist.River	Dist.Settlement	Lineament Density	Wterbody Density	SUM	AVG
Soil Texture	0.292	0.445	0.426	0.380	0.291	0.169	0.145	0.150	0.175	0.163	0.178	0.139	2.953	0.246
Geomorphology	0.097	0.148	0.256	0.272	0.174	0.169	0.145	0.150	0.175	0.098	0.127	0.083	1.895	0.158
Geology	0.058	0.049	0.085	0.163	0.174	0.169	0.145	0.150	0.125	0.098	0.076	0.083	1.377	0.115
Slope	0.042	0.030	0.028	0.054	0.174	0.169	0.145	0.150	0.125	0.098	0.076	0.083	1.175	0.098
Rainfall Intensity	0.058	0.049	0.028	0.018	0.058	0.169	0.145	0.150	0.075	0.098	0.076	0.083	1.009	0.084
LULC	0.097	0.049	0.028	0.018	0.019	0.056	0.145	0.090	0.125	0.098	0.127	0.083	0.937	0.078
Drainage Density	0.097	0.049	0.028	0.018	0.019	0.019	0.048	0.090	0.075	0.098	0.076	0.083	0.702	0.059
Dist.Roads	0.058	0.030	0.017	0.011	0.012	0.019	0.016	0.030	0.075	0.098	0.076	0.083	0.525	0.044
Dist.River	0.042	0.021	0.017	0.011	0.019	0.011	0.016	0.010	0.025	0.098	0.076	0.083	0.430	0.036
Dist.Settlement	0.058	0.049	0.028	0.018	0.019	0.019	0.016	0.010	0.008	0.033	0.076	0.083	0.419	0.035
Lineament Density	0.042	0.030	0.028	0.018	0.019	0.011	0.016	0.010	0.008	0.011	0.025	0.083	0.303	0.025
Wterbody Density	0.058	0.049	0.028	0.018	0.019	0.019	0.016	0.010	0.008	0.011	0.008	0.028	0.274	0.023

The weighted values derived from the pairwise comparison matrix was integrated into ArcGIS. Each criterion was represented as a thematic map, standardized and reclassified to align with the analysis framework. The reclassification ensured consistency across datasets and facilitated the use of weighted overlay techniques. Using these weighted overlays, a composite map was generated to illustrate zones of varying groundwater recharge potential, reflecting the combined influence of all criteria.

4. RESULTS AND DISCUSSION:

The results section presents the spatial distribution of various factors influencing groundwater recharge potential, including different parameters. The pairwise comparison matrix method was employed to assign weights to these factors. The discussion section delves into the interpretation of these maps, highlighting areas of high and low groundwater recharge potential. It analyses the factors contributing to groundwater recharge and discusses



the implications of the findings for sustainable groundwater management.

Soil texture map:

The soil texture map of Bankura Sadar provides critical insights into the infiltration potential across the region. Dominated by loamy and sandy loamy soils, these areas exhibit high permeability, making them suitable for groundwater recharge. Fine loamy and clay loamy regions, found in smaller pockets, present moderate to low permeability, which can hinder water percolation. The spatial variability of soil types highlights the interplay between texture and recharge efficiency. Sandy skeletal areas, often associated with rapid infiltration, dominate the northwestern parts, whereas fine-textured soils in the eastern zones restrict infiltration. This map effectively delineates zones where soil characteristics favour or impede groundwater recharge, serving as a primary input for multi-criteria analysis.

Geomorphology map:

The geology map captures the lithological diversity of Bankura Sadar, significantly influencing groundwater movement and storage. The region predominantly features Lower Gondwana formations, with their sedimentary nature supporting moderate groundwater storage. The Upper Gondwana strata, scattered in the central parts, exhibit low permeability due to compact sandstone layers. The northern areas are characterized by crystalline rocks, including gneiss and unclassified metamorphics, which restrict recharge due to their low porosity. However, fractures and weathered zones in these hard rock regions create localized recharge opportunities. The detailed representation of lithology underscores the critical role of geological characteristics in shaping recharge potential.

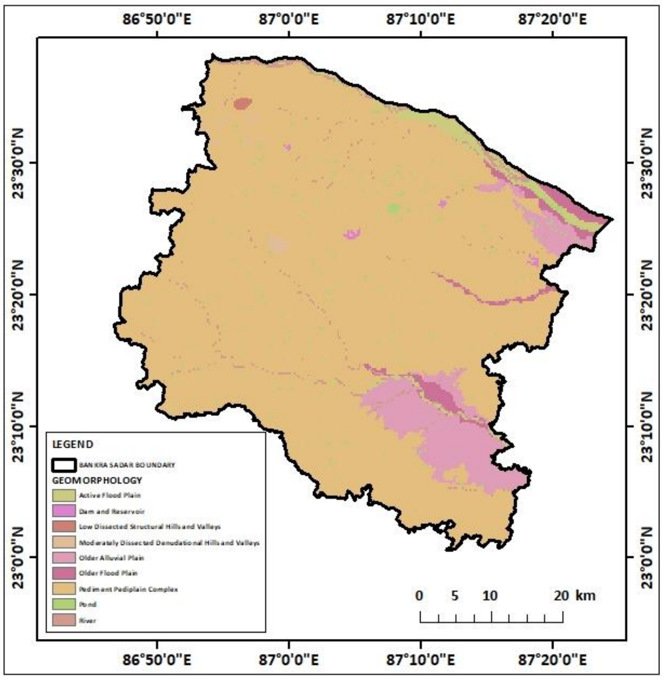


Figure 89: Geomorphology map

Geology map:

The geology map captures the lithological diversity of Bankura Sadar, significantly influencing groundwater movement and storage. The region predominantly features Lower

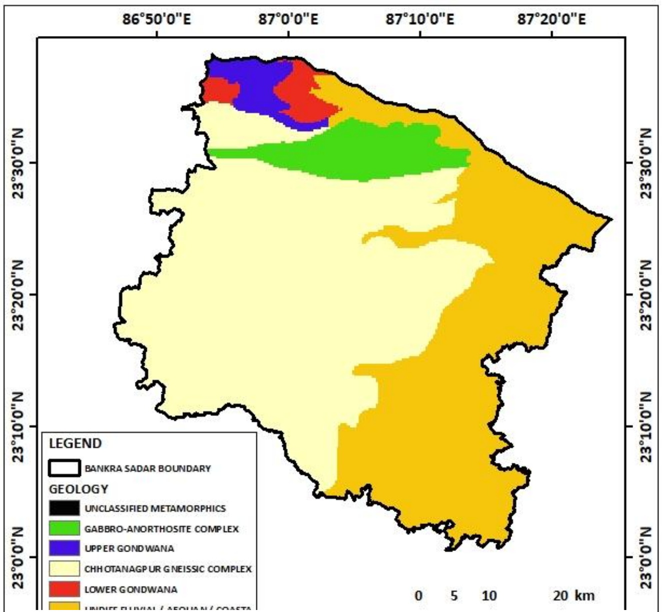
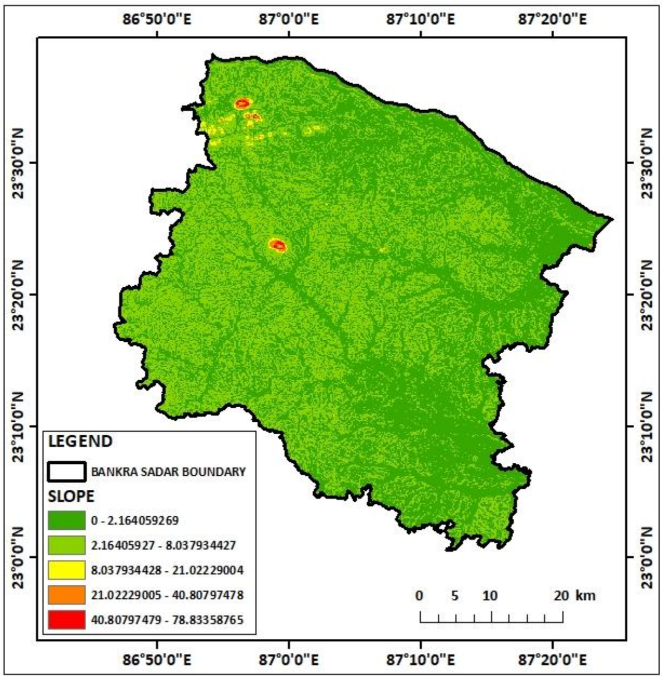


Figure 90: Geology map

Gondwana formations, with their sedimentary nature supporting moderate groundwater storage. The Upper Gondwana strata, scattered in the central parts, exhibit low permeability due to compact sandstone layers. The northern areas are characterized by crystalline rocks, including gneiss and unclassified metamorphics, which restrict recharge due to their low porosity. However, fractures and weathered zones in these hard rock regions create localized recharge opportunities. The detailed representation of lithology underscores the critical role of geological characteristics in shaping recharge potential.



Slope map:

The slope map illustrates the spatial variability of terrain steepness within the Bankura Sadar subdivision. A wide range of slopes is observed, from gentle to moderately steep. Areas characterized by gentle slopes are more favorable for groundwater recharge as they allow for greater infiltration of rainwater into the subsurface. In contrast, steeper slopes may lead to increased surface runoff, reducing the potential for groundwater replenishment. By analyzing the slope map, we can identify potential recharge zones and areas where soil conservation measures may be necessary to enhance groundwater recharge.

Rainfall intensity map:

This map illustrates the spatial distribution of rainfall intensity within the Bankura Sadar subdivision. The map reveals a variation in rainfall patterns, with areas of higher rainfall intensity concentrated in specific regions. These areas likely benefit from higher rainfall, which can contribute to groundwater recharge. However, it's important to consider other factors like soil permeability and topography to assess the actual recharge potential. Understanding the spatial variability of rainfall intensity is crucial for identifying potential groundwater recharge zones and for

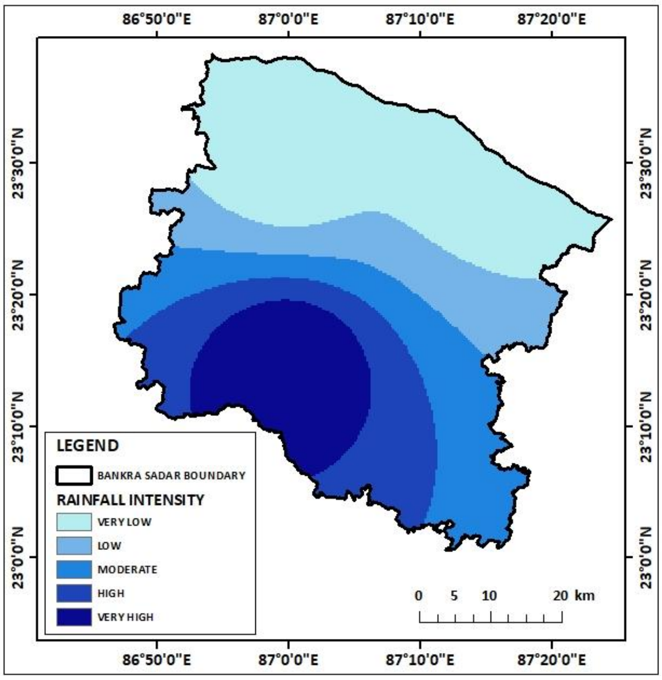


Figure 92: Rainfall intensity map

implementing appropriate water resource management strategies.

LULC map:

This map presents the land use and land cover (LULC) classification of the Bankura Sadar subdivision. The LULC categories depicted include dense forest, flooded vegetation, agricultural land, built-up areas, bare land, and scrub field. The spatial distribution of these LULC classes plays a significant role in groundwater recharge. Permeable surfaces like forests and agricultural land can facilitate infiltration of rainwater, whereas impervious surfaces like built-up areas hinder infiltration. Understanding the LULC pattern is crucial for identifying potential groundwater recharge zones and for implementing land use planning strategies that promote groundwater recharge.

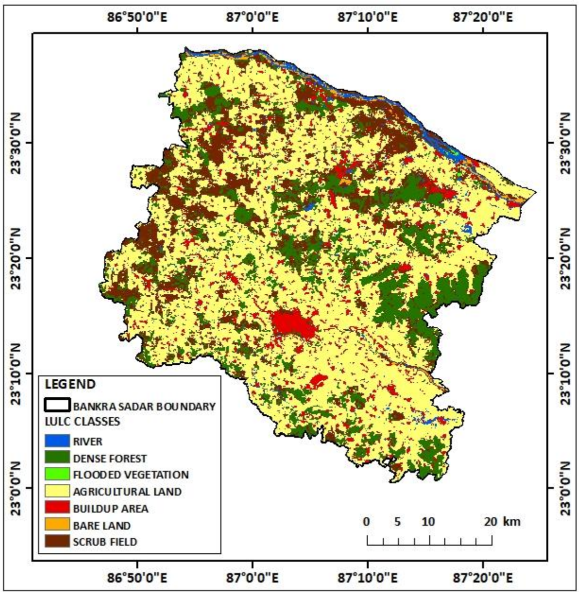


Figure 93: LULC map

Drainage density map:

This map illustrates the spatial distribution of drainage density within the Bankura Sadar subdivision. Drainage density is a measure of the total length of streams in a given area, and it provides insights into the hydrological processes and landscape characteristics. The map reveals a variation in drainage density, with higher values concentrated in certain regions. Areas with higher drainage density may have a greater potential for groundwater recharge as they facilitate the infiltration of rainwater. However, it's important to consider other factors like soil permeability and topography to assess the actual recharge potential. Understanding the spatial pattern of drainage density is crucial for identifying potential groundwater recharge zones and for implementing appropriate water resource management strategies.

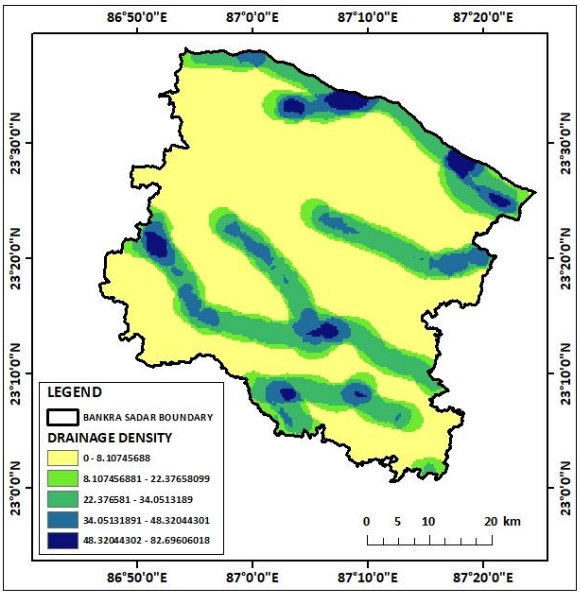


Figure 94: Drainage density map

Distance from road map:

This map illustrates the spatial distribution of distances from roads within the Bankura Sadar subdivision. Areas closer to roads may have increased accessibility for water extraction and distribution, potentially impacting groundwater recharge. However, it's important to consider other factors like soil permeability, topography, and land use to assess the actual recharge potential. Understanding the spatial pattern of distances from roads is crucial for identifying potential groundwater recharge zones and for implementing appropriate water resource management strategies.

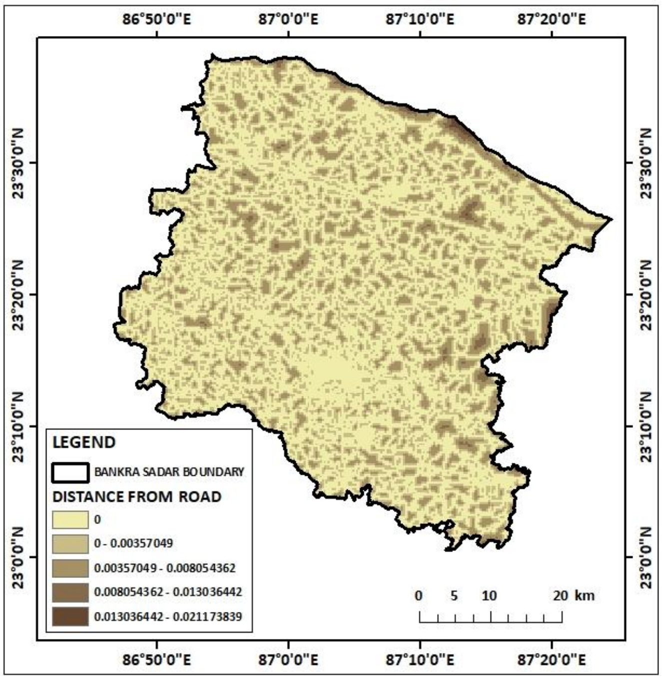


Figure 95: Distance from road map

Distance from river map:

This map illustrates the spatial distribution of distances from rivers within the Bankura Sadar subdivision. Areas closer to rivers may have increased groundwater recharge potential due to riverbank filtration and increased moisture availability. However, it's important to consider other factors like soil permeability, topography, and land use to assess the actual recharge potential. Understanding the spatial pattern of distances from rivers is crucial for identifying potential groundwater recharge zones and for implementing appropriate water resource management strategies.

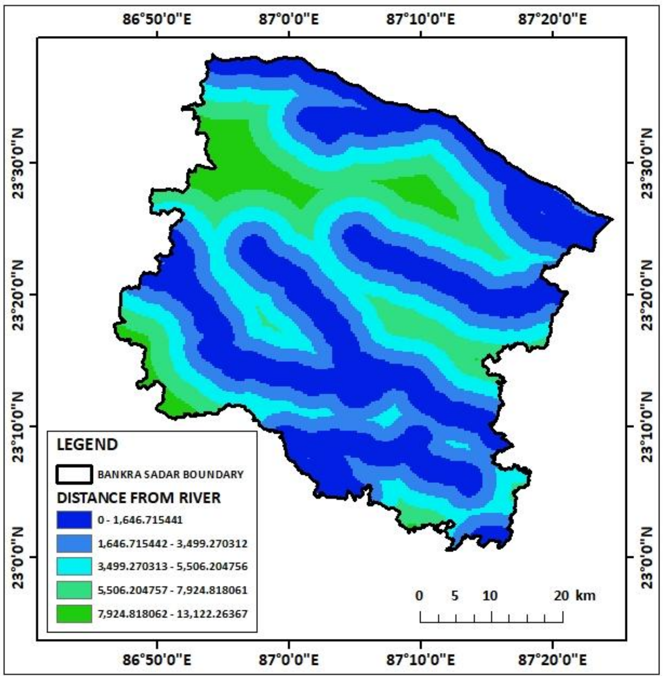


Figure 96: Distance form river map

Distance from settlement map:

This map illustrates the spatial distribution of distances from settlements within the Bankura Sadar subdivision. Areas closer to settlements may experience increased anthropogenic activities, such as groundwater extraction and pollution, which can impact groundwater recharge. However, it's important to consider other factors like soil permeability, topography, and land use to assess the actual recharge potential. Understanding the spatial pattern of distances from settlements is crucial for identifying potential groundwater recharge zones and for implementing appropriate water resource management strategies.

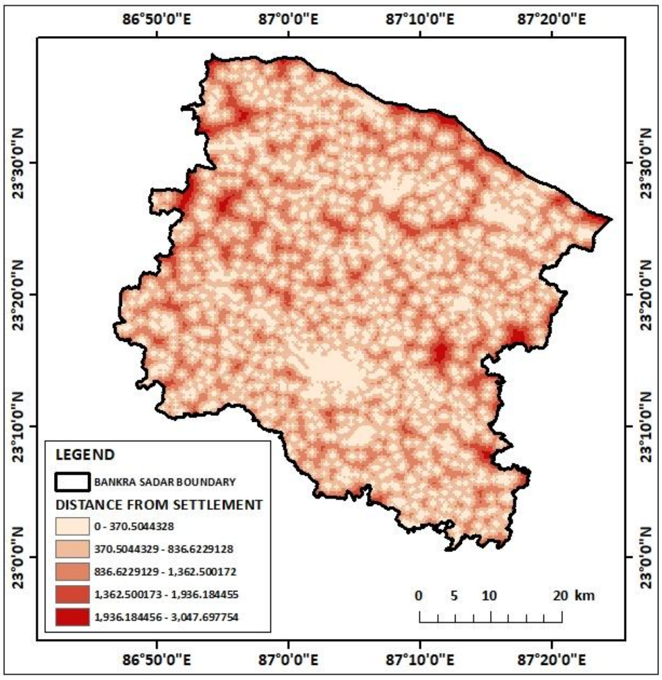


Figure 97: distance from settlement map

Lineament density map:

This map illustrates the spatial distribution of lineament density within the Bankura Sadar subdivision. Lineaments are linear features on the Earth's surface, such as faults and fractures, which can influence groundwater flow and recharge. Areas with higher lineament density may have increased groundwater potential due to the presence of fractures that can facilitate water infiltration. However, it's important to consider other factors like soil permeability, topography, and land use to assess the actual recharge potential. Understanding the spatial pattern of lineament density is crucial for identifying potential groundwater recharge zones and for implementing appropriate water resource management strategies.

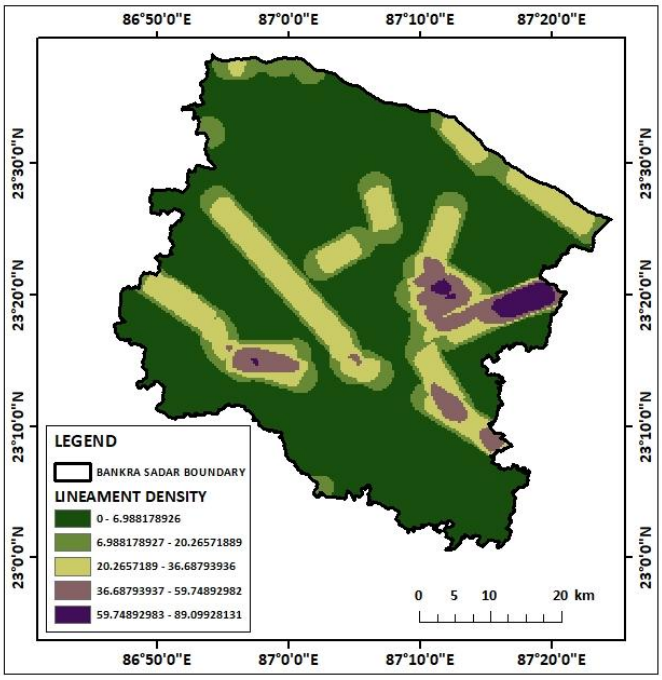


Figure 98: Lineament density map

Waterbody density map:

This map illustrates the spatial distribution of water body density within the Bankura Sadar subdivision. Water bodies, such as rivers, lakes, and ponds, can significantly contribute to groundwater recharge through infiltration. Areas with higher water body density may have increased groundwater recharge potential. However, it's important to consider other factors like soil permeability, topography, and land use to assess the actual recharge potential. Understanding the spatial pattern of water body density is crucial for identifying potential groundwater recharge zones and for implementing appropriate water resource management strategies.

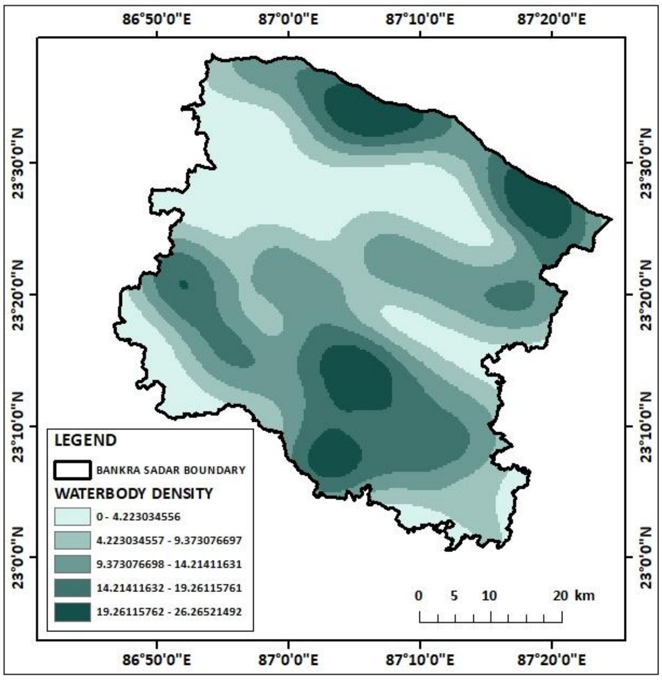


Figure 99: Waterbody density map

Ground water recharge potential zones map:

The groundwater recharge potential map of Bankura Sadar subdivision reveals a spatial variation in recharge potential zones. The map identifies five distinct zones: very low, low, moderate, high, and very high potential zones.

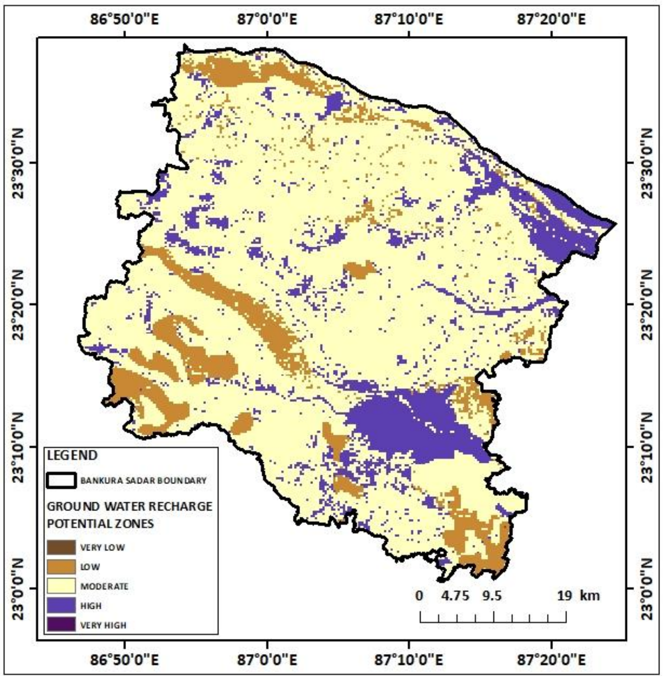


Figure 100: Ground water recharge potential zones map

Very Low Potential Zones: These areas are characterized by impervious surfaces like built-up areas and barren lands, limiting infiltration. The presence of steep slopes and high drainage density further reduces the potential for groundwater recharge in these zones.

Low Potential Zones: These zones often exhibit moderate slopes and moderate drainage density, hindering infiltration. The presence of a mix of permeable and impervious surfaces also contributes to the low potential for groundwater recharge in these areas.

Moderate Potential Zones: These zones are characterized by moderate slopes, moderate drainage density, and a mix of permeable and impervious surfaces. While the potential for groundwater recharge is moderate, these areas can still contribute significantly to overall groundwater storage.

High Potential Zones: These zones are characterized by gentle slopes, low drainage density, and high permeability, facilitating infiltration. The presence of dense forest cover and high waterbody density further enhances the potential for groundwater recharge in these areas.

Very High Potential Zones: These zones are often associated with dense forest cover, high water body density, and favourable geological and geomorphological conditions, promoting high infiltration rates. These areas have the highest potential for groundwater recharge and should be prioritized for conservation and protection.

Understanding the spatial distribution of these zones is crucial for sustainable groundwater management and for prioritizing areas for groundwater conservation and recharge interventions.

5. CONCLUSION:

The study successfully assessed the groundwater recharge potential of Bankura Sadar subdivision using a combination of geospatial techniques and multiple factors. The analysis revealed a significant variation in groundwater recharge potential across the study area, with certain regions exhibiting higher potential due to favourable factors like gentle slopes, permeable soils, and abundant vegetation. The pairwise comparison matrix method effectively prioritized these factors, leading to the generation of a comprehensive groundwater recharge potential map.

The results of this study have important implications for sustainable groundwater management in the region. By identifying areas of high groundwater recharge potential, policymakers and stakeholders can prioritize these areas for conservation and protection. Additionally, the study can inform decision-making regarding groundwater extraction, irrigation practices, and other water resource management strategies. It is crucial to implement appropriate measures to enhance groundwater recharge, such as afforestation, rainwater harvesting, and soil conservation practices, to ensure the long-term sustainability of groundwater resources. Additionally, it is crucial to involve local communities in groundwater management and conservation efforts. By raising awareness about the importance of groundwater and its sustainable use, communities can play a significant role in protecting and conserving groundwater resources.

However, it is important to note that this study has certain limitations, including the availability and accuracy of input data. Future research can further refine the groundwater recharge potential assessment by incorporating additional factors, such as soil moisture content, evapotranspiration rates, and climate change projections, and by using more advanced modelling techniques.